# Small Language Models as Judges: A Survey

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**Website:** anishh15.github.io/Awesome-SLM-as-a-Judge/
**GitHub:** github.com/anishh15/Awesome-SLM-as-a-Judge

## Abstract

Automated evaluation of language model outputs has shifted from costly human annotation toward model-based judging, yet the field has relied mostly on large proprietary models whose cost, opacity, and data-privacy risks limit scalable deployment. This survey examines *Small Language Models* (SLMs, $\leq$14B parameters) as judges, synthesizing **105 works** spanning the full pipeline from supervised fine-tuning and preference optimization through reinforcement learning with verifiable rewards to representation-based probing. We organize the field into five dimensions and distill five insights, each graded by the strength of its evidence: (**I1**) evaluation-specific training matters more than raw scale; (**I2**) reasoning tokens help only when they surface genuinely new discriminative signals; (**I3**) cross-family ensemble diversity decorrelates errors; (**I4**) debate value is bounded by information novelty, not round count; and (**I5**) SLM judge reliability tracks the verifiability gradient of the domain. In every surveyed paper that reports a trained judge next to its own base model under one protocol, evaluation-specific training improves the base model, with relative gains of 15–63%. Position-consistent accuracy protocols further show that much apparent judge performance is heuristic-driven, and that all judges, including frontier models, degrade sharply on hard evaluation pairs. Across surveyed studies, fine-tuned 3–8B judges match proprietary-model accuracy on standard benchmarks, and heterogeneous panels of smaller models reach comparable quality at over 7$\times$ lower API cost; this parity narrows in subjective domains that lack external verification. Small judges also make private, on-device evaluation practical for continuous Reinforcement Learning from Human Feedback (RLHF) loops. We close with a research agenda on calibration, adaptive inference, and generalization.

## 1 Introduction

Evaluating language model outputs reliably and at scale remains a central open problem in NLP. Human annotation, while the gold standard, fails to scale: assembling expert annotators for the thousands of iterative experiments required by modern RLHF pipelines (Ouyang et al., 2022) and constitutional alignment approaches (Bai et al., 2022) is prohibitively expensive. This bottleneck motivated the *LLM-as-a-Judge* paradigm (Zheng et al., 2023), in which a powerful model, typically GPT-4 (Achiam et al., 2023), serves as an automated proxy for human preference. MT-Bench demonstrated over 80% agreement between GPT-4 and expert humans; JudgeLM (Zhu et al., 2025) and Prometheus (Kim et al., 2024a) then extended this capability to open-source fine-tuned models.

The proprietary judge paradigm, however, carries structural costs that constrain its use. At evaluation scale $N \gg 10^4$, total API expenditure scales linearly in $N$, making continuous RLHF loops financially untenable for most research groups. Cloud-based judges also create data-privacy exposure, version-drift opacity, and latency that prevents real-time feedback loops. These constraints, combined with the rapid capability gains of recent small models (Abdin et al., 2024; Yang et al., 2025; Grattafiori et al., 2024; Guo et al., 2025), raise the question this survey sets out to answer: *can small language models reliably serve as judges, and if so, under what conditions?* The answer is not a simple yes or no. As we show, it depends on how the judge is trained, how much it is allowed to reason, whether it acts alone or in a panel, and above all on whether the domain admits external verification.

**The moment is right for this survey.** Two concurrent pressures converge: (1) SLMs in 2025–2026 achieve reasoning performance previously

requiring 70B+ models (Abouelenin et al., 2025; Guo et al., 2025); and (2) the LLM-as-a-Judge literature has matured, with dedicated benchmarks (Tan et al., 2025; Bi et al., 2026), training paradigms (Whitehouse et al., 2025; Deshpande et al., 2024), and systematic failure analyses (Wynn et al., 2025; Zhang et al., 2025a), to support a synthesis.

Existing surveys address either LLM judges broadly (Gu et al., 2024; Li et al., 2025a, 2024a), SLMs generally (Lu et al., 2024), or efficient LLMs from a systems perspective (Wan et al., 2023). Table 1 makes the gap concrete: none of them dedicates a section to sub-14B judge training, ensemble economics, reasoning-budget analysis, or domain-verifiability effects. To our knowledge this is the first survey specifically examining *SLMs as judges*, which raises distinct questions that fall outside the scope of general LLM-judge surveys: how does parameter budget constrain judging capability? when do ensemble strategies with sub-10B models pay off? and can representation-based evaluation bypass generation entirely? The question of whether automated judges need validation (Shankar et al., 2024) becomes acute for smaller models with narrower training distributions.

Table 5 (Appendix A) further delineates what is already known from the general LLM-judge literature and what this survey contributes as SLM-specific findings for each insight.

**Scope and Methodology.** We survey papers through mid-2026 from Semantic Scholar, arXiv, ACL Anthology, and Google Scholar. Inclusion required that (i) the work involve models with $\leq$14B parameters in an evaluative role, or (ii) it provide methodology or baselines adopted by SLM judge papers. From $\sim$320 initial candidates, title-and-abstract screening and full-text assessment yielded the 105 works cited here, spanning individual judging (§3), ensemble and debate strategies (§4), reliability (§5), and future directions (§6). Appendix M documents the search strategy, inclusion criteria, and corpus statistics.

**Contributions.** (1) A five-dimension taxonomy (Figure 1) unifying 105 works; (2) five insights graded by evidence strength, summarized in Table 2 and detailed per insight in Appendix N; (3) a cross-system comparison on shared benchmarks (Table 10) and a 19-system master table; (4) a deployment decision tree and coverage-gap analysis; and (5) an agenda on calibration, adaptive inference, and domain generalization.

## 2 Background

We first fix the paradigm and vocabulary the survey builds on: how model-based judging is defined, and why a parameter budget became the central axis. This grounds the judge function and deployment economics used throughout.

### 2.1 The LLM-as-a-Judge Paradigm

LLM-based evaluation was systematized by Zheng et al. (2023) with MT-Bench, a static benchmark of 80 multi-turn questions across eight categories. The paradigm defines a *judge* $f_{\mathcal{J}}$ mapping a (question, response) or (question, response-A, response-B) tuple to a verdict $v \in \mathcal{V}$, where $\mathcal{V}$ is either a scalar (*pointwise direct assessment*) or a preference direction (*pairwise ranking*). A third mode, *listwise ranking*, orders multiple candidates simultaneously. GPT-4's agreement with expert humans exceeds 80%, comparable to human-to-human agreement, which established automated judging as a viable proxy (Zheng et al., 2023). Appendix A provides full historical context.

Subsequent work diversified both architectures and supervision: distilled judges (PandaLM, JudgeLM), rubric-conditioned scorers (Prometheus 2), human-preference-trained reward models (FLAMe), and compact generalists (GLIDER, 3.8B) progressively demonstrated that capable evaluation does not require frontier scale. Appendix A traces this full history.

### 2.2 Why SLMs? Definitions and Deployment Economics

Motivated by recent SLM surveys (Lu et al., 2024) that target the 100M–5B range, we broaden the operational definition: a *Small Language Model* has $\leq$14B *dense* parameters, deployable on a single consumer GPU (24GB VRAM class) at 8-bit precision without model parallelism: at one byte per parameter its weights take about 14GB, against about 28GB at 16-bit. For mixture-of-experts (MoE) architectures, we apply this threshold to *active* (per-token) parameters and report the total next to it. Prometheus 2's 8$\times$7B variant, for example, activates 12.9B of its 46.7B parameters per token, so it falls inside our scope while needing the memory of a much larger model. Throughout the survey, sizes are totals for dense models and active/total for MoE models.

The threshold is a deployment regime, not a capability claim. What fits on one GPU stays well

| Dimension | Gu et al. (2024) | Li et al. (2024a) | Li et al. (2025a) | Lu et al. (2024) | Wan et al. (2023) | Ours |
|---|---|---|---|---|---|---|
| Model scope | Any | Any | Any | 100M–5B | Any | ≤14B |
| Published | 2024 | 2024 | 2025 | 2024 | 2023 | 2026 |
| LLM judge overview | ✓ | ✓ | ✓ | – | – | ✓ |
| SLM judge training | – | – | – | ∼ | – | ✓ |
| Reasoning budget | – | – | – | – | ∼ | ✓ |
| Ensemble economics | ∼ | ∼ | ∼ | – | – | ✓ |
| Debate failure modes | – | – | ∼ | – | – | ✓ |
| Domain verifiability | – | – | – | – | – | ✓ |
| Bias taxonomy | ✓ | ✓ | ✓ | – | – | ✓ |
| Deployment framework | – | – | – | – | ∼ | ✓ |

Table 1: Comparison with prior surveys. **Model scope** is the parameter range each survey restricts itself to; Lu et al. (2024) covers 70 open SLMs at 100M–5B, while the judging surveys place no size bound. Feature rows indicate whether a survey dedicates a section (✓) or gives only passing mention (∼). The judging surveys predate the 2025–2026 literature on reinforcement-learned and representation-based judges that this survey covers, and none of them treats parameter budget as an organizing axis.

| Insight | Evidence | Main support | Boundary conditions | Design |
|---|---|---|---|---|
| **I1** Evaluation-specific training matters more than scale | CONSENSUS: 5 independent lines; across related benchmarks | Selene Mini, J1, GLIDER, JudgeLRM, J4R, GenPRM; the trained judge beats its own base model in 8 of 8 papers (+15–63%) | Smaller margins on fine-grained rubric correlation (FLASK) and on the hardest pairs (JudgeBench) | Same backbone and cross-study |
| **I2** Reasoning helps only when it adds new signal | CONSENSUS: 7 studies; across tasks | Thinking-mode gains in 0.6–4B judges; NoThinking wins at low budgets; accuracy collapses past a complexity threshold | The best budget depends on both the judge and the task difficulty | Budget sweeps within each study |
| **I3** Cross-family panels decorrelate errors | REPLICATED: 3 studies; QA, math, code | PoLL (over 7× cheaper than GPT-4), JudgeBoard, COSMosFL | Each member must beat chance; members should come from different families | Panel vs. single judge within each study |
| **I4** Debate value is bounded by information novelty | CONSENSUS: 5 studies; across tasks | Gains when agents hold distinct evidence or assigned roles; MAST failure analysis | Shared-context debate loses to single agents given equal compute | Debate vs. single agent within each study |
| **I5** Reliability tracks domain verifiability | CONSENSUS: 8 domains; across domains | Near parity in math and code, good agreement in factual QA | Widest gap in creative and medical evaluation | Cross-study |

Table 2: The five insights at a glance. **Evidence** gives the grade (§3) and how widely the result has been shown: within one benchmark, across related benchmarks, or across tasks and domains. **Design** records whether the supporting comparisons hold the backbone fixed or compare systems across studies. Appendix N lists every study behind each row.

defined as models improve, whereas a capability-anchored boundary erodes: defining SLMs as "models that cannot match GPT-4 at judging" would already exclude most systems surveyed here. Resource-anchored scoping is standard, with Lu et al. (2024) anchored to edge deployment and Wan et al. (2023) to systems-level efficiency. What changes is what fits inside the envelope, and that change is this survey's subject rather than a threat to its scope. Appendix B details the SLM landscape and MoE boundary cases.

Let $f_{\mathcal{J}}^{(B)} : \mathcal{Q} \times \mathcal{R} \to \mathcal{V}$ denote a judge operating under token budget $B$, and let the *Instruction Following Rate* (IFR) be the fraction of evaluations producing a parseable verdict. Verga et al. (2024) report that a heterogeneous panel of small and mid-tier models is over 7× less expensive than a single GPT-4 judge at API list prices, which brings continuous evaluation at the scale of millions of judgments within reach.

## 3 Taxonomy and Individual SLM Judges

**Taxonomy.** Figure 1 organizes SLM judging across five orthogonal dimensions, each yielding one of our five insights: **background** (§2); **individual SLM judges** (this section); **multi-agent evaluation strategies** (§4); **reliability and meta-evaluation** (§5); and **challenges and future directions** (§6). We begin with the simplest dimension, the single judge acting alone.

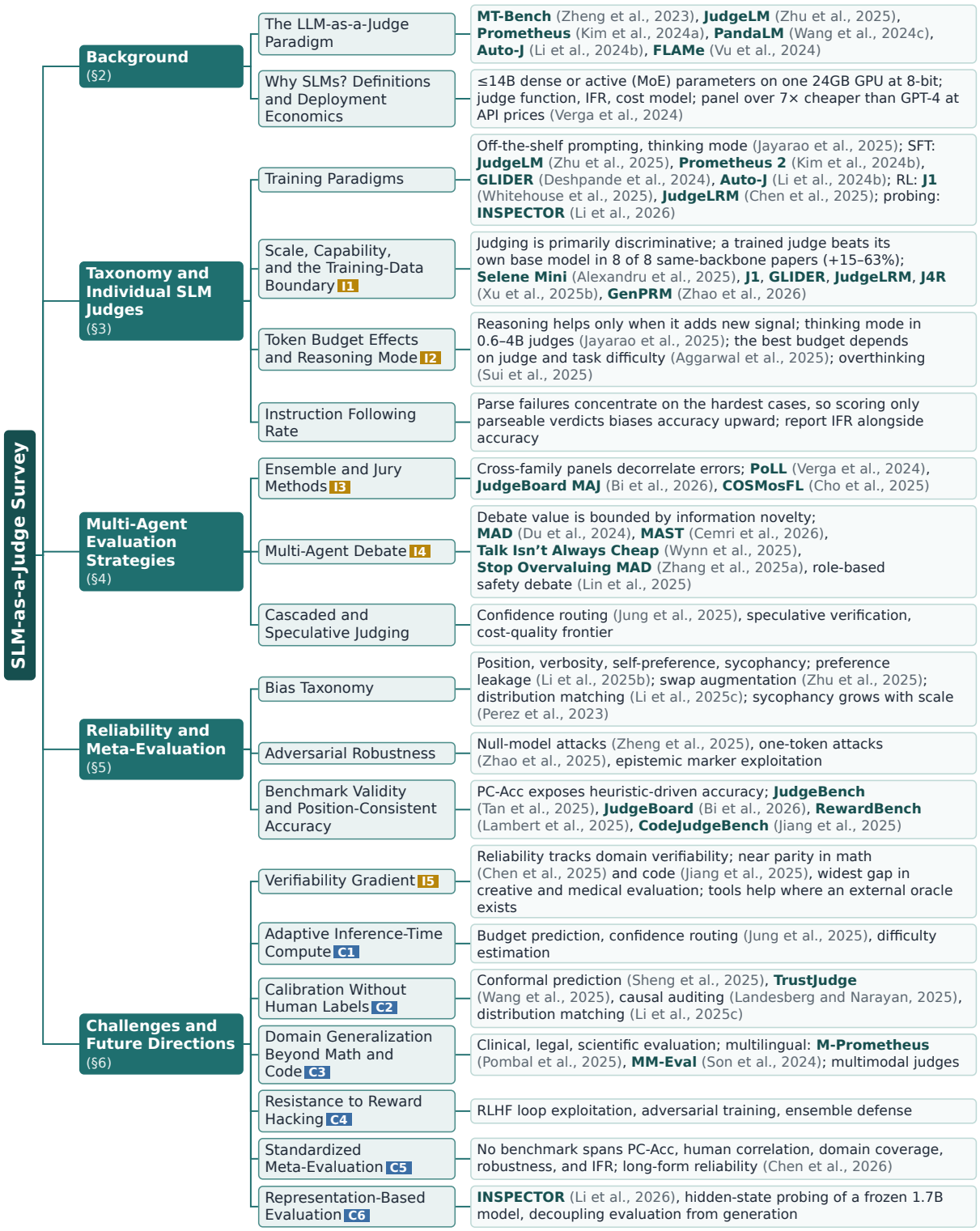



Figure 1: Taxonomy of the SLM-as-a-Judge survey. The five dimensions map to main paper sections (§2–§6), with insights I1–I5 and challenges C1–C6 positioned at their primary locations. Each leaf node lists representative papers and key concepts. Appendices A–L provide extended coverage of all branches.

**The individual judge.** An individual SLM judge realizes the map $f_{\mathcal{J}} : \mathcal{Q}\times\mathcal{R}^{(1:k)} \to \mathcal{V}$ of §2 with no auxiliary models, ensemble voting, or inter-agent communication. Its design space factorizes along four axes that interact in task-dependent ways,

$$f_{\mathcal{J}} = g\big(\underbrace{\pi}_{\text{train}}, \underbrace{B}_{\text{budget}}, \underbrace{m}_{\text{mode}}, \underbrace{\theta}_{\text{scale}}\big), \qquad (1)$$

the training paradigm $\pi$ (§3.1), the token budget $B$ and inference mode $m \in \{\text{THINK}, \text{DIRECT}\}$ (§3.3), and the parameter count $\theta$ (§3.2); the first two dominate quality. Cascaded evaluation (§4.3) optimizes $B$ dynamically. We take each axis in turn.

### 3.1 Training Paradigms

**Off-the-shelf prompting.** General-purpose SLMs can serve as zero-shot or few-shot judges without task-specific training; Jayarao et al. (2025) find that thinking-mode models (Qwen 3, 0.6B–4B) already achieve ∼10% higher accuracy than non-thinking counterparts through innate chain-of-thought. But off-the-shelf judges face three structural failure modes: inconsistent format adherence (frequent parse failures on complex rubrics for sub-3B models), unmitigated positional and verbosity biases, and susceptibility to adversarial phrases that inflate scores regardless of content.

**Supervised fine-tuning (SFT).** SFT on evaluation-specific data is the dominant paradigm. JudgeLM (Zhu et al., 2025) trains on GPT-4 judgments with *swap augmentation* (both orderings) to reduce position bias at the data level; Prometheus 2 (Kim et al., 2024b) adds rubric-conditioned Likert scoring; and Auto-J (Li et al., 2024b), a 13B generative judge, generalizes across 58 real-world scenarios (Appendix C).

**RL-based judge training.** RL offers a second-order improvement by optimizing the judge directly against evaluation accuracy. J1 (Whitehouse et al., 2025) trains thinking judges via Group Relative Policy Optimization (GRPO), casting both verifiable and open-ended tasks into a unified verifiable reward, with a consistency reward that penalizes position-dependent verdicts *at the objective level*. JudgeLRM (Chen et al., 2025) trains 3B–14B judges with outcome-driven GRPO rewards and finds a negative correlation between SFT gains and the share of reasoning-demanding samples, confirming that SFT is structurally insufficient for reasoning-intensive judgment.

**Representation-based judging.** A different paradigm bypasses generation entirely. Li et al. (2026) introduce INSPECTOR, probing hidden states of a frozen small LM to predict aspect-level scores via lightweight linear probes. Its basis is the *Semantic Capacity Asymmetry Hypothesis*: evaluation is a discriminative task accessible in early-to-mid layers, whereas generation requires the full forward pass plus sampling (Appendix C).

### 3.2 Scale, Capability, and the Training-Data Boundary

A prevalent assumption in early LLM judging work was that larger models make better judges. The record is more nuanced once evaluation-specific training is controlled for. Each insight carries an evidence tag: [CONSENSUS] for a finding replicated across four or more independent sources, [REPLICATED] for two or three, or one study with strong controls, and [PRELIMINARY] for a single study or an untested mechanism. Sources count as independent when they come from different groups and do not share both backbone and benchmark; systems above 14B serve only as reference points. The same levels grade rows in Appendix N.

EVALUATION-SPECIFIC TRAINING OUTWEIGHS SCALE BECAUSE JUDGING IS PRIMARILY DISCRIMINATIVE. [CONSENSUS: 5 independent lines of evidence] *Across surveyed studies, a 3.8B judge trained on evaluation data can outperform a far larger proprietary model such as GPT-4o under reported protocols, because recognizing quality differences requires pattern-matching against internalized evaluation criteria, not the full generative capacity needed for open-ended text production.* Eight surveyed papers report a trained judge next to its own base model under one protocol, which holds the backbone fixed, and in all eight, evaluation-specific training improves the base model, with relative gains from 15% to 63% (Table 33). In the clearest case, Selene Mini adds evaluation training to LLaMA-3.1-8B-Instruct and lifts RewardBench accuracy from 0.750 to 0.891, a 19% gain (Alexandru et al., 2025). Table 3 summarizes nine key systems, and Figure 2(a) plots reported RewardBench accuracy against size: among trained judges the curve does not climb with scale (8B: 0.891; 24B: 0.878).

Table 10 extends this across shared benchmarks. On FLASK, GLIDER (3.8B) reaches Pearson $r \approx 0.615$, above GPT-4o's 0.610 and GPT-4o-mini's 0.565 under the same evaluation run (Desh-

| System | Base | Params | Train | Mode | Strength | Open |
|---|---|---|---|---|---|---|
| JudgeLM | Vicuna | 7–33B | SFT | DA/PR | Speed | ✓ |
| Prometheus 2 | Mistral | 7B; 12.9B/46.7B | SFT+Merge | DA+PR | Rubric | ✓ |
| Selene Mini | LLaMA-3.1 | 8B | SFT+DPO | DA/PR | OOD Gen. | ✓ |
| GLIDER | Phi-3.5-mini | 3.8B | SFT | DA | Coverage | ✓ |
| FLAMe-RM | PaLM 2 | 24B† | SFT-Human | ORM | Fairness | – |
| Auto-J | LLaMA-2 | 13B | SFT | DA/PR | Diversity | ✓ |
| J1 | LLaMA-3.1 | 8B | GRPO | DA/PR | Reasoning | ✓ |
| JudgeLRM | Qwen | 3–14B | GRPO | DA | F1 Score | ✓ |
| INSPECTOR | frozen SLM | 1.7B | Probe | DA | Efficiency | ✓ |

Table 3: Selected SLM judge systems. DA = direct assessment, PR = pairwise ranking, ORM = outcome reward model. **Params** are totals for dense models and active/total for the 8×7B mixture-of-experts variant of Prometheus 2. †FLAMe-RM exceeds 14B and is shown as a reference point. Table 10 (Appendix C) compares systems on shared benchmarks.

pande et al., 2024). On PandaLM, JudgeLRM at 3B exceeds GPT-4 and at 7B outperforms DeepSeek-R1 by 2.6 F1 (Chen et al., 2025). J1-8B matches the prompted Llama-3.3-70B on RewardBench (85.7 against 85.4) (Whitehouse et al., 2025), CompassJudger-2 (7B) reports judgment accuracy competitive with DeepSeek-V3, which activates 37B of its 671B parameters (Zhang et al., 2025c), J4R (7B) beats GPT-4o by 6.7% on its overall score across PPE, JudgeBench, and ReasoningJudgeBench (Xu et al., 2025b), and with test-time scaling GenPRM (7B) surpasses a 72B scalar process reward model on ProcessBench (Zhao et al., 2026). The crossover depends on the benchmark. Under JudgeBench's original protocol, J1-8B reaches 42% position-consistent accuracy and no SLM reaches Claude-3.5-Sonnet's 64.3% (Whitehouse et al., 2025; Tan et al., 2025), and on fine-grained rubric correlation Prometheus 2 scores 0.470 Pearson on FLASK against GPT-4o-mini's 0.615 in the same run (Alexandru et al., 2025). Appendix N.1 lists supporting and contrary results side by side.

*Within the ≤14B regime, what a judge is trained on matters more than how large it is.*

**Threats to comparability.** Most cross-paper comparisons pit fine-tuned SLMs against proprietary baselines prompted zero-shot. That asymmetry *is* the claim, since I1 concerns what training buys rather than inherent capacity, and the same-backbone comparisons above isolate the training variable directly. Table 14 (Appendix D.1) breaks down six sources of asymmetry and the controls available for each.

### 3.3 Token Budget Effects and Reasoning Mode

The reasoning budget $B$ interacts with judging capability in a task-dependent, non-monotone way: $B^\star(\theta, d) = \arg\max_B \mathrm{Acc}(B \mid \theta, d)$ depends on both judge scale $\theta$ and task difficulty $d$.

REASONING TOKENS IMPROVE EVALUATION ONLY WHEN THEY SURFACE NEW DISCRIMINATIVE SIGNALS. [CONSENSUS: 7 studies] *Across surveyed studies, extended chain-of-thought (Wei et al., 2022) improves judgment for complex tasks but degrades verdict quality for simple ones, because the evaluative signal saturates and additional tokens introduce noise through self-contradiction.* Figure 2(b) places each study by its reported effect. Thinking models reach about 10% higher accuracy than non-thinking counterparts (Jayarao et al., 2025; Jiang et al., 2025), and in 0.6–4B judges the extra compute stays under 3× while consistency under positional and verbosity biases also improves (Jayarao et al., 2025). Because the gain is about the same at every size, it matters most for the smallest judges: about 19% in relative terms at 0.6B, against 13% at 4B. Sweet spots vary by (model, difficulty) pair (Aggarwal et al., 2025), and over-reasoning degrades performance (Sui et al., 2025; Shojaee et al., 2026). Budget-constrained prompting (Han et al., 2025) preserves accuracy while cutting compute, so adaptive budgets or dual-mode judges give the best cost-quality trade-off. See Appendix N.2 for the evidence table and Appendix D for analysis.

### 3.4 Instruction Following Rate

IFR (§2) is underreported and its failures are not random: they concentrate on the hardest cases, so scoring only parseable verdicts over-weights easy ones and biases accuracy upward (Appendix D, Eq. 14). Sub-3B non-thinking judges fail to produce a parseable verdict far more often on complex rubrics, while judges at 7B and above with format-reinforced training rarely fail. IFR must therefore be reported alongside accuracy.

## 4 Multi-Agent Evaluation Strategies

Individual judges, however capable, carry systematic biases and failure modes (§5). Multi-agent strategies pool judgments across models to address these limitations. Two approaches with distinct motivations and empirical profiles dominate, ensemble/jury voting and multi-agent debate, with




(a) I1: training vs. scale. (b) I2: effect of more reasoning. (c) I3: panel vs. GPT-4. (d) I5: verifiability gradient.

Figure 2: *The four quantitative trends behind the survey's insights.* **(a)** Published RewardBench accuracy against model size. Trained judges do not improve monotonically with scale, and on a fixed LLaMA-3.1-8B backbone the trained judge leads the prompted one by 14.1 points (I1). **(b)** Reasoning-budget studies by the effect of more reasoning they report (I2). **(c)** PoLL's agreement with human labels against cost. Its panel beats a single GPT-4 judge on three QA datasets at 7–8× lower cost (I3). **(d)** Reported quality by domain, ordered by the four-level framework of Appendix J, which places medical last because clinical judgement is contextual and no deterministic oracle settles it (I5). I4 is discussed in §4. *Panels (a) and (c) plot reported numbers; (b) and (d) place each study by its reported outcome.* Appendices N and O give the underlying comparisons.

Figure 8 (Appendix F) giving the full taxonomy.

### 4.1 Ensemble and Jury Methods

Ensemble judging aggregates the verdicts of $n$ independently queried judges, relying on the Condorcet Jury Theorem: if each juror is correct with probability $p > \frac{1}{2}$ and errors are independent, the majority-vote accuracy $\sum_{k>n/2} \binom{n}{k} p^k (1-p)^{n-k} \to 1$ as $n$ grows, so independent errors cancel.

CROSS-FAMILY JURY DIVERSITY DECORRELATES ERRORS THROUGH ARCHITECTURAL INDEPENDENCE. [REPLICATED: 3 studies] *Under reported protocols, heterogeneous panels of smaller judges outperform a single large judge at over 7× lower API cost, because distinct pretraining corpora and attention mechanisms yield more weakly correlated failure modes, better approximating the independence assumption that Condorcet aggregation relies on.* Figure 2(c) situates this result on the cost-quality plane. Verga et al. (2024) formalize the approach as PoLL (Panel of LLM Evaluators): a heterogeneous ensemble of Command R, GPT-3.5-Turbo, and Claude Haiku outperforms a single GPT-4 judge on multiple QA and preference benchmarks and shows less of the intra-model bias that leads a single judge to favor its own family. Bi et al. (2026) demonstrate that Multi-Agent Judging (MAJ) with SLM backbones can match standalone LLM judges on MATH even when each individual SLM falls substantially short. Cho et al. (2025) confirm on fault localization that diverse small-model panels match large-model performance. Appendix N.3 collects the evidence.

*Family* diversity matters more than stochastic diversity: distinct pretraining corpora, architectures, and alignment objectives leave members of different families failing on different inputs, which is what majority voting needs and what resampling one model cannot supply (Appendix E). *Draw jury members from distinct families rather than sampling one model repeatedly.*

### 4.2 Multi-Agent Debate

Multi-agent debate (MAD) (Du et al., 2024) runs multiple LLM instances that exchange and critique responses over multiple rounds. The motivation is information aggregation: diverse agents surface distinct evidence, and cross-criticism enables error correction beyond single-agent reasoning.

DEBATE VALUE IS BOUNDED BY INFORMATION NOVELTY ACROSS ROUNDS, NOT BY ROUND COUNT. [CONSENSUS: 5 studies] *Across surveyed studies, multi-agent debate improves evaluation when agents contribute genuinely different evidence or problem-solving strategies, but degrades performance in shared-context settings because additional rounds amplify sycophantic conformity rather than surfacing new information.* Wynn et al. (2025) and Zhang et al. (2025a) independently find MAD underperforms single-agent baselines on binary correctness judgment, and the MAST taxonomy (Cemri et al., 2026) attributes ∼37% of multi-agent failures to *Inter-Agent Misalignment*, agents preferring consensus to accuracy (Appendix G). Debate helps when asymmetry is high, whether from distinct evidence fragments (Du et al., 2024) or from opposing roles assigned to the agents (Liang et al., 2024). Small models benefit too. In safety evaluation, a critic, a defender, and a judge built on Qwen3-14B come within 0.03

Cohen's $\kappa$ of GPT-4o at about 46% of its cost (Lin et al., 2025). Table 36 (Appendix N.4) sorts these studies by asymmetry and the outcomes split cleanly along it, not by round count. *The critical design variable is information asymmetry.*

### 4.3 Cascaded and Speculative Judging

An alternative is cascaded judging: route evaluations to a small SLM judge, accept high-confidence verdicts, and escalate uncertain ones to a larger model (Jung et al., 2025). *Speculative judging* pushes this: a small judge proposes a verdict and a large judge verifies only on disagreement. On the cost-quality plane (Figure 7, Appendix D), cascade and speculative judging sit at the frontier's knee; self-consistency lies below it.

## 5 Reliability and Meta-Evaluation

Whether using individual or multi-agent judges, evaluation reliability requires analysis along three axes: structural biases, adversarial vulnerabilities, and benchmark validity.

### 5.1 Bias Taxonomy

We highlight the four most studied bias categories here; Appendix H gives the full six-category taxonomy with mechanisms and mitigations. **Position bias** favors responses by presentational order rather than quality (Zheng et al., 2023); it is quantified by *position-consistent accuracy*, $\text{PC} = \mathbb{1}[v_{ab} = v_{ba}]$, which credits a verdict only when it survives swapping the A/B order, and is mitigated by swap augmentation (Zhu et al., 2025), consistency-based RL rewards (Whitehouse et al., 2025), or inference-time swapping; fine-tuning 7–9B judges to match the distribution of human preferences raises consistency under swapped order by 7–81% (Li et al., 2025c). **Verbosity bias** rewards longer responses regardless of accuracy (Chen et al., 2024; Szymanski et al., 2025), conflating information volume with quality. **Self-preference** favors outputs with low perplexity under the judge's own distribution, and extends to related lineages as *preference leakage* (Li et al., 2025b): a judge trained on data from the evaluated model's family inflates its scores. **Sycophancy bias** aligns the verdict with expressed user expectations or authority cues rather than substantive quality.

SLM judges are more exposed to surface cues such as position and length (Appendix H, Table 22), but they are less prone to self-preference, and sycophancy grows with model size (Perez et al., 2023; Wei et al., 2024).

### 5.2 Adversarial Robustness

Judges also face structured adversarial attacks exploiting superficial signals. Zheng et al. (2025) show that constant, content-free responses achieve disproportionately high win rates; Zhao et al. (2025) find that a single token such as a colon can elicit false-positive rewards; and epistemic markers ("clearly", "definitively") inflate judged quality without improving content. SLM judges are particularly vulnerable due to weaker instruction following; heterogeneous ensembles provide partial protection since attacks tuned to one architecture transfer poorly to a diverse panel (Appendix K).

### 5.3 Benchmark Validity and Position-Consistent Accuracy

POSITION-CONSISTENT ACCURACY REVEALS THAT MUCH APPARENT JUDGE PERFORMANCE IS HEURISTIC-DRIVEN. [REPLICATED: 3 studies] *Under position-consistent accuracy protocols, even frontier judges degrade sharply on hard evaluation pairs, sometimes approaching random-choice baselines, indicating that much apparent performance relies on positional heuristics rather than substantive quality discrimination.* Tan et al. (2025) construct challenging pairs where PC-Acc is dramatically lower than naive accuracy; Bi et al. (2026) and Jiang et al. (2025) confirm this on mathematics and code. The practical implication is to always report PC-Acc alongside naive accuracy, since the gap between them measures how much apparent performance derives from heuristic exploitation.

This reconciles with I1: evaluation-specific training delivers parity on standard benchmarks, but PC-Acc reveals that this parity narrows on the hardest evaluation pairs and in subjective domains (I5), where all judges, including frontier models, degrade toward chance. Additional validity concerns include limitations of the human gold standard (Shankar et al., 2024), contamination risks (Bedemariam et al., 2025), and *preference leakage*: Li et al. (2025b) show that when judge and generator share a model family, the judge systematically overscores, inflating apparent performance on benchmarks like AlpacaEval 2.0 (Appendix I).

## 6 Challenges and Future Directions

SLM JUDGE RELIABILITY FOLLOWS THE VERIFIABILITY GRADIENT OF THE EVALUATION

DOMAIN. [CONSENSUS: 8 domains] *Across surveyed studies, SLMs approach LLM-judge quality in high-verifiability domains (math, code) where external verification compensates for limited parametric knowledge, but degrade in subjective domains where internalized knowledge is the irreducible bottleneck.* The gap narrows as domain verifiability $\nu(d)$ rises (Figure 2(d)): SLM judges reach near parity on mathematics (Bi et al., 2026; Chen et al., 2025) and code (Jiang et al., 2025), but drop sharply for creative tasks where no external oracle exists, though principle-guided judges can drive creative-writing improvement (Wei et al., 2025). Table 37 in Appendix N.5 maps the gradient across eight domains with evidence grading. The challenges below stem from this gradient; Appendix L expands them.

Six open challenges follow, each expanded into a full research direction in Appendix L.

- **C1: Adaptive inference-time compute.** $B^\star(\theta, d)$ varies sharply with difficulty (§3.3), but confidence-based routing (Jung et al., 2025) needs calibration SLM judges lack: they are overconfident where least accurate.
- **C2: Calibration without human labels.** Conformal prediction (Sheng et al., 2025), TrustJudge (Wang et al., 2025), causal surrogate auditing (Landesberg and Narayan, 2025), and fine-tuning 7–9B judges to match human preference distributions (Li et al., 2025c) are starting points, not general frameworks.
- **C3: Domain generalization beyond math and code.** Only 5% of 650 judge papers address multilingual or low-resource settings (Doğruöz et al., 2026); M-Prometheus (Pombal et al., 2025) is a first step, MM-Eval (Son et al., 2024) now tests judges in 122 languages, and multimodal SLM judges are unexplored.
- **C4: Resistance to reward hacking.** Generators trained against an SLM judge produce superficially convincing reasoning, and SLM robustness is less studied than LLM robustness.
- **C5: Standardized meta-evaluation.** No benchmark spans PC-Acc, human correlation, domain coverage, robustness, and IFR; long-form evaluation is weakest (Chen et al., 2026).
- **C6: Representation-based evaluation.** Probing the hidden states of a frozen 1.7B model closely approximates full LLM judges on GSM8K, MATH, and GPQA (Li et al., 2026); open-ended evaluation remains untested.

**From insights to deployment.** Read together, the insights give a simple recipe for choosing a judge (Figure 14, Appendix L.2). For verifiable, low-stakes evaluation such as math, code, and factual QA, a single fine-tuned 3–8B judge giving a direct verdict is a practical default, paired with a tool wherever an external oracle exists (I1, I5). When a domain is only partly verifiable, the budget decides the design: a cascade that escalates uncertain cases to a larger judge keeps spending low (I2, C1), while a panel of three judges from different families buys reliability when quality matters most (I3). Hard comparisons can go to thinking mode, which raises a small judge's accuracy by 13–19% (I2). Debate earns its cost only when the agents hold different evidence or roles (I4), and subjective, high-stakes evaluation keeps a human in the loop. Whatever the design, two numbers belong next to accuracy in every report: position-consistent accuracy, which exposes positional heuristics, and IFR, which exposes silent parse failures (Table 31). Figure 13 (Appendix L.1) shows where the evidence behind these choices concentrates: Qwen and LLaMA on RewardBench, with Gemma, DeepSeek-Distill, and safety underexplored.

## 7 Conclusion

This survey organizes 105 works on SLMs as judges into a five-dimension taxonomy with five insights. The convergent finding is that **reliable automated evaluation needs far fewer resources than the field has assumed**: across surveyed studies, fine-tuned 3–8B judges match proprietary models on standard benchmarks (I1), reasoning tokens help only when they carry new signal (I2), and heterogeneous juries decorrelate errors (I3), while the information-novelty boundary (I4) and the verifiability gradient (I5) mark where human oversight remains necessary. Calibration, multilingual coverage, and contamination resistance remain prerequisites for high-stakes deployment. These findings make SLM judges the practical default for verifiable, low-stakes evaluation rather than a budget compromise, with frontier models and human review reserved for the subjective, high-stakes end of the gradient. The throughline is that judging is discriminative and discrimination is cheaper than generation: scale buys knowledge, but most of what a judge needs comes from targeted training, verification, or panel diversity, so progress lies in better signals rather than larger judges.

## Limitations

This survey synthesizes findings from 105 papers published through mid-2026, but several limitations should be noted. **First**, we have not conducted new experiments; all quantitative claims are drawn from cited sources and subject to original study biases. **Second**, the survey focuses primarily on English-language judges; generalizing to low-resource languages risks underestimating domain-specific failure modes. **Third**, rapid field evolution means coverage through mid-2026 may not capture later developments. **Fourth**, our literature selection may underrepresent domain-specific research outside major NLP venues (Appendix M documents the search protocol). **Fifth**, panels (b) and (d) of Figure 2 place each study by the outcome it reports, so they show the direction of each result but not its size, and Figures 5, 6, and 7 have ordinal or schematic axes rather than measured quantities. Panels (a) and (c) plot published numbers, and Appendix O documents the source of every plotted element. **Finally**, IFR figures and cost estimates use varying setups, making cross-paper comparisons approximate.

**Potential Risks.** This work does not pose direct risks, but reliance on automated evaluation may introduce systematic biases that propagate into downstream model training. Deployment of SLM judges in critical applications (e.g., medical, legal, safety evaluation) should carefully consider robustness limitations documented in this survey. Over-reliance on a single judge architecture risks creating evaluation monocultures where shared failure modes go undetected. Practitioners should document judge identity, model version, and known biases to reduce evaluation drift over time.

## Ethical Considerations

This survey evaluates existing research on small language models as judges using publicly available papers, benchmarks, and datasets, ensuring transparency and reproducibility. No private or sensitive data was used, and all surveyed systems are assessed based on their published results under fair conditions.

Automated judges influence model alignment through RLHF and data curation. Poor-quality judges can systematically corrupt training signals, amplifying biases at scale. Practitioners should: calibrate SLM judges against human-annotated sets before deployment; disclose judge identity, provenance, and known biases; avoid automated judges as sole arbiters of high-stakes decisions without human review; and audit probe features for demographic bias. The information-novelty boundary of debate (§4) shows that consensus does not guarantee correctness.

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# Appendix

This appendix provides supplementary material including extended background, model landscape details, training methodology analysis, individual and ensemble judging results, debate mechanisms, bias documentation, benchmark descriptions, domain-specific evaluation, adversarial robustness analysis, and a research agenda.

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## Glossary of Specialized Terms

Table 4 collects the specialized terms this survey relies on, together with the section that introduces each.

The appendix follows the same arc as the main paper. We first establish *where the field came from* (the evaluation problem and the SLM landscape: §A–B), then *how SLM judges are built and behave alone* (training and individual judging: §C–D), then *how they are combined* (ensembles, the strategy taxonomy, and debate: §E–G), then *how far we can trust them* (biases, benchmarks: §H–I), and finally *where they break and where to go next* (domains, robustness, and the research agenda: §J–L). Each section expands a claim made in the main paper with full mechanistic detail.

## A Background: The LLM-as-a-Judge Paradigm

This section expands §2, tracing the evaluation problem from n-gram metrics to model-based judges and formalizing the judge function that the rest of the survey builds on.

Table 5 delineates what is already known from the general LLM-judge literature and what this survey contributes as SLM-specific findings.

### A.1 The Evaluation Problem in NLP

Language model evaluation has always faced a tension between *scalability* and *validity*. Human evaluation, the undisputed gold standard, is slow, expensive, and difficult to reproduce: assembling expert annotators for the thousands of iterative model comparisons required by modern RLHF pipelines (Ouyang et al., 2022) is infeasible at production scale. The field compensated with automatic metrics, but the growing capability of generative models exposed structural flaws in each successive paradigm.

**The n-gram era (pre-2022).** BLEU (Papineni et al., 2002) and ROUGE measure n-gram overlap between candidate and reference responses. These metrics suffer from four fundamental deficiencies:

- **Semantic insensitivity**: Valid paraphrases sharing no words with the reference receive near-zero scores.
- **Hallucination blindness**: Fluent, factually fabricated responses can achieve high BLEU scores if they share surface vocabulary with the reference.
- **Goodhart's Law susceptibility**: Optimizing for BLEU causes models to shorten outputs and repeat reference phrases, inflating scores while degrading quality.
- **Reference dependency**: Performance varies with reference text quality and fails to represent the diversity of valid responses for open-ended tasks.

BERTScore (Zhang et al., 2019) and MoverScore partially addressed the semantic gap by replacing n-gram matching with contextual embedding similarity, but they still cannot evaluate logical reasoning, factual accuracy, or multi-step argumentation.

| Term | Definition | First App. |
|---|---|---|
| **IFR** | **Instruction Following Rate**: The fraction of evaluations producing a parseable verdict that fits the expected output format. | §2 |
| **PC-Acc** | **Position-Consistent Accuracy**: A metric for pairwise evaluation that credits a verdict only when the judge maintains its choice after swapping the A/B presentation order of candidates. | §5 |
| $B^\star$ | **Optimal Token Budget**: The number of reasoning (Chain-of-Thought) tokens that maximizes evaluation accuracy for a specific model scale $\theta$ and task difficulty $d$. | §3 |
| **Verifiability gradient** | The spectrum of evaluation domains ranging from objectively verifiable (e.g., math, code execution) to purely subjective (e.g., creative writing, tone), which strongly predicts SLM judge reliability. | §6 |
| **PoLL** | **Panel of LLM Evaluators**: A heterogeneous ensemble of independent models from different families used to aggregate judgments via majority voting (Condorcet aggregation). | §4 |
| **GRPO** | **Group Relative Policy Optimization**: A reinforcement learning algorithm used to train judges by optimizing directly for evaluation accuracy using group-normalized rewards, often incorporating consistency penalties. | §3 |
| **MAD** | **Multi-Agent Debate**: An evaluation strategy where multiple LLM instances exchange arguments and critique responses over multiple rounds to reach a consensus verdict. | §4 |
| **MAST** | **Multi-Agent System Failure Taxonomy**: An empirically grounded taxonomy of 14 multi-agent failure modes in three categories, notably *Inter-Agent Misalignment* where agents prioritize consensus over accuracy. | §4 |

Table 4: Glossary of specialized terms used throughout the survey, with the section where each is introduced.

| | Known (general) | SLM-specific (this survey) |
|---|---|---|
| **I1** | Larger models are not always better judges | Training lifts the base model in all 8 papers that hold the backbone fixed (+15–63%); the crossover with larger prompted judges appears below 8B |
| **I2** | Over-reasoning can degrade accuracy | Thinking mode adds about 10 points at 0.6B, 1.7B, and 4B alike, for under 3× the compute |
| **I3** | Ensemble voting improves reliability | Practical aggregation is viable only when jurors are cheap enough to run in parallel |
| **I4** | Debate does not always help | Small-model debate with assigned roles beats matched small-model baselines at lower cost than GPT-4o |
| **I5** | Some domains are harder than others | Verifiability gradient calibrated to $\leq$14B; quantitatively sharper degradation on subjective tasks |

Table 5: General LLM-judge knowledge vs. SLM-specific findings from this survey. Each insight builds on a known general phenomenon but adds a quantitatively distinct dimension visible only at the $\leq$14B scale.

**The learned metric era (2021–2022).** The rise of instruction-tuned models created evaluation demands that no n-gram or embedding metric could meet. Open-ended questions have no single correct reference answer; quality depends on tone, depth, factual accuracy, and coherence simultaneously. COMET (Rei et al., 2020) and BLEURT demonstrated that learned metrics correlate substantially better with human judges than n-gram metrics for translation, but both still required reference translations and operated only within constrained task types.

**The RLHF era and human bottleneck (2022–2023).** RLHF (Ouyang et al., 2022) aligned language models with human preferences through a three-stage pipeline: supervised fine-tuning, reward model training on human preference pairs, and PPO optimization. This framework was transformative for model quality but created a new bottleneck: each training run required thousands of fresh human preference labels; each domain required domain-expert annotators; each model update required a new batch of comparison pairs. Human data curation became the primary rate-limiting factor in alignment research.

THE N-GRAM-TO-NEURAL-TO-MODEL PROGRESSION FOLLOWS AN ABSTRACTION GRADIENT. [CONSENSUS: 4 sources] *Each generation of evaluation methodology increased the abstraction level of the quality signal being captured: n-gram metrics measure lexical overlap, embedding metrics measure semantic similarity, and model-*

*based judges measure functional quality against multi-dimensional criteria.* This progression is not coincidental. As language models became capable of producing fluent, contextually appropriate text, lower-abstraction metrics lost discriminative power. N-gram metrics cannot distinguish between a factually correct and a factually fabricated response if both are fluent; embedding metrics cannot distinguish between a logically valid and an invalid argument if both use similar vocabulary. Only model-based evaluation operates at the abstraction level required to assess the functional properties (correctness, helpfulness, safety) that define modern response quality. The implication is that future evaluation demands, particularly for reasoning-heavy and domain-specialized tasks, will require judges that operate at even higher abstraction levels, explaining the push toward tool-augmented and representation-based evaluation.

### A.2 The LLM-as-a-Judge Paradigm: Formalization

The modern formalization was established by Zheng et al. (2023) with MT-Bench and Chatbot Arena. The paradigm defines a *judge function* $f_{\mathcal{J}}$ that maps evaluation inputs to verdicts:

$$f_{\mathcal{J}} : \mathcal{Q} \times \mathcal{R}^{(1:k)} \to \mathcal{V} \tag{2}$$

where $\mathcal{Q}$ is the space of evaluation questions, $\mathcal{R}^{(1:k)}$ is a tuple of $k$ candidate responses ($k$=1 for pointwise, $k$=2 for pairwise, $k$>2 for listwise), and $\mathcal{V}$ is the verdict space. For pointwise evaluation, $\mathcal{V} = \{1, 2, 3, 4, 5\}$ (Likert scale) or $\mathbb{R}$ (continuous). For pairwise, $\mathcal{V} = \{\texttt{A}, \texttt{B}, \texttt{Tie}\}$. For listwise, $\mathcal{V}$ is a permutation of the $k$-element response set.

A critical extension is the *rubric-conditioned* judge:

$$f_{\mathcal{J}}^{(\rho)} : \mathcal{Q} \times \mathcal{R}^{(1:k)} \times \rho \to \mathcal{V} \tag{3}$$

where $\rho \in \mathcal{P}$ is a rubric that conditions the judge's output on user-specified evaluation dimensions. Prometheus (Kim et al., 2024a) and GLIDER (Deshpande et al., 2024) operationalize this form, enabling fine-grained multi-criterion evaluation.

Beyond rubric conditioning, some systems (Li et al., 2024b) produce structured *critiques*: explanatory rationales identifying specific aspects that satisfy or violate evaluation criteria:

$$f_{\mathcal{J}}^{(\text{crit})} : \mathcal{Q} \times \mathcal{R}^{(1:k)} \times \rho \to (\mathcal{V}, \mathcal{C}) \tag{4}$$

where $\mathcal{C}$ is the space of natural-language critiques. Critique-conditioned judges provide interpretable, auditable evaluation and can serve as natural-language feedback in iterative refinement loops.

**MT-Bench: The first systematic evaluation.** MT-Bench (Zheng et al., 2023) consists of 80 multi-turn questions across eight categories (writing, roleplay, extraction, reasoning, math, coding, STEM, humanities). GPT-4-as-judge achieved >80% agreement with expert human annotators, comparable to human-to-human agreement. A key finding was that GPT-4's preference flipped in roughly one-third of cases when response presentation order was swapped, revealing the prevalence of position bias even in frontier models.

**Chatbot Arena: Crowdsourced preference at scale.** Chatbot Arena (Zheng et al., 2023) collected >30,000 human pairwise preferences from crowdsourced users, establishing a gold-standard ELO-style ranking. This ranking became the authoritative external measure against which automated judge systems compare their alignment, providing a living, human-annotated benchmark that is resistant to overfitting.

### A.3 Key Judge Systems: A Comprehensive Timeline

Table 6 summarizes the key milestones. We organize the timeline into three phases.

**2023: The fine-tuning wave.** The first wave of dedicated judge models trained LLMs specifically on evaluation data.

**JudgeLM.** Zhu et al. (2025) trained models at 7B, 13B, and 33B scales on GPT-4-generated judgment data. The 7B variant achieves >90% agreement with GPT-4 and can evaluate 5,000 response pairs in approximately 3 minutes on 8×A100 GPUs. JudgeLM introduced *swap augmentation*: evaluating both orderings of response pairs during training to reduce position bias. It also introduced *reference support* (providing a gold reference) and *reference drop* (randomly omitting the reference during training to improve reference-free generalization).

**PandaLM.** Wang et al. (2024c) distilled evaluation capability from GPT-3.5 into a LLaMA-based model, demonstrating competitive performance on a 1,000-sample human-annotated test set. PandaLM was the first judge model to directly tackle

automated hyperparameter search for instruction-tuned models.

**Shepherd.** Wang et al. (2023b) developed a critic model that generates natural-language feedback identifying errors, inconsistencies, and missing information in model outputs. Shepherd demonstrated that training on community feedback data (Stack Exchange, human annotations) produces critic models that outperform ChatGPT on feedback quality, establishing the distinction between *scoring* (producing a verdict) and *critiquing* (producing diagnostic feedback) as complementary evaluation modes.

**Prometheus.** Kim et al. (2024a) introduced rubric-conditioned evaluation with fine-grained, customizable evaluation criteria. Trained on the Feedback Collection dataset, Prometheus demonstrated that conditioning the judge on explicit rubrics substantially improves evaluation validity compared to unconditioned scoring.

**Auto-J.** Li et al. (2024b) developed a 13B generative judge covering 58 diverse, real-world scenarios. Auto-J supports both pairwise comparison and pointwise evaluation, provides detailed natural-language critiques, and demonstrates reduced positional bias compared to non-specialized baselines.

**2024: Unified evaluation, multi-task breadth, and public data.**

**Prometheus 2.** Kim et al. (2024b), building on the original Prometheus (Kim et al., 2024a), unified direct assessment and pairwise ranking through weight merging: two separate models (Mistral-7B-Instruct-v0.2 and Mixtral-8x7B-Instruct-v0.1) trained on the Feedback Collection (pointwise) and Preference Collection (pairwise) datasets are merged into a single evaluator. Prometheus 2 achieves GPT-4-level correlation on MT-Bench, VicunaBench, and FLASK while supporting custom evaluation rubrics.

**FLAMe.** Vu et al. (2024) trained on $>$100 quality assessment tasks encompassing $>$5M human judgments drawn exclusively from permissively licensed datasets. FLAMe-RM achieves 87.8% accuracy on RewardBench while exhibiting lower bias than popular proprietary judges on the CoBBLEr benchmark.

**Skywork-Critic.** The Skywork-Critic-Llama-3.1 models, trained on heterogeneous open-source data including HelpSteer2 and OffsetBias (a dataset specifically targeting bias-robust evaluation training), achieve top performance on RewardBench. OffsetBias provides contrast pairs that train against categorical evaluation bias, where judges exhibit different error rates for different semantic categories.

**GLIDER.** Deshpande et al. (2024) developed a 3.8B judge based on Phi-3.5-mini-instruct, trained on 183 distinct evaluation metrics across 685 domains. GLIDER generates bullet-point reasoning chains and text highlights for each decision. Crucially, GLIDER outperforms GPT-4o on FLASK despite being far smaller (and is the smallest model reported to surpass GPT-4o-mini as an evaluator), establishing that evaluation-specific training breadth can compensate for substantial parameter reductions.

**Self-Taught Evaluators.** Wang et al. (2024b) proposed generating synthetic contrast pairs for unlabeled instructions and training a judge on these pairs without any human labels. An iterative self-improvement loop improves RewardBench score from 75.4 to 88.3 for a Llama-3 70B seed model, demonstrating that a judge's own evaluative capacity can bootstrap reliable training signal.

**2025–2026: RL-trained and reasoning judges.**

**J1.** Whitehouse et al. (2025) developed an RL-trained thinking-judge using GRPO. J1 converts both verifiable and non-verifiable judgment tasks into a unified verifiable reward format. J1-Llama-8B reports state-of-the-art results among 8B judges and matches the prompted Llama-3.3-70B on RewardBench (85.7 against 85.4). The consistency-based reward provides additional positive signal when the judge produces the same verdict across both positional orderings of the same pair.

**JudgeLRM.** Chen et al. (2025) developed a family of RL-based judge models (3B–14B) demonstrating that RL training for judgment consistently outperforms SFT-initialized judges of equal size.

**J4R.** Xu et al. (2025b) addressed discriminative precision challenges in pairwise evaluation using equivalent-initial-state group relative policy optimization.

**CompassJudger-2.** Zhang et al. (2025c) developed a generalist judge trained with verifiable rewards, rejection sampling, and a margin policy-

| Year | System | Size | Key Contribution |
|---|---|---|---|
| 2022 | InstructGPT | 175B | RLHF paradigm + reward model |
| 2023 | MT-Bench | N/A | >80% human agreement |
| 2023 | JudgeLM | 7B–33B | Swap augmentation |
| 2023 | PandaLM | 7B–70B | Auto-hyperparameter judge |
| 2024 | Prometheus 2 | 7B, 12.9B/ 46.7B | Weight merging DA+PR |
| 2024 | FLAMe | Varied | 5M+ human annotations |
| 2024 | GLIDER | 3.8B | 183 metrics; beats GPT-4o |
| 2025 | J1 | 8B, 32B, 70B | GRPO + consistency reward |
| 2025 | JudgeLRM | 3B–14B | RL judge family |

Table 6: Key milestones in the LLM-as-a-Judge paradigm (2022–2026). Systems are ordered chronologically within each year. Mixture-of-experts sizes are given as active/total parameters.

gradient loss; its 7B model reports judgment accuracy competitive with DeepSeek-V3 (37B active / 671B total parameters).

THE FIELD HAS PROGRESSED THROUGH THREE DISTINCT TRAINING PARADIGM SHIFTS. [CONSENSUS: 6 systems] *Judge model training evolved from distillation (2023, GPT-4 as teacher), through supervised multi-task learning on human data (2024, FLAMe and GLIDER), to reinforcement learning with structured reward signals (2025, J1 and JudgeLRM), because each successive paradigm addresses a limitation of the previous one.* Distillation transfers the teacher's biases alongside its capabilities; multi-task SFT reduces teacher dependency but cannot optimize for evaluation accuracy directly; RL with verifiable rewards enables direct optimization of the judge's output against ground truth. The pattern suggests that future advances will come from self-improving or tool-augmented judge training, where the reward signal itself is generated by external verification rather than human annotation.

### A.4 Evaluation Modalities: A Structured Comparison

The LLM-as-a-Judge paradigm supports three primary evaluation modalities with distinct tradeoffs (Table 7).

PAIRWISE EVALUATION CREATES AN INHERENT TRADEOFF BETWEEN DISCRIMINATIVE POWER AND POSITION BIAS. [PRELIMINARY: proposed mechanism] *Pairwise comparison provides higher discriminative power than pointwise scoring because relative comparison is cognitively easier than absolute calibration, but this advantage comes at the structural cost of position bias, because the judge must process both responses in a fixed serial order.* This tradeoff explains why position bias mitigation (swap augmentation, consistency rewards) is a central concern for pairwise judge training, while pointwise judges face calibration challenges instead. The tradeoff is fundamental: any evaluation format that presents multiple items in sequence will be susceptible to order effects, while any format that scores items independently requires calibration against an absolute scale.

| Property | Pointwise | Pairwise | Listwise |
|---|---|---|---|
| Input | Single response | Response pair | $k$-tuple |
| Output | Score (Likert) | Preference | Ranking |
| Position bias | Low | High | Moderate |
| Transitivity | Guaranteed | Not guaranteed | By construction |
| Scalability | $O(N)$ | $O(N^2)$ | $O(N)$ |
| Use case | Scoring, RLHF | Ranking | Leaderboard |

Table 7: Comparison of pointwise, pairwise, and listwise evaluation modalities.

The **listwise paradigm** presents all $k$ candidate responses simultaneously and asks the judge to rank them. This produces a consistent total ordering (transitivity guaranteed by construction) and eliminates the $O(N^2)$ pairwise tournament cost. Bradley-Terry models can convert pairwise win rates into global ELO rankings. The tradeoff is context length: presenting $k$ long responses simultaneously can exceed SLM context windows, making listwise evaluation more challenging for small models.

### A.5 Cost Model and Deployment Tradeoffs

Let $f_{\mathcal{J}}^{(B)}$ denote a judge operating under token budget $B$, with per-token costs $\alpha$ (input) and $\beta$ (output). For a proprietary LLM judge processing $N$ evaluations:

$$C_{\text{total}} = N \cdot \left(\alpha \cdot B_{\text{in}} + \beta \cdot B\right) \tag{5}$$

where $B_{\text{in}}$ is the prompt length and $B$ is the output (reasoning + verdict) budget. At $N = 10^5$ evaluations, GPT-4 API costs can reach thousands of dollars. Verga et al. (2024) empirically document that the PoLL ensemble achieves $>7\times$ cost reduction over single-GPT-4 judging while increasing human correlation. Locally deployed SLM judges eliminate per-token API costs entirely, with cost

determined solely by hardware amortization and energy consumption.

Beyond direct API cost, the proprietary judge paradigm introduces structural constraints for production deployment:

- **Privacy exposure**: Sensitive inputs must leave local infrastructure when sent to cloud-based judge APIs.
- **Version drift**: Model providers silently update judge models, breaking evaluation reproducibility.
- **Latency**: Production RLHF loops require low-latency feedback; cloud API round-trips create bottlenecks at large batch sizes.
- **Output opacity**: Proprietary judges provide no access to logits, intermediate representations, or reasoning traces, blocking calibration and mechanistic analysis.
- **Rate limits**: API rate limits cap evaluation throughput, making simultaneous evaluation of many model checkpoints impractical.

THE PROPRIETARY JUDGE BOTTLENECK DRIVES THE SLM TURN THROUGH OPERATIONAL CONSTRAINTS, NOT ACCURACY DEFICITS. [PRELIMINARY: analytical argument] *The transition from proprietary to small-model judges is driven by a convergence of operational constraints (cost, privacy, reproducibility, latency) that make proprietary judging structurally incompatible with production ML pipelines, not primarily by accuracy concerns.* GPT-4 remains a capable judge; the motivation for SLM judges is that specialized 3–8B models match proprietary accuracy on domain-specific evaluation while eliminating the structural constraints. This distinction is important because it predicts that SLM judges will be adopted even in scenarios where proprietary judges are marginally more accurate, because the operational benefits dominate.

### A.6 Constitutional AI and the Self-Critique Paradigm

Bai et al. (2022) introduce Constitutional AI (CAI): a framework in which a model evaluates its own outputs against an explicit set of principles and revises responses found to violate those principles. CAI operates through a multi-round critique-revision pipeline:

1. The generator produces an initial response.
2. A *critique judge* evaluates the response against each constitutional principle.
3. A *revision judge* proposes corrections based on the critique.
4. The process repeats for $K$ rounds.

CAI is the most direct demonstration of a model serving as its own judge. The self-critique paradigm connects to the Semantic Capacity Asymmetry Hypothesis (Li et al., 2026): the same model's evaluative capacity (lower-dimensional classification) supervises its generative capacity (higher-dimensional distribution), anticipating the finding that evaluation requires less computational capacity than generation.

### A.7 LLM Judging and Scalable Oversight

A theoretically important connection exists between LLM-as-a-Judge and the *scalable oversight* problem in AI safety. Burns et al. (2023) demonstrate *weak-to-strong generalization*: when a strong pretrained model is fine-tuned on labels generated by a weak supervisor, it often *outperforms* the weak supervisor, because the strong model already possesses latent knowledge that weak supervision elicits rather than teaches. The implication is bidirectional:

1. Small models fine-tuned on large-model judgment data demonstrate the same elicitation mechanism: latent evaluative capacity is being surfaced, not instilled.
2. An SLM judge successfully overseeing a larger model's outputs provides empirical evidence of scalable oversight in the evaluation domain.

The 2025–2026 W2SG refinement techniques (trajectory trees, selective alignment, graph smoothing) further validate that weak judge supervision can elicit strong-model generalization beyond the supervisor's own capacity boundary (Appendix L).

### A.8 Surveys and Related Work

Three prior surveys characterize the LLM-as-a-Judge landscape. Gu et al. (2024) provide formal definitions of evaluation dimensions (helpfulness, harmlessness, honesty, factuality), catalog bias types, and discuss mitigation. Li et al. (2025a) frame the landscape around the transition from generation to judgment, and Li et al. (2024a) review LLM-based evaluation methods more broadly.

None of them addresses the specific considerations arising when the judge population is constrained to $\leq$14B parameters: ensemble feasibility with sub-10B models, the role of thinking-mode efficiency, representation-based judging, or cascade architectures uniquely enabled by SLMs. The SLM survey of Lu et al. (2024) covers SLM architectures, training, and deployment but does not examine the judging role. This survey fills the intersection of all three domains.

## B The SLM Landscape

Having established *why* the field turned to small models, we now survey *which* models are available, detailing the families (Phi, Gemma, Qwen, LLaMA, R1-Distill) and architectural trends that make sub-14B judging feasible.

### B.1 Defining "Small": The Parameter Boundary Debate

"Small Language Model" is a contextual, not absolute, designation. The community has not converged on a single definition, and the practical boundary has shifted as hardware efficiency and training methodology have improved. Three competing framings exist:

- **Hardware feasibility**: Models deployable on a single consumer GPU (24GB VRAM class, e.g., NVIDIA RTX 4090) in native or quantized (INT4/INT8) form. At two bytes per parameter, a 14B model needs about 28GB in 16-bit precision but about 14GB at 8-bit, so 14B is the largest size such a card holds at 8-bit, and roughly 30B fits with 4-bit quantization.
- **Edge deployment**: Models that run on battery-powered devices (smartphones, laptops, embedded systems) without cloud infrastructure. This tightens the boundary to 1B–7B parameters, with 3.8B being a practical sweet spot for the performance-power tradeoff.
- **Relative capability**: Models characterized as "small" relative to concurrent frontier systems. By this definition, the boundary shifts upward over time: what is "small" in 2026 encompasses models that were frontier-class in 2022.

For this survey, we adopt the hardware-feasibility boundary: **SLMs are models with $\leq$14B parameters deployable on a single A100/H100 or consumer-class GPU without model parallelism**. This threshold corresponds to the reach of typical single-node research computing infrastructure, making it a natural dividing line between "local" and "distributed" deployment.

**Why 14B and not 7B?** The 14B boundary coincides with commercially available instruction-tuned models that demonstrably match 2022-era frontier capability (GPT-3 class) at a fraction of inference cost. JudgeLM and Prometheus 2 both treat 7B as their primary small variant; GLIDER operates at 3.8B; INSPECTOR probes at 1.7B. The key empirical contribution of SLM-as-a-Judge research is showing that reliable judging does *not* require the 70B+ models initially assumed necessary.

THE "SMALL" BOUNDARY IS HARDWARE-DEFINED, NOT CAPABILITY-DEFINED, BECAUSE CAPABILITY BOUNDARIES SHIFT WITH TRAINING INNOVATION. [PRELIMINARY: analytical argument] *Defining SLMs by parameter count rather than capability is necessary because evaluation-specific training continuously pushes the capability frontier downward: a 3.8B judge trained on evaluation data in 2024 already outscores GPT-4o on FLASK (Deshpande et al., 2024).* If the boundary were defined by capability (e.g., "models that cannot match GPT-4 on judging"), the SLM category would progressively shrink as training improves. A hardware-feasibility definition provides a stable, deployment-relevant boundary that captures the operational advantages (cost, privacy, latency) motivating the SLM judge research program.

### B.2 The Key SLM Model Families (Early 2026)

Five families account for nearly all SLM judge research. We review each in turn, focusing on the design choices (training-data philosophy, attention pattern, dual-mode inference, distillation strategy) that bear directly on judging.

#### B.2.1 Microsoft Phi-4 Family

The Phi series represents the most studied example of the "data quality over data quantity" training philosophy, deliberately prioritizing *reasoning-dense synthetic data* over raw token scale.

**Phi-4 (14B).** Abdin et al. (2024) trained on high-quality "textbook-grade" synthetic data, curated from public web content via quality classifiers, and supplemented with SFT and DPO preference data. Phi-4 achieves STEM and mathematical reasoning

scores rivaling models 3–5× larger, establishing that structured, high-quality data is a powerful substitute for massive compute.

**Phi-4-mini (3.8B).** Released February 2025 (Abouelenin et al., 2025), trained on 5 trillion tokens with a 128K token context window, 200K vocabulary, and Grouped-Query Attention. The training curriculum includes: (1) filtered public documents; (2) synthetic textbook-style data targeting mathematics, coding, and scientific knowledge; and (3) SFT and DPO preference data. Phi-4-mini frequently outperforms models twice its size on mathematical reasoning benchmarks. The same lineage's strong instruction following and structured-reasoning capacity is what made the earlier Phi-3.5-mini the base model for GLIDER (Deshpande et al., 2024).

**Phi-4-reasoning variants.** Abdin et al. (2025) trained these specifically for multi-step STEM reasoning via reinforcement learning on verifiable mathematical problems. The reasoning variant achieves state-of-the-art results for its size on competition mathematics, demonstrating that small generic models can be specialized toward deliberate stepwise evaluation.

THE PHI-4 FAMILY VALIDATES THAT TRAINING DATA COMPOSITION DOMINATES PARAMETER COUNT FOR EVALUATION TASKS. [PRELIMINARY: single study] *GLIDER's exceptional judging performance on FLASK, built on the 3.8B Phi-3.5-mini, demonstrates that for tasks requiring precision and domain-specific reasoning, including evaluation, what the model is trained on matters more than how many parameters it has.* This is because evaluation (discriminative classification against criteria) requires internalized pattern-matching capabilities that can be instilled through targeted training, whereas generation (open-ended text production) requires the broader distributional knowledge that scales with parameters. The Phi-4 family's "textbook-grade" training philosophy is particularly well-aligned with evaluation: rubric-conditioned scoring is structurally similar to grading against a reference standard, exactly the kind of structured reasoning that curated training data teaches well.

#### B.2.2 Google Gemma 2 and Gemma 3 Families

Google's Gemma series (Team et al., 2024a) represents the most architecturally innovative SLM family, with each generation introducing design choices that directly benefit judge deployment.

**Gemma 2.** Team et al. (2024b) introduced two critical architectural innovations:

- **Alternating local-global attention**: Each layer alternates between local sliding window attention (window = 4,096 tokens) and global attention (span = 8,192 tokens). Local attention handles the majority of tokens efficiently; global attention periodically integrates information from the full context. This pattern processes long evaluation inputs (question + multiple responses + rubric) more efficiently than full-attention models at equivalent parameter budgets.
- **Knowledge distillation training**: Gemma 2-9B and 2-2B were trained using probability distributions from the 27B teacher as the training signal, rather than one-hot next-token prediction. This provides richer gradients and enables smaller models to consistently outperform next-token-trained models of the same size.

**Gemma 3 (March 2025).** Gemma 3 adds multimodal capability while maintaining efficiency optimizations:

- **Multimodal judge capability**: Gemma 3 (4B, 12B, 27B) integrates native vision-language understanding via a SigLIP vision encoder, enabling evaluation of image-response pairs.
- **128K context window**: All Gemma 3 variants (4B+) support 128K token contexts, enabling evaluation of long responses and multi-turn conversations.
- **Multilingual coverage**: Pretrained on an expanded multilingual dataset supporting 140+ languages, using the Gemini tokenizer for improved cross-lingual token efficiency.
- **KV-cache efficiency**: Interleaved local/global attention reduces KV-cache memory requirements, enabling higher batch sizes for evaluation workloads on consumer hardware.

**Note on Gemma 4 (April 2026).** Shortly before submission, Google released Gemma 4 in Effective-2B, Effective-4B, 26B MoE (3.8B active), and 31B Dense configurations (Manik and Wang, 2026). Gemma 4 adds native audio input for the edge models and extends multimodal capabilities, with

Apache 2.0 licensing. Full evaluation of Gemma 4 as a judge backbone is deferred to future work.

ARCHITECTURAL INNOVATIONS IN THE GEMMA FAMILY DIRECTLY ADDRESS THE CONTEXT-LENGTH BOTTLENECK FOR EVALUATION. [PRELIMINARY: proposed mechanism] *The alternating local-global attention pattern is particularly well-suited to evaluation tasks because judge inputs consist of a long context (question, rubric, candidate responses) followed by a short generation (verdict and rationale), and local attention efficiently processes the long input while global attention captures cross-response comparisons.* Standard full-attention models allocate equal computation to every token pair, most of which are within the same response and do not contribute to cross-response comparison. Gemma's alternating pattern concentrates global computation on the periodic cross-response integration steps that are most relevant to pairwise and listwise evaluation, while using cheaper local attention for within-response processing.

#### B.2.3 Alibaba Qwen 3 Family

Qwen 3 (Yang et al., 2025) introduces the most architecturally ambitious design among studied SLM families for judging applications: a unified dual-mode architecture.

**Dual-mode operation.** Qwen 3 supports a "thinking budget" mechanism: a per-query parameter that controls computational resources allocated to chain-of-thought reasoning before the final response. In *thinking mode*, the model generates an explicit reasoning trace before producing its answer. In *non-thinking mode*, it responds directly in a single pass. This enables the same model to serve as both a fast verdict generator and a deliberative evaluator depending on task requirements.

**Distillation from flagship.** Smaller Qwen 3 models (0.6B, 1.7B, 4B, 8B) are trained using distillation from the 32B and 235B-A22B flagship models. This strong-to-weak distillation transfers reasoning patterns that would typically require much larger parameter budgets, enabling even the 0.6B model to exhibit explicit chain-of-thought behavior.

**Multilingual breadth.** Qwen 3 supports 119 languages and dialects natively, making it the broadest-coverage multilingual SLM family with judge capability, directly relevant to the M-Prometheus multilingual direction (Appendix J).

**MoE architecture (Qwen 3-235B-A22B).** The largest Qwen 3 uses Mixture-of-Experts activating 22B of 235B parameters per forward pass, achieving frontier-level performance at substantially reduced inference cost. This MoE approach is increasingly relevant as "large" SLM deployments shift toward activated-parameter rather than total-parameter budgets.

Jayarao et al. (2025) study Qwen 3 judges explicitly, finding that thinking-mode operation ($\leq$4B parameters) achieves approximately 10% higher judging accuracy than non-thinking counterparts on pairwise evaluation tasks.

QWEN 3'S NATIVE DUAL-MODE DESIGN RESOLVES THE REASONING-COST TRADEOFF AT THE ARCHITECTURAL LEVEL. [PRELIMINARY: analytical argument] *Unlike models where thinking mode requires a separate fine-tuned variant or external scaffolding, Qwen 3 integrates both modes into a single set of weights, enabling per-query mode selection based on task difficulty without model switching.* This is significant for adaptive judge deployment (§6, C1) because it eliminates the need to maintain, load, and route between separate thinking and non-thinking models. A single Qwen 3 deployment can serve as a fast direct-verdict judge for simple tasks and a deliberative CoT judge for complex tasks, with the mode selection controllable via a single parameter. The $\sim$10% accuracy improvement from thinking mode (Jayarao et al., 2025) comes at a token-budget cost that is worthwhile only for tasks where the additional reasoning tokens surface genuinely new discriminative signals (I2).

#### B.2.4 Meta LLaMA 3 and LLaMA 4 Families

LLaMA 3 (Grattafiori et al., 2024) provides the most widely deployed SLM backbone in judge research and production. The LLaMA 3.1-8B and LLaMA 3.2-3B/1B variants are standard baselines across most judgment work due to their open weights, broad community support, and compatibility with all major inference frameworks.

Key architectural choices include standard Grouped-Query Attention with a 128K context window (LLaMA 3.1+), RoPE positional embeddings, and a vocabulary of 128K tokens. J1 (Whitehouse et al., 2025) trains judge variants on LLaMA 3 bases, with J1-Llama-8B achieving state-of-the-art among open-source 8B judges on JudgeBench and RewardBench.

**LLaMA 4 (April 2025).** Meta released LLaMA 4 (Scout and Maverick) in April 2025, using a native multimodal MoE architecture. LLaMA 4 Scout (109B total parameters with 17B activated per forward pass, 16 experts) supports up to 10 million tokens of context. Maverick (400B total, 17B active, 128 experts) was pretrained on approximately 22 trillion tokens, while Scout was pretrained on approximately 40 trillion tokens. While Scout's 17B active parameters are within the compute range relevant to SLM deployment, its 109B total parameter footprint means all weights must reside in memory for expert routing, exceeding the single-consumer-GPU constraint of strict SLM definitions. Systematic evaluation of LLaMA 4 models as judges is an open research task.

#### B.2.5 DeepSeek-R1-Distill Family

DeepSeek-R1 (Guo et al., 2025) introduced a 671B reasoning model trained using large-scale RL on verifiable rewards. The R1-distill models (1.5B, 7B, 14B, 32B, 70B) fine-tune Qwen2.5-Math (for the 1.5B and 7B variants) and Qwen 2.5 (for the 14B and 32B variants) (Qwen et al., 2025), alongside Llama-3.1-8B and Llama-3.3-70B-Instruct bases, on 800K high-quality reasoning traces generated by DeepSeek-R1 itself, a chain-of-thought distillation approach requiring no separate RL training.

Performance on AIME 2024 competition mathematics scales clearly with size (Figure 3): 1.5B achieves approximately 28.9% Pass@1, 7B achieves 55.5%, and 14B achieves 69.7%. These numbers are remarkable for models of their size: the 14B distill frequently outperforms non-reasoning models 4–5× larger on mathematical reasoning. For SLM judging, this demonstrates that combining architectural base capacity with high-quality reasoning-trace fine-tuning produces models capable of the multi-step verification required for evaluating complex responses. The dependence of this verification substrate on base size is precisely why the verifiability gradient (Appendix J) bends downward for the smallest judges in knowledge-intensive domains.

### B.3 Architectural Trends Relevant to Judging

**Thinking vs. non-thinking modes.** The field is converging on hybrid inference: models choose between fast (non-thinking) and deliberate (thinking) inference modes. Qwen 3 implements this natively; DeepSeek-R1-distill models always reason; Phi-4-reasoning is dedicated to thinking mode. For judge deployment, this choice has first-order implications for the accuracy-efficiency tradeoff (Appendix D).

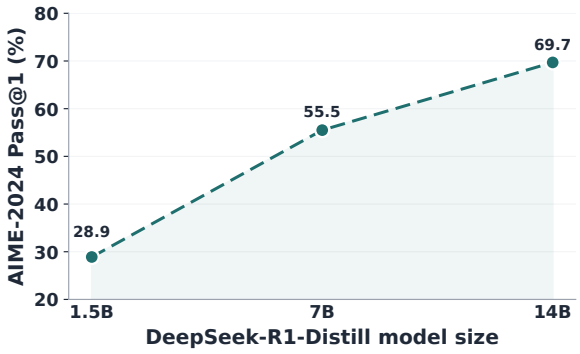



Figure 3: Distilled reasoning capacity scales with base size on AIME-2024 (Pass@1) for the DeepSeek-R1-Distill family. Because multi-step verification draws on this same capacity, the smallest distilled judges have the least headroom for reasoning-intensive evaluation.

**Mixture of Experts.** MoE architecture (Qwen 3-235B-A22B, LLaMA 4, DeepSeek-V3) decouples total parameter count from per-token inference cost. A 235B MoE model activating 22B parameters per forward pass may be "small" in inference cost terms while "large" in knowledge storage terms. This complicates the parameter-count-based SLM definition but creates new opportunities: MoE judges can encode broader evaluation knowledge than their activated compute budget would suggest.

**Context length expansion.** 128K context windows are now standard across LLaMA 3, Phi-4-mini, and Qwen 3. This enables judges to reason over very long candidate responses, multi-turn conversations, and multi-document grounding contexts, significantly expanding the range of evaluation tasks accessible to SLM judges.

**Multimodal capability.** Gemma 3, LLaMA 4, and Qwen 3-VL now provide multimodal SLMs capable of evaluating image-response pairs. As AI-generated content increasingly includes images, charts, and diagrams, multimodal judge capability will become essential. The SLM judge literature has not yet systematically addressed multimodal evaluation; this represents an important open frontier.

**Vocabulary size and tokenization.** Phi-4-mini's 200K vocabulary and Qwen 3's multilingual tokenizer contrast with LLaMA 3's 128K vocabulary. For judge tasks involving code evaluation, mathematical notation, or multilingual content, vocabulary coverage directly affects the model's ability to

| Family | Sizes | Ctx | Think | Multiling | Judge Str. |
|---|---|---|---|---|---|
| Phi-4 | 3.8B, 14B | 128K | Yes | Med | Reasoning |
| Gemma 3 | 4B–27B | 128K | No | High | Multimodal |
| Qwen 3 | 0.6B–32B | 128K | Yes | High | Dual-mode |
| LLaMA 3 | 1B–70B | 128K | No | Med | Ecosystem |
| R1-Distill | 1.5B–70B | 128K | Always | Med | STEM |

Table 8: SLM family comparison relevant to judging (early 2026). Context lengths and language counts reflect the maximum available variant within each family.

represent the nuances of what it is evaluating.

### B.4 SLM Capability Characterization for Judging

Not all SLM capabilities are equally relevant to judging. We distinguish four capability dimensions:

1. **Instruction following (IFR)**: The ability to adhere to evaluation format constraints. SLMs below ∼1B show markedly lower IFR, while most instruction-tuned SLMs above 3B follow standard rubric formats reliably.

2. **Evaluative reasoning**: The ability to assess logical consistency, factual correctness, and quality criteria. This requires understanding the content being evaluated, not just its lexical properties.

3. **Calibration**: Whether the model's expressed confidence in its verdict correlates with its actual accuracy. Uncalibrated judges produce misleading confidence signals that distort downstream aggregation. Calibration assessment for SLM judges is largely an open research problem.

4. **Knowledge depth**: For domain-specific evaluation tasks, the judge must possess sufficient domain knowledge to assess content correctness. This dimension is irreducible by training: an SLM without the relevant domain knowledge cannot accurately evaluate domain-specific responses regardless of judge training.

THE SLM CAPABILITY THRESHOLD FOR RELIABLE JUDGING IS 3B, NOT 7B, BECAUSE IFR IS THE BINDING CONSTRAINT. [REPLICATED: 3 studies] *The field's initial assumption that reliable judging requires 7B+ parameters has been refuted by three converging findings: GLIDER (3.8B) surpasses GPT-4o on FLASK, JudgeLRM (3B) exceeds GPT-4 on PandaLM, and Qwen3-4B in thinking mode achieves ∼10% higher accuracy than non-thinking counterparts.* What sets the floor is instruction following reliability (IFR) more than raw parameter count: below approximately 3B parameters, models struggle to consistently produce verdicts in the required format, which makes evaluation harnesses unreliable. Above 3B, with evaluation-specific training, judge quality scales primarily with training data quality rather than model size. Table 8 summarizes the key families and their judge-relevant properties, and Figure 4 visualizes their relative strengths across the capability dimensions most relevant to judging. No family leads on every dimension: Gemma 3 and Qwen 3 are strongest on multilingual coverage, Gemma 3 alone on multimodal input, Phi-4, Qwen 3, and R1-Distill on reasoning density, while R1-Distill is the weakest on instruction following. This complementarity is exactly what makes cross-family panels (Appendix E) effective.

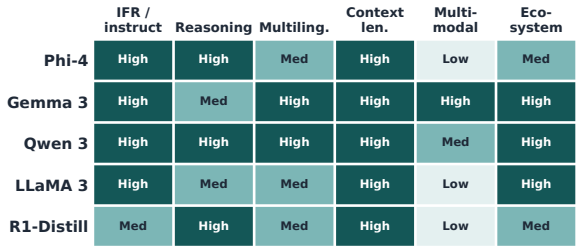



Figure 4: Relative strengths of the major SLM families across judge-relevant capability dimensions. Darker cells indicate stronger capability. Instruction following is rated from IFEval, reasoning from MMLU-Pro and MATH, and multilingual coverage from M-RewardBench and MGSM; context length, multimodal input, and ecosystem maturity are read from the respective model technical reports and release notes. Ratings are ordinal (low, medium, high) rather than measurements on a common scale. The absence of a uniformly dominant family motivates heterogeneous panels.

MOE ARCHITECTURES DECOUPLE THE KNOWLEDGE-COST TRADEOFF FOR JUDGE DEPLOYMENT. [PRELIMINARY: proposed mechanism] *Mixture-of-Experts models store more evaluation-relevant knowledge than their per-token compute cost would suggest, because different experts can specialize in different evaluation domains while sharing the routing and attention infrastructure.* A Qwen 3-235B-A22B model activates only 22B parameters per forward pass but has access to 235B parameters of stored knowledge, enabling it to evaluate across a broader range of domains than a 22B dense model. For judge deployment, this means MoE models can approach the knowl-

edge depth of large models while maintaining the latency characteristics of small models. The practical implication is that the parameter-count boundary ($\leq$14B) may need refinement to account for activated vs. total parameters as MoE architectures become standard.

### B.5 Inference Infrastructure for SLM Judge Deployment

Practical deployment of SLM judges at scale requires infrastructure beyond raw model capability.

**vLLM.** Kwon et al. (2023) developed PagedAttention for high-throughput batched decoding with dynamic KV cache management. vLLM is the de facto standard for SLM judge inference in research evaluation pipelines, supporting tensor parallelism, speculative decoding, and model quantization. It achieves 2–4$\times$ higher GPU utilization than naive inference.

**llama.cpp / Ollama.** Pure CPU and consumer GPU inference, enabling fully local SLM judge deployment on unmodified consumer hardware. Supports GGUF-format quantized models (INT4, INT8); at 4-bit, a 14B model's weights take about 7GB, so it runs on a 12–16GB consumer GPU.

**Hugging Face TGI.** Production-grade serving supporting continuous batching, multi-GPU tensor parallelism, and custom tokenization. Standard for enterprise deployments with SLA requirements.

**Quantization.** Post-training quantization (INT4 via GPTQ or AWQ) reduces memory requirements by $\sim$4$\times$, making 14B models deployable on a single 16GB consumer GPU. For 7B models, INT4 quantization enables deployment on a single 8GB VRAM GPU.

### B.6 The Emerging SLM Judge Ecosystem

The period from 2023 to 2026 has seen rapid consolidation of a dedicated SLM judge ecosystem distinct from general-purpose SLM deployment:

- **Open judge model hubs**: Prometheus-Eval repository, GLIDER on Hugging Face, J1 checkpoints, all providing ready-to-deploy judge models with documented training recipes and license terms.
- **Evaluation harnesses**: lm-evaluation-harness adapted for judge evaluation, enabling reproducible meta-evaluation across systems.
- **Judgment-specific benchmarks**: JudgeBench, JudgeBoard, and RewardBench provide standardized evaluation targets (Appendix I).
- **Training datasets**: Feedback Collection, Preference Collection (Prometheus), SFT judge datasets (JudgeLM), RL judge preference datasets (J1), OffsetBias, and HelpSteer2.
- **Tool-augmented reward modeling**: An emerging trend toward agentic judge systems that invoke external tools (code executors, search engines, document retrievers) for fact-grounded, verifiable assessment.

This ecosystem makes the development of new SLM judges substantially more accessible than even two years prior. The combination of large context windows (128K+), thinking-mode capability, and open training datasets means that a research lab with a single high-end GPU can now train and deploy a competitive SLM judge in days, a capability that required frontier-model API access as recently as 2023.

## C Training Methodologies for SLM Judges

A capable base model is necessary but not sufficient. This section expands the training paradigms of §3, showing how the supervision signal, not parameter count, determines judge quality.

### C.1 A Consolidated Map of SLM Judge Systems

Before turning to individual training paradigms, Table 9 consolidates the major SLM and SLM-relevant judge systems surveyed across this paper into a single reference, organized by training paradigm and annotated with base model, parameter scale, evaluation mode, target domain, the headline empirical result, and the survey insight (I1–I5) each system most directly supports. The table shows three field-level patterns. First, a clear progression down the rows: from distillation (2023) through human-supervised and rubric SFT (2024) to RL with verifiable rewards and representation probing (2025–2026), tracing the decreasing-supervision gradient developed in this appendix. Second, the *scale compression*: headline results at or above proprietary parity are achieved by 3–8B models throughout, with no monotone improvement from the larger entries (I1). Third,

the *domain concentration*: the strongest verifiable-domain results (math, code) cluster among RL and tool-augmented systems, while subjective-domain coverage remains sparse (I5). The remainder of this appendix expands each row's training methodology in turn.

### C.2 Overview of Training Paradigms

The capability of a language model to serve as a reliable judge depends critically on how it was trained, not just what data it saw, but what training *objective* it was optimized for. Table 11 summarizes seven training paradigms for SLM judges, separating the three supervised variants that Table 9 groups together.

A critical insight from the 2025–2026 literature is that the bottleneck for high-performance SLM judges is not the *volume* of training data but the *quality and structure* of the supervision signal.

### C.3 Off-the-Shelf Judging: Baselines and Limits

The most accessible deployment strategy uses general-purpose instruction-tuned SLMs as zero-shot or few-shot judges. The model receives a structured evaluation prompt and produces a verdict via standard inference.

**Why this sometimes works.** Instruction-following models trained on diverse human preference data internalize approximate quality signals across many tasks. A well-prompted 7B model can reliably detect obvious factual errors, gross formatting violations, or blatant hallucinations. Jayarao et al. (2025) find that thinking-mode SLMs (0.6B–4B, Qwen 3 family) achieve approximately 10% higher judging accuracy in pairwise evaluation than non-thinking counterparts, because explicit chain-of-thought traces encode evaluation criteria before issuing a verdict. This is the "baseline ceiling" for off-the-shelf judging.

**Why this fails for precision evaluation.** Off-the-shelf models were not optimized for the discriminative precision that expert evaluation demands. Key failure modes:

- **Low Instruction Following Rate (IFR)**: The model may produce freeform prose instead of the required verdict format. For models below 3B, IFR can drop below 80% on structured rubrics.
- **Unmitigated position bias**: Without explicit training to counter position bias, models default to primacy or recency heuristics.
- **Calibration failure**: High expressed confidence for incorrect verdicts; no correlation between certainty and accuracy.
- **Task-specific deterioration**: Performance on code evaluation, mathematical reasoning, and multilingual tasks is substantially lower than on general text quality.
- **Adversarial vulnerability**: Off-the-shelf judges are susceptible to meaningless phrases or specific punctuation sequences that trigger high reward scores regardless of response quality.

OFF-THE-SHELF JUDGES FAIL ON PRECISION TASKS BECAUSE PRETRAINING OPTIMIZES FOR GENERATION, NOT DISCRIMINATION. [REPLICATED: 2 studies] *General-purpose SLMs can detect gross quality differences (obvious errors, blatant hallucinations) but fail on subtle discriminations (partially correct reasoning, nuanced rubric criteria) because their pretraining objective (next-token prediction) rewards fluent continuation, not quality classification against criteria.* The gap between off-the-shelf and trained judges is largest on tasks requiring rubric-conditioned reasoning, where the judge must simultaneously parse evaluation criteria, map response features to rubric dimensions, and produce a calibrated score. This multi-step discriminative operation is absent from standard pretraining curricula. The $\sim$10% accuracy improvement from thinking mode (Jayarao et al., 2025) partially compensates by forcing explicit criterion enumeration, but it cannot substitute for evaluation-specific training.

### C.4 Supervised Fine-Tuning for Judging

SFT on judgment-specific datasets is the most widely studied training paradigm and the foundation of nearly all dedicated open-source judge models.

#### C.4.1 Data Construction Strategies

**Distillation from proprietary judges.** The dominant approach annotates evaluation data using a teacher model (GPT-4, GPT-4o) and fine-tunes a smaller student to replicate the teacher's verdict and reasoning. JudgeLM (Zhu et al., 2025) assembles task seeds, model responses, and GPT-4 judgments and trains models at 7B, 13B, and 33B

| System | Base model | Size | Mode | Domain | Headline result | Insight |
|---|---|---|---|---|---|---|
| ***Off-the-shelf prompting*** | | | | | | |
| Qwen3-Judge (Jayarao et al., 2025) | Qwen 3 | 0.6–4B | PR | General | ∼10% over non-thinking mode | I2 |
| ***Supervised fine-tuning (distillation / human / rubric)*** | | | | | | |
| PandaLM (Wang et al., 2024c) | LLaMA | 7B | PR | General | Auto hyperparameter selection | I1 |
| JudgeLM (Zhu et al., 2025) | Vicuna | 7–33B | DA/PR | General | >90% agreement with GPT-4 | I1 |
| Auto-J (Li et al., 2024b) | LLaMA-2 | 13B | DA/PR/CR | 58 scenarios | Strong cross-scenario transfer | I1 |
| Prometheus 2 (Kim et al., 2024b) | Mistral | 7B, 12.9B/46.7B | DA+PR | Rubric | GPT-4-level corr. (MT-Bench/FLASK) | I1 |
| FLAMe (Vu et al., 2024) | PaLM 2 | Varied | ORM | 100+ tasks | 87.8% RewardBench; low bias | I1 |
| GLIDER (Deshpande et al., 2024) | Phi-3.5-mini | 3.8B | DA | 685 domains | Beats GPT-4o on FLASK (far smaller) | I1 |
| Selene Mini (Alexandru et al., 2025) | LLaMA-3.1 | 8B | DA/PR | General (11 OOD) | Beats GPT-4o-mini; top 8B on RewardBench | I1 |
| M-Prometheus (Pombal et al., 2025) | Qwen2.5 | 3–14B | DA/PR | Multilingual (20+) | SOTA open on M-RewardBench | I5 |
| ***Self-supervised / preference optimization*** | | | | | | |
| Self-Taught Eval. (Wang et al., 2024b) | LLaMA-3 | 70B seed | PR | General | RewardBench 75.4 → 88.3, no labels | I1 |
| ***Reinforcement learning (verifiable rewards)*** | | | | | | |
| J1 (Whitehouse et al., 2025) | LLaMA-3.1/3.3, Qwen3 | 8B, 32B, 70B | DA/PR | Verifiable+open | SOTA 8B on JudgeBench/RewardBench | I1 |
| JudgeLRM (Chen et al., 2025) | Qwen | 3–14B | DA | Reasoning | 3B > GPT-4; 7B > DeepSeek-R1 (+2.6 F1) | I1 |
| CompassJudger-2 (Zhang et al., 2025c) | Qwen2.5 | 7B | DA/PR | Generalist | Matches DeepSeek-V3 on a 4-benchmark avg. | I1 |
| J4R (Xu et al., 2025b) | Qwen2.5 | 7B | PR | Reasoning | 7B judge surpasses GPT-4o | I1 |
| GenPRM (Zhao et al., 2026) | R1-Distill | 1.5–32B | PRM | Math steps | 7B gen. PRM > 72B scalar PRM (test-time scaling) | I1 |
| StepWiser (Xiong et al., 2025) | Qwen2.5 | 1.5B | PRM | Reasoning steps | Stepwise generative judge via RL | I1 |
| TIR-Judge (Xu et al., 2026) | Qwen3 | 4–8B | DA/PR/LR | Verifiable+open | 8B listwise ≈ Claude-Opus-4 | I5 |
| ***Representation-based (decoding-free)*** | | | | | | |
| INSPECTOR (Li et al., 2026) | frozen SLM | 1.7B | DA (probe) | Structured | Beats prompted SLM at equal size | I1 |
| LAGER (Lai et al., 2026) | frozen LM | Varied | DA (probe) | Scoring | Up to +7.5% Spearman, no retraining | I1 |

Table 9: Master comparison of SLM and SLM-relevant judge systems surveyed in this paper, grouped by training paradigm. **Mode**: DA = direct assessment, PR = pairwise ranking, LR = listwise ranking, CR = critique, ORM = outcome reward model, PRM = process reward model. **Size** is total parameters for dense models and active/total for mixture-of-experts variants. **Insight** links each system to the field-wide insight (I1–I5) it most directly supports. Headline results are reported by the cited source under that source's protocol; cross-row comparison is therefore indicative, not controlled (see Limitations).

scales. PandaLM (Wang et al., 2024c) uses GPT-3.5-generated verdicts with human-crafted contexts. The effective assumption: the teacher's judgment quality is bottlenecked only by its cost, and distillation can transfer the teacher's evaluative capacity at one-time training cost.

**Human-annotated preference data.** FLAMe (Vu et al., 2024) avoids teacher-LLM distillation entirely, training on >5M human annotations across >100 quality assessment tasks drawn exclusively from permissively licensed datasets. Human annotation provides more reliable training labels and eliminates the risk of propagating teacher-model biases. The tradeoff is substantial annotation infrastructure; FLAMe achieves 87.8% on RewardBench while exhibiting lower CoBBLEr bias than proprietary judges.

**Rubric-conditioned evaluation data.** Prometheus (Kim et al., 2024a) introduces datasets where each evaluation instance includes a task, candidate response, human-written rubric specifying evaluation criteria, and a reference answer. The model learns to condition its verdict on the rubric rather than producing generic assessments. The Feedback Collection (pointwise) and Preference Collection (pairwise), released alongside Prometheus 2, have become standard resources for judge fine-tuning.

**Diversity-driven data.** GLIDER (Deshpande et al., 2024) scales the data dimension differently: rather than increasing label quality, it trains across 183 distinct evaluation metrics and 685 domains, developing broad coverage over the evaluation space. Auto-J (Li et al., 2024b) achieves a similar effect by training across 58 diverse scenario types.

**Bias-targeted data: OffsetBias.** OffsetBias specifically targets the failure mode where judges exhibit different error rates for different semantic categories of response. OffsetBias provides contrast pairs chosen to train against systematic favoritism toward particular response types (e.g., overly verbose responses, responses with confident-sounding hedges). Skywork-Critic models trained on OffsetBias demonstrate substantially reduced bias across multiple evaluation categories.

TRAINING DATA DIVERSITY GOVERNS CROSS-TASK GENERALIZATION, NOT TRAINING DATA VOLUME. [REPLICATED: 3 studies] *The most important lesson from comparing GLIDER, FLAMe, and Prometheus 2 is that the number of distinct evaluation dimensions covered during training, not the total number of training examples, determines whether a judge*

| System | Base | Size | Training | RB | JB | FLASK | Source |
|---|---|---|---|---|---|---|---|
| Selene Mini | LLaMA-3.1 | 8B | S+D | 0.891 | n/a | 0.613 | Alexandru et al. (2025) |
| J1 (Pairwise) | LLaMA-3.1 | 8B | R | 0.857 | 0.420[†] | n/a | Whitehouse et al. (2025) |
| J4R | Qwen2.5 | 7B | R | n/a | +6.7%[‡] | n/a | Xu et al. (2025b) |
| FLAMe-RM | PaLM-2 | 24B[*] | S | 0.878 | n/a | n/a | Vu et al. (2024) |
| GLIDER | Phi-3.5-mini | 3.8B | S | 0.784 | n/a | 0.615 | Deshpande et al. (2024) |
| Prometheus 2 | Mistral | 7B | S | 0.720 | n/a | 0.470[a] | Kim et al. (2024b) |
| CompassJudger-2 | Qwen2.5 | 7B | R | n/a | n/a | n/a | Zhang et al. (2025c) |
| M-Prometheus | Qwen2.5 | 3–14B | S | n/a | n/a | n/a | Pombal et al. (2025) |
| *LLM baselines (for reference; not fine-tuned for judging)* | | | | | | | |
| GPT-4o-mini | n/a | n/a | P | 0.801 | n/a | 0.615 | Alexandru et al. (2025) |
| GPT-4o (vanilla) | n/a | n/a | P | n/a | 0.509 | n/a | Tan et al. (2025) |
| GPT-4-0125 | n/a | n/a | P | 0.859 | n/a | n/a | Vu et al. (2024) |
| Claude-3.5-Sonnet | n/a | n/a | P | n/a | 0.643 | n/a | Tan et al. (2025) |
| LLaMA-3.1-8B-Inst. | LLaMA-3.1 | 8B | P | 0.750 | n/a | 0.448 | Alexandru et al. (2025) |

[*] FLAMe-RM-24B exceeds the ≤14B scope; included because it is the only non-proprietary system reporting RewardBench above 87%. [†] Position-consistent accuracy. [‡] Relative improvement over GPT-4o on J4R's overall score across PPE, JudgeBench, and ReasoningJudgeBench. [a] Prometheus 2 (7B) FLASK Pearson from Selene Mini's Table 1, where all judges share one evaluation run; Prometheus 2's own paper reports higher correlations under its own protocol.

Table 10: Cross-system comparison on shared benchmarks. Each number is reported by the cited source under that source's evaluation protocol. **RB** = RewardBench (selection accuracy), **JB** = JudgeBench (overall accuracy), **FLASK** = Pearson correlation with human scores. Cells marked "n/a" indicate the system was not evaluated on that benchmark; CompassJudger-2 and M-Prometheus report only on benchmarks no other row shares, which is itself the protocol fragmentation behind challenge C5. Rows above the line are judge systems trained for evaluation, all ≤14B except FLAMe-RM as noted; rows below are baselines evaluated by prompting alone. The **Training** column gives the supervision type: P = off-the-shelf prompting, S = supervised fine-tuning, D = DPO or self-supervised, R = reinforcement learning (GRPO). **Caution:** cross-row numbers are *indicative, not controlled*, since the trained systems were fine-tuned for judging while the baselines were not. The one controlled pair is Selene Mini against LLaMA-3.1-8B-Instruct, which share a backbone and a parameter count. This asymmetry is the subject of Insight I1 (§3).

| Paradigm | Signal | Cost | Human | Systems |
|---|---|---|---|---|
| Off-shelf | Prompt only | None | None | Qwen3, Phi-4 |
| SFT (dist.) | Teacher verd. | Med | Indirect | JudgeLM |
| SFT (rubric) | Human+syn. | Med | Rubrics | Prometheus 2 |
| SFT (self) | Self-gen. | Low | None | Self-Taught |
| DPO | Pref. pairs | Low | Pref. | Selene Mini |
| RL (GRPO) | Verif. rew. | High | None | J1, JudgeLRM |
| Repr. probe | Probe labels | V.Low | Small set | INSPECTOR |

Table 11: Overview of judge training paradigms. Compute cost is relative to off-the-shelf deployment. Human data indicates whether human annotation is required.

*generalizes to unseen evaluation tasks.* GLIDER's 183-metric coverage is what enables its surprising cross-task transfer to FLASK, a benchmark not in its training distribution. FLAMe's 100-task human-annotation breadth achieves the same generalization through human label diversity. Both outperform single-task distillation approaches that use orders of magnitude more training examples but cover fewer evaluation dimensions. The mechanism is coverage of the evaluation space: each new evaluation dimension during training teaches the model a new discriminative pattern, whereas additional examples from the same dimension provide diminishing returns after the pattern is learned.

| Resource | Mode | Label | Used by |
|---|---|---|---|
| Feedback Collection | DA | Synthetic | Prometheus |
| Preference Collection | PR | Synthetic | Prometheus 2 |
| JudgeLM data | DA/PR | GPT-4 | JudgeLM |
| FLAMe (100+ tasks) | DA/PR | Human | FLAMe |
| HelpSteer2 | DA | Human | Skywork-Critic |
| OffsetBias | PR | Contrast | Skywork-Critic |
| Self-Taught pairs | PR | Self-gen. | Self-Taught |
| J1 synthetic (∼22K) | PR | Verifiable | J1 |

Table 12: Public training and meta-evaluation resources for SLM judges. "Mode" indicates direct assessment (DA), pairwise (PR), or critique (CR); "Label" indicates the supervision source.

Table 12 consolidates the public training resources that underpin these data-construction strategies, which together form the practical starting point for any new SLM judge.

### C.4.2 Anti-Bias Training Techniques

**Swap augmentation.** JudgeLM (Zhu et al., 2025) evaluates both orderings of response pairs and includes both in the training set, forcing the model to reach the same verdict regardless of positional order. This directly penalizes position-dependent behavior during training at the data level.

**Reference support and reference drop.** JudgeLM alternates between providing a gold reference answer and withholding it during training. This dual-mode training produces a judge that can operate reference-free (generalizable to open-ended tasks) while maintaining higher accuracy when references are available.

**Synthetic contrast pairs.** Self-Taught Evaluators (Wang et al., 2024b) generate contrasting response pairs for unlabeled instructions: one high-quality response and one subtly worse variant. Training on these pairs without any human labels improves RewardBench from 75.4 to 88.3 for a Llama-3 70B seed model. The key mechanism: the bootstrapping loop (train judge, use judge to generate better training data, retrain judge) allows the model's own evaluative capacity to self-correct, eliminating the teacher-bias propagation risk of distillation approaches.

**Dynamic difficulty curricula.** A 2025 trend in alignment training replaces static training sets with dynamic curricula that adapt the training distribution to the model's current capability frontier. The curriculum progressively introduces harder evaluation pairs as the model demonstrates competence on simpler ones, preventing early overfitting to easy evaluation patterns.

### C.4.3 Weight Merging for Multi-Mode Judges

Prometheus 2 (Kim et al., 2024b) trains two separate models for pointwise direct assessment (DA) and pairwise ranking (PR), then merges their weights. The authors experiment with multiple merging strategies (Task Arithmetic, TIES, DARE-TIES, DARE-Linear, SLERP) and find that DARE-Linear achieves the best performance. DARE-Linear first sparsifies the delta parameters (differences between fine-tuned and base weights) by randomly dropping a proportion and rescaling the remainder, then merges linearly:

$$\theta_{\text{merged}} = \theta_{\text{base}} + \alpha \cdot \hat{\Delta}_{\text{DA}} + (1-\alpha) \cdot \hat{\Delta}_{\text{PR}} \quad (6)$$

where $\hat{\Delta}$ denotes the sparsified, rescaled delta vectors. Experiments show the merged model outperforms both single-task variants, suggesting that evaluation "skills" from the two modes are complementary in parameter space and do not exhibit catastrophic interference.

WEIGHT MERGING SUCCEEDS BECAUSE POINTWISE AND PAIRWISE EVALUATION ACTIVATE COMPLEMENTARY SUBSPACES. [PRELIMINARY: single study] *The success of Prometheus 2's weight merging indicates that pointwise and pairwise evaluation skills are not in competition for the same parameters; instead, they develop in orthogonal regions of the weight space, allowing lossless combination.* This is a stronger claim than "merging works": it implies that the two evaluation modalities require different internal circuitry. Pointwise scoring requires absolute calibration against internalized quality standards; pairwise comparison requires relative feature matching between two responses. These computations naturally engage different attention patterns and projection weights, explaining why merging produces a model better than either constituent. The practical implication is that modality-specific fine-tuning followed by weight merging may be more efficient than joint multi-task training, which forces both modalities to share parameters that they would otherwise specialize.

## C.5 Reinforcement Learning for Judge Training

The third major paradigm applies RL directly to the judging objective, optimizing evaluation accuracy rather than imitation of a teacher's outputs.

### C.5.1 Direct Preference Optimization (DPO)

Rafailov et al. (2023) reformulate the traditional PPO-RLHF pipeline as a direct classification objective on preference pairs, eliminating the need for a separate reward model. Given prompt $x$, preferred response $y_w$, and rejected response $y_l$, DPO minimizes:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right] \quad (7)$$

where $\pi_\theta$ is the policy being trained, $\pi_{\text{ref}}$ is a frozen reference policy, and $\beta$ controls the KL penalty. Selene Mini (Alexandru et al., 2025) combines a DPO loss with supervised fine-tuning to train a general-purpose 8B judge. DPO's key advantage is stability: no reward model instability, no PPO

hyperparameter sensitivity, and no critic network memory overhead.

#### C.5.2 Group Relative Policy Optimization (GRPO)

GRPO was introduced in DeepSeek-R1 (Guo et al., 2025) as a computationally efficient alternative to PPO. For each input prompt, GRPO samples a group of outputs $\{o_1, \ldots, o_G\}$, computes rewards $\{r_1, \ldots, r_G\}$, and normalizes within the group:

$$A_i = \frac{r_i - \mu_{\text{group}}}{\sigma_{\text{group}}} \tag{8}$$

The group-relative normalization eliminates the need for a learned value function, substantially reducing memory and compute vs. PPO. The policy gradient loss is:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}_i \left[\min(\rho_i A_i,\ \text{clip}(\rho_i, 1\pm\epsilon)A_i) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\right] \tag{9}$$

where $\rho_i = \pi_\theta(o_i|q)/\pi_{\text{ref}}(o_i|q)$ is the probability ratio.

**GRPO for judge training.** J1 (Whitehouse et al., 2025) applies GRPO with verifiable rewards to train thinking-capable judge models. The key innovation is a unified reward formulation:

- For **verifiable tasks** (math, code): Rule-based correctness reward.
- For **non-verifiable tasks**: Synthetic preference pairs provide pseudo-verifiable rewards.
- **Consistency reward**: Additional reward when the judge produces the same verdict across both positional orderings, directly penalizing position bias during RL training.

J1-Llama-8B and J1-Qwen-32B report state-of-the-art results for their sizes while trained on only ∼22K synthetic preference examples.

#### C.5.3 J4R: Equivalent Initial State Optimization

J4R (Xu et al., 2025b) addresses a subtle flaw in standard pairwise RL training: when positive and negative samples are drawn from different response spaces, the model may learn shortcuts based on response provenance rather than genuine quality discrimination. J4R uses equivalent initial state group relative policy optimization, enforcing that both preferred and rejected responses emerge from the same input distribution. This is particularly important for small SLM judges, whose limited capacity makes them more susceptible to shortcut learning.

RL Training Outperforms SFT For Judging Because It Optimizes The Evaluation Objective Directly. [Replicated: 2 studies] *SFT optimizes for imitation of a teacher's outputs, bounding the student's evaluative behavior by the teacher's; RL with verifiable rewards optimizes directly against judgment accuracy, allowing the student to discover evaluation strategies that the teacher never demonstrated.* J1-Llama-8B, trained on ∼22K synthetic examples, matches the prompted Llama-3.3-70B on RewardBench (85.7 against 85.4), demonstrating that a well-designed RL objective provides vastly more useful training signal per example than teacher-distilled verdicts. The mechanism is that SFT treats each verdict as a token-prediction target, whereas RL treats the entire verdict as a binary outcome (correct/incorrect) and adjusts the full generation distribution to maximize correctness. This outcome-level optimization avoids inheriting token-level biases from the teacher's reasoning style.

J1's Consistency Reward Is The First Training-Level Solution To Position Bias. [Preliminary: single study] *Prior position bias mitigations (swap augmentation, inference-time averaging) operate at the data or inference level; J1's consistency reward operates at the training objective level, providing gradient signal that directly penalizes the model's internal representations for producing position-dependent verdicts.* The distinction matters because data-level mitigations (swap augmentation) teach the model that both orderings should produce the same verdict, but they do so through example memorization. Objective-level mitigation (consistency reward) teaches the model that position-dependent behavior reduces reward regardless of the specific evaluation pair, producing a more general and robust position-invariance property.

### C.6 Reward Model Training: ORM vs. PRM

A related family of judge models are *reward models*: discriminative models trained to score candidate responses, used primarily in RLHF pipelines.

#### C.6.1 Outcome Reward Models (ORM)

ORMs evaluate only the final response, producing a scalar reward $r \in \mathbb{R}$:

$$\text{ORM} : (x, y) \to r \tag{10}$$

Training requires preference pairs $(y_w, y_l)$; the model learns to assign $r(y_w) > r(y_l)$. Key properties:

- **Low annotation cost**: Labels can be obtained automatically (checking math answer values, running code tests).
- **Process-agnostic**: Cannot identify which reasoning step caused a failure.
- **Reward hackable**: A model can reach a correct final answer through flawed reasoning and receive full ORM reward.

RewardBench (Lambert et al., 2025) is the primary benchmark for ORM evaluation.

#### C.6.2 Process Reward Models (PRM)

PRMs assign step-level rewards to each reasoning step (Lightman et al., 2024):

$$\text{PRM} : (x, [s_1, s_2, \ldots, s_T]) \rightarrow [r_1, r_2, \ldots, r_T] \tag{11}$$

where $s_t$ is the $t$-th reasoning step and $r_t \in \{-1, 0, +1\}$ indicates step correctness. PRMs provide:

- **Error localization**: Precisely identifies which step failed, enabling targeted correction.
- **Search guidance**: Can prune incorrect reasoning paths in Best-of-N sampling early.
- **Alignment quality**: Incentivizes correct reasoning processes, not just lucky correct answers.

The primary cost is annotation: step-level labels require human annotators to validate each step independently. Automated process supervision, using Monte Carlo rollouts to generate step-level labels automatically, reduces this bottleneck. A recent shift replaces scalar PRMs with *generative* process judges: GenPRM (Zhao et al., 2026) reasons explicitly about each step with code verification, and StepWiser (Xiong et al., 2025) trains stepwise generative judges via RL. Both report that small generative PRMs can surpass much larger scalar PRMs under test-time scaling, reinforcing I1 (targeted training over scale) for step-level evaluation specifically.

PRMs Are Particularly Valuable For SLM Judges Because They Compensate For Limited Parametric Knowledge. [Preliminary: proposed mechanism] *SLMs lack the broad domain knowledge of frontier models, making them more prone to the "lucky correct answer" failure mode that ORMs cannot detect; PRM-style step-level evaluation catches flawed reasoning even when the final answer is correct, providing a verification mechanism that compensates for the SLM's knowledge limitations.* This insight connects to the verifiability gradient (I5): in high-verifiability domains (math, code), step-level verification provides the strongest SLM judge signal because each step can be checked against external criteria. In low-verifiability domains, step-level evaluation degrades because the judge lacks the domain knowledge to assess intermediate reasoning quality. Appendix J discusses domain-specific implications.

### C.7 Knowledge Distillation for Judges

Knowledge distillation from a large, capable judge into a small, deployable one is the standard approach when ground-truth annotation is infeasible.

**Reasoning trace distillation.** DeepSeek-R1 (Guo et al., 2025) demonstrates that fine-tuning a small base on high-quality reasoning traces from a large teacher transfers chain-of-thought patterns without additional RL. Applied to judging, the same recipe fine-tunes a small judge on a large teacher's judgment traces, passing on multi-step evaluative reasoning that is hard to learn from verdict labels alone.

**Capacity ceiling.** Distilled judges cannot surpass the teacher's evaluative capacity. If the teacher cannot discriminate subtle quality differences, the distilled student inherits this limitation. This motivates combining distillation with RL: use distillation to bootstrap initial capability, then refine with task-specific RL rewards.

**Preference leakage risk.** When the distillation teacher (e.g., GPT-4) is also used to generate training data for the models being evaluated, the judge may systematically favor stylistically similar outputs, a manifestation of preference leakage (Li et al., 2025b). Training judges on human-annotated data (FLAMe) or self-generated contrasting pairs (Self-Taught Evaluators) avoids this family-lineage contamination.

Distillation Creates A Teacher-Bias Propagation Risk That RL Training Avoids. [Preliminary: proposed mechanism] *When a student judge is distilled from a teacher, it inherits not only the teacher's evaluation capability*

*but also the teacher's systematic biases (position preference, verbosity preference, self-preference), because the training objective cannot distinguish between the teacher's correct judgments and the teacher's biased ones.* This is the fundamental limitation of distillation-based judge training: the student's error profile is a subset of the teacher's error profile. RL with verifiable rewards breaks this coupling because the reward signal comes from ground truth (was the verdict correct?), not from the teacher's judgment. Self-Taught Evaluators (Wang et al., 2024b) partially mitigate this by using the model's own iteratively improved judgments, but they remain bounded by the model's own biases unless external verification is introduced.

### C.8 Representation-Based Judging: Beyond Generation

The paradigms above all produce a verdict by generating tokens. A final line of work questions that assumption entirely, extracting the verdict from a model's internal states without any generation. Two systems, INSPECTOR and LAGER, anchor this direction.

#### C.8.1 The INSPECTOR Framework

The most fundamental rethinking of judge training shifts from generative to representation-based evaluation. Li et al. (2026) introduce INSPECTOR (INternal Signal Probing and EvaluaTion Of Representations), which probes hidden states of a frozen small LM to predict evaluation scores directly.

The **Semantic Capacity Asymmetry Hypothesis** states: the computation required to evaluate response quality is substantially smaller than the computation required to generate that response. Evaluation is a classification problem defined over the model's world representation; generation is a sequential decoding problem over the full output distribution.

INSPECTOR operates as:

1. Forward-pass a (question, candidate response) pair through a frozen small LM ($\leq$1.7B), collecting hidden states $\{h_1^{(l)}, \ldots, h_T^{(l)}\}$ at layer $l$.
2. Pool hidden states across relevant token positions.
3. Apply a lightweight linear probe $\psi_a : \mathbb{R}^d \to \mathbb{R}$ per evaluation aspect $a$.
4. Report aspect-level scores $\{s_a = \psi_a(h_{\text{pool}})\}$ as the verdict.

Experiments on GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), and GPQA show that INSPECTOR substantially outperforms prompting-based SLM judges of the same size and closely approximates full LLM judges.

#### C.8.2 LAGER: Cross-Layer Logit Aggregation

LAGER (Lai et al., 2026) replaces single-layer final logit scoring with cross-layer aggregation:

$$\hat{s} = \sum_{\text{layers } l \in L} w_l \sum_{d \in \mathcal{D}} d \cdot P_l(\text{token} = d) \qquad (12)$$

where $\mathcal{D}$ is the set of digit tokens (1–5) and $w_l$ are learnable layer-weighting parameters. Mid-layer representations carry more semantically stable evaluative signals than the final layer. Because the backbone stays frozen and the aggregation is applied after the forward pass, LAGER is a post-hoc addition to an existing judge rather than a new model: it reports gains of up to 7.5% in Spearman correlation over the strongest baseline on FLASK, HelpSteer, and BIGGen with no retraining.

MID-LAYER REPRESENTATIONS CARRY STRONGER EVALUATIVE SIGNALS THAN FINAL-LAYER REPRESENTATIONS. [PRELIMINARY: single study] *LAGER's finding that mid-layer aggregation outperforms final-layer scoring implies that evaluative information is encoded earlier in the transformer forward pass and is partially overwritten by the generation-oriented computation in later layers.* This is consistent with the mechanistic interpretability literature showing that middle layers encode semantic content while final layers encode output distribution. For evaluation, the semantic representation ("is this response logically sound?") is the relevant computation; the output distribution ("what token should come next?") adds noise. Probing mid-layers extracts the evaluative signal before it is contaminated by generation-oriented processing, explaining why representation-based evaluation can match generative evaluation at a fraction of the compute.

#### C.8.3 Implications: Probing as Judge Training

INSPECTOR and LAGER collectively shift the question from "how to train a better SLM judge"

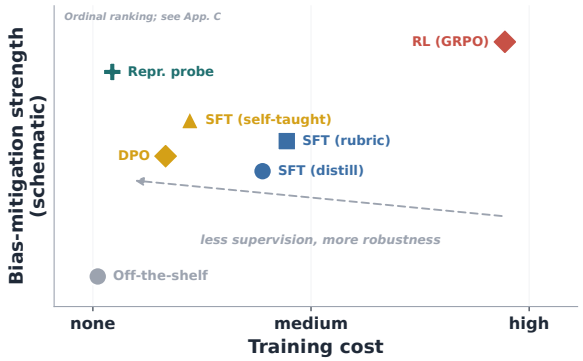



Figure 5: Judge training paradigms positioned by training cost and bias-mitigation strength. Both axes are ordinal rankings argued in this appendix, not measured quantities. Marker shape encodes the supervision source. The dashed guide marks the direction the field has moved, away from expensive teacher-dependent training toward paradigms that reach comparable bias resistance for less, with representation probing at the extreme. RL with verifiable rewards is the exception on cost: it buys the strongest bias mitigation at the highest training expense.

toward "how to better extract the evaluative capacity that small models already encode." Two consequences:

1. **Probe training is far cheaper than full fine-tuning**: A lightweight linear probe trained on a small labeled set (<1,000 examples) from a frozen model reduces training cost to near-negligible.

2. **Frozen SLMs are viable judges for structured tasks**: Rather than fine-tuning a new judge per domain, domain-specific probes can be trained atop a frozen general-purpose SLM, enabling rapid judge adaptation to new evaluation settings.

### C.9 Summary: Training Paradigm Comparison

Table 13 provides a comprehensive comparison of all training paradigms along key deployment dimensions, and Figure 5 positions them on the training-cost versus bias-mitigation plane. The plot makes the central trade-off visible: RL with verifiable rewards achieves the strongest bias mitigation but at the highest training cost, while representation probing reaches comparable bias resistance at almost no training cost, and off-the-shelf prompting sits alone in the low-cost, low-robustness corner.

THE TRAINING PARADIGM PROGRESSION FROM DISTILLATION TO RL TO PROBING FOLLOWS A DECREASING-SUPERVISION GRADIENT.

| Paradigm | Data | Cost | Bias Mit. | Systems |
|---|---|---|---|---|
| Off-shelf | None | None | None | Qwen3, Phi-4 |
| SFT (dist.) | Teacher | Med | Swap aug. | JudgeLM |
| SFT (rubric) | Hum.+syn. | Med | Ref. drop | Prometheus 2 |
| SFT (self) | Self-gen. | Low | Iterative | Self-Taught |
| DPO | Pref. pairs | Low | Implicit | Selene Mini |
| RL (GRPO) | Verif. rew. | High | Consist. r. | J1, JudgeLRM |
| Repr. probe | Probe lab. | V.Low | N/A | INSPECTOR |

Table 13: Comparison of SLM judge training paradigms. Bias mitigation indicates the primary mechanism for reducing systematic evaluation biases during training.

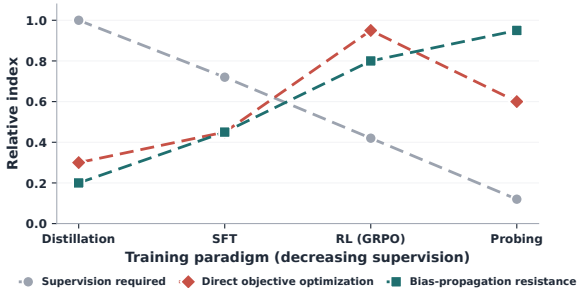



Figure 6: The training-paradigm progression follows a decreasing-supervision gradient. Moving from distillation to SFT to RL to probing, the external supervision required falls monotonically and resistance to teacher-bias propagation rises, while direct optimization of the evaluation objective peaks at RL and drops at probing, which optimizes nothing but reads out what the frozen model already encodes. The three indices are ordinal encodings of the qualitative ranking argued in this appendix, not measured quantities.

[CONSENSUS: 4 systems] *Each successive training paradigm requires less external supervision: distillation needs a capable teacher, SFT needs annotated data, RL needs only a reward signal, and probing needs only a small calibration set from a frozen model (Figure 6).* This gradient has a direct practical implication: as the field moves toward less-supervised paradigms, the barrier to entry for training new SLM judges decreases. A researcher with a single GPU can now train a competitive judge via probing (INSPECTOR) or RL on synthetic data (J1), without access to proprietary teacher models or large-scale human annotation infrastructure. The decreasing supervision requirement also implies decreasing risk of bias propagation: probing from a frozen model cannot inherit the teacher's evaluation biases because there is no teacher in the loop.

# D Individual Judging: Token Budget and Scale

With training fixed, we turn to inference-time decisions: how many reasoning tokens to spend, when thinking mode helps, and how scale interacts with capability. This is the full account behind §3 and Figure 2.

## D.1 Threats to Comparability for Insight I1

Table 14 decomposes the six sources of asymmetry behind the comparisons in §3. The first is the one most easily mistaken for the second: when a trained judge and a proprietary baseline differ in base model, scale, and training all at once, an improvement cannot be attributed to any one of them. Only one comparison in the surveyed literature closes this. Alexandru et al. (2025) report Selene Mini at 0.891 on RewardBench against 0.750 for LLaMA-3.1-8B-Instruct, the model it was fine-tuned from, holding architecture, scale, and pre-training fixed so that evaluation-specific training is the only variable.

The second source, judges fine-tuned against baselines merely prompted, is deliberate: I1 concerns what training buys, not inherent capacity. Three further observations bound how far that generalizes. Selene Mini holds its advantage on 11 benchmarks outside its training distribution (Alexandru et al., 2025) and across six RewardBench prompt templates, which addresses both benchmark overlap and template sensitivity. J1 enforces position-consistent evaluation through a consistency reward rather than post-hoc swapping (Whitehouse et al., 2025). JudgeLRM's crossover holds on both GPT-4-annotated and human-annotated data (Chen et al., 2025), so it does not depend on a teacher model's idiosyncrasies.

The last two rows have only partial mitigations: IFR reporting (§3.4) exposes parsing differences, and position-consistent accuracy on JudgeBench controls for presentation order.

| Asymmetry | Impact on I1 | Mitigation |
|---|---|---|
| Base model and scale differ | Confounds training with capacity | Selene Mini vs. its own base model at 8B |
| Training data: judge fine-tuned, baseline prompted | This IS I1's claim | By design |
| Benchmark overlap with training set | Could inflate judge scores | Selene Mini: 11 OOD benchmarks |
| Prompting template varies across studies | Uncontrolled | Selene Mini: 6 RewardBench templates |
| IFR/parsing differences | Sub-3B models fail to parse more often | IFR analysis in §3.4 |
| Pairwise vs. pointwise protocol | Studies differ | JudgeBench uses PC-Acc to control for position |

Table 14: Threats to comparability for Insight I1. Each row lists an asymmetry source, its impact on the claim, and the mitigation available in the surveyed literature. The last two rows have only partial mitigations.

## D.2 The Individual Judge: Scope and Design Space

An individual SLM judge is a single language model deployed to evaluate candidate responses without auxiliary models, ensemble voting, or inter-agent communication. It operates according to Eq. 2 and Eq. 3 from Appendix A, consuming a (question, response, optional rubric) tuple and producing a verdict. The evidence base for this section draws primarily on seven studies in three threads: the overthinking analyses of Sui et al. (2025), Shojaee et al. (2026), and Srivastava et al. (2025b); the reasoning-budget framework of Han et al. (2025) and Aggarwal et al. (2025); and the thinking-mode evaluation of Jayarao et al. (2025) and Ma et al. (2025). The training-beats-scale crossover evidence (GLIDER, Prometheus 2, JudgeLM, Selene Mini, JudgeLRM, J1) is detailed in the main paper's I1 discussion (§3) and the evidence tables in Appendix N.1.

The design space has four primary axes:

1. **Training paradigm**: How the model acquired its judging capability (Appendix C).
2. **Token budget**: How many reasoning tokens to allocate per evaluation.
3. **Inference mode**: Thinking vs. non-thinking; chain-of-thought vs. direct verdict.
4. **Architectural scale**: How parameter count interacts with judging capability given a fixed training budget.

This appendix examines axes 2–4 in depth, with particular focus on the non-obvious finding that "more reasoning" does not monotonically improve verdict quality.

## D.3 Token Budget Analysis for SLM Judges

The single highest-leverage operational decision for an individual judge is how many tokens to spend per evaluation. We decompose that budget, then

derive the scaling laws that govern its optimal setting.

#### D.3.1 Components of the Total Token Budget

For an individual judge evaluating a single (question, response) pair, the total token budget $B$ comprises:

$$B = B_{\text{sys}} + B_{\text{ctx}} + B_{\text{rationale}} + B_{\text{verdict}} \quad (13)$$

where $B_{\text{sys}}$ is the system prompt (evaluation instructions, rubric), $B_{\text{ctx}}$ is the context (question + candidate response), $B_{\text{rationale}}$ is the optional chain-of-thought reasoning the judge generates before its verdict, and $B_{\text{verdict}}$ is the final verdict token(s). For typical judge deployments:

- $B_{\text{sys}} \approx 200$–$800$ tokens (rubric complexity determines this)
- $B_{\text{ctx}} \approx 500$–$3{,}000$ tokens (varies with response length and task)
- $B_{\text{rationale}} \approx 0$–$2{,}000$ tokens (0 for verdict-only, up to full CoT)
- $B_{\text{verdict}} \approx 1$–$50$ tokens (score digit(s) or pairwise label)

The operational cost per evaluation scales with $B$. For $N = 10^5$ evaluations during a training run, $B_{\text{rationale}}$ becomes the dominant cost driver, making its optimization the highest-leverage operational decision in SLM judge deployment.

#### D.3.2 Optimal CoT Scaling Laws: The Sweet Spot

A foundational finding of 2025 reasoning research, with direct implications for judging, is that reasoning performance follows a non-monotonic curve as a function of chain-of-thought length:

1. **Below the sweet spot**: The model underreasons, missing evaluation criteria that additional analysis would have caught.
2. **At the sweet spot**: Reasoning tokens provide genuine evaluative signal; each step either confirms or revises the current assessment.
3. **Above the sweet spot**: The model overthinks, generating redundant elaborations that introduce error cascades and entropy accumulation.

The optimal CoT length depends on two factors:

- **The judge**: models differ in how many tokens they need before extra reasoning stops helping (Aggarwal et al., 2025).
- **The task**: Harder evaluation tasks warrant more reasoning tokens before the degradation curve begins.

The practical implication: there is no universally optimal $B_{\text{rationale}}$. Each (model, task-difficulty) pair has its own sweet spot.

EACH JUDGE AND TASK HAS ITS OWN SWEET SPOT, SO REASONING BUDGETS MUST BE SET PER SETTING. [PRELIMINARY: proposed mechanism] *A judge stops gaining from extra reasoning once its chain of thought has surfaced the evidence that decides the verdict; how many tokens that takes depends on how much of the relevant signal the model already encodes and on how hard the comparison is.* A single fixed budget is therefore wrong for most settings, which motivates the adaptive allocation methods below, and the sweet spot has to be measured for each judge and task rather than read off model size. This connects the sweet-spot finding to I1 (judging is discriminative) and I2 (reasoning tokens help only when they surface new signal).

#### D.3.3 Budget-Aware Reasoning Allocation

Han et al. (2025) introduce token-budget-aware LLM reasoning: models are prompted with an explicit token budget constraint derived from estimated task complexity. Key findings:

- Tasks within a model's clear competency can be evaluated accurately with zero rationale tokens.
- For tasks near the competency boundary, rationale generation helps, but only up to approximately 200–400 tokens.
- For tasks beyond the model's competency, extended reasoning actively harms accuracy.

Wang et al. (2024a) provide a complementary cost-performance frontier analysis: different reasoning strategies (zero-shot direct, zero-shot CoT, few-shot CoT, self-consistency) occupy distinct positions on the token budget vs. accuracy curve. The optimal strategy is task-dependent.

Techniques for adaptive token allocation include:

- **TALE / IBPO**: Token-budget-aware training methods that teach models to dynamically adjust reasoning effort based on perceived query difficulty.

- **Length-filtered voting**: When generating multiple candidate verdicts, filter out those with reasoning chains deviating from the established optimal range.
- **SelfBudgeter**: A lightweight classifier that predicts optimal CoT length before generation begins.

BUDGET-AWARE JUDGING REDUCES COST WITH LITTLE ACCURACY LOSS BECAUSE WITHIN-COMPETENCY RATIONALES ARE POST-HOC, NOT COMPUTATIONALLY PRIOR. [REPLICATED: 2 studies] *For tasks well within a model's competency, the verdict is effectively determined in the forward pass before any reasoning tokens are generated; the subsequent rationale is post-hoc justification, not genuinely prior reasoning.* Explicitly constraining the budget forces the model to commit to its top-layer representation, avoiding the waste of extended text generation for an already-decided verdict. This mechanism explains why budget-aware prompting cuts evaluation cost substantially with little accuracy loss on within-competency tasks: the removed tokens were never contributing to the verdict in the first place.

#### D.3.4 Self-Consistency for Individual Judges

Self-consistency (Wang et al., 2022) runs the same judge multiple times with temperature $T > 0$ and takes the majority verdict. For judges:

- **Variance reduction**: Reduces random generation noise without increasing model size.
- **Pseudo-ensemble**: A single model producing diverse reasoning paths weakly approximates an ensemble.
- **Cost**: Scales linearly with the number of samples $k$; typically $k = 3$–$5$ is optimal.

SELF-CONSISTENCY REDUCES VARIANCE BUT CANNOT REDUCE BIAS, MAKING IT CATEGORICALLY DIFFERENT FROM CROSS-FAMILY ENSEMBLES. [PRELIMINARY: proposed mechanism] *Self-consistency is most effective when individual verdict samples are high-variance but independently unbiased; it is least effective when the judge has a systematic bias, because all samples share the same directional error and majority voting reinforces the bias rather than canceling it.* This is the critical distinction from cross-family ensemble methods (Appendix E), which decorrelate systematic biases because different model families encode different systematic errors. A practitioner should use self-consistency when the primary concern is generation noise (stochastic variation between runs) and cross-family ensembles when the primary concern is systematic bias (consistent directional error). Using self-consistency to address systematic bias will waste compute without improving accuracy.

#### D.3.5 Comparative Context as Inference-Time Signal

A third inference-time strategy adds information rather than repeating the same query. Crowd Comparative Reasoning (Zhang et al., 2025b) generates additional *crowd* responses to the same prompt and asks the judge to evaluate the candidate pair against that wider field rather than in isolation, using the resulting crowd judgment as context for the final verdict. The reported effect is a more complete chain of thought and accuracy that continues to improve as inference scales.

The contrast with self-consistency is the point. Self-consistency resamples the same conditioning and therefore cannot introduce evidence the judge did not already have; comparative context changes the conditioning itself. This is the same boundary that governs I2 and I4: extra inference-time compute pays only when it surfaces discriminative signal that was previously absent, whether that signal comes from a tool, a differently trained panel member, or an additional point of comparison.

### D.4 Thinking Modes in SLM Judges

How a judge spends its token budget matters as much as how large the budget is. We now compare explicit thinking modes against direct verdicts, covering the empirical gains, the architectural mechanisms that enable them, and the always-thinking alternative.

#### D.4.1 Thinking vs. Non-Thinking: Empirical Evidence

Jayarao et al. (2025) evaluate Qwen 3 models (0.6B to 4B) in thinking and non-thinking modes on pairwise judge tasks. Key findings:

- Thinking mode adds about **10 points** of accuracy over non-thinking mode at every size tested (0.6B, 1.7B, and 4B).
- The largest gains appear on the Reasoning and Chat Hard categories of RewardBench, and the smallest on Safety.

- The extra compute is modest: thinking mode uses **1.3–2.9**$\times$ the FLOPs of a direct verdict.
- Thinking judges also stay more consistent under positional, bandwagon, and verbosity biases, by about 6 points on average for the 4B model.

THINKING MODE PROVIDES THE LARGEST RELATIVE GAIN AT THE SMALLEST MODEL SIZES BECAUSE IT COMPENSATES FOR SPARSER INTERNAL REPRESENTATIONS. [PRELIMINARY: single study] *Because the absolute gain is about the same at every size, the relative gain is largest for the smallest judge: about 19% at 0.6B against 13% at 4B on the RewardBench average (Jayarao et al., 2025).* A plausible reading is that explicit reasoning writes out the comparison that larger models carry out internally, so it helps most where internal representations are weakest. This implies a practical deployment rule: thinking mode is most cost-effective for the smallest deployable judge size, and its relative value shrinks as the base model grows.

#### D.4.2 The Thinking Budget Mechanism (Qwen 3)

Qwen 3 implements a thinking budget as a per-query inference parameter $B_{\text{think}}$:

- $B_{\text{think}} = 0$: Non-thinking mode (fast inference, direct verdict).
- $B_{\text{think}} > 0$: Thinking mode with the specified ceiling; the model self-regulates, terminating early on simple tasks.

For judge deployment, this enables difficulty-adaptive evaluation: route simple evaluations to $B_{\text{think}} = 0$ and complex evaluations to $B_{\text{think}} =$ 1,000–2,000. A lightweight classifier can estimate task difficulty from rubric complexity and response length, routing accordingly. Because thinking mode costs up to about 3$\times$ the compute of a direct verdict, sending only hard comparisons to thinking mode keeps most of the savings while preserving accuracy where deliberation matters.

#### D.4.3 Always-Thinking Models: DeepSeek-R1-Distill

DeepSeek-R1-Distill models always generate reasoning traces enclosed in `<think>...</think>` tags, with no non-thinking mode. This design is advantageous for high-complexity evaluation tasks (verifiable STEM) where deliberation provides genuine uplift. For open-ended evaluation (writing quality, helpfulness), the always-thinking mode often produces long, redundant reasoning chains with no accuracy benefit over non-thinking SLMs.

This tradeoff motivates a hybrid deployment strategy: use DeepSeek-R1-Distill judges for structured, verifiable evaluation domains, and non-thinking Qwen 3 or Phi-4-mini for open-ended, preference-based evaluation.

#### D.4.4 Reasoning Without Thinking

A distinct approach (Ma et al., 2025) directs the model to evaluate the response without generating an explicit reasoning trace, relying on internalized evaluation heuristics. The model's "reasoning" happens implicitly in its internal forward pass rather than in generated tokens. This is most effective for tasks clearly within the model's competency and eliminates the entropy accumulation risk of extended generation entirely.

### D.5 Scale-Capability Relationship for Individual Judges

Token budget and inference mode interact with a third axis: raw parameter count. This subsection establishes where scale does and does not matter for judging, bounding the regime from below (a capability floor) and above (a knowledge ceiling).

#### D.5.1 The Core Finding: Targeted Training Dominates Scale

The most consistent finding across the individual SLM judge literature is that training composition and evaluation-specific optimization dominate raw parameter count. Three converging lines of evidence:

1. **GLIDER vs. GPT-4o**: GLIDER (3.8B, Phi-3.5-mini base) outperforms GPT-4o on FLASK despite being far smaller (Deshpande et al., 2024).
2. **Selene Mini vs. its own base model**: with the backbone fixed at LLaMA-3.1-8B, evaluation training lifts RewardBench accuracy from 0.750 to 0.891, a 19% gain (Alexandru et al., 2025).
3. **Same-backbone comparisons**: in all eight surveyed papers that report a trained judge next to its own base model under one protocol, training improves the base model, by 15% to 63% in relative terms (Table 33).

TARGETED TRAINING DOMINATES SCALE BECAUSE JUDGING REQUIRES DISCRIMINATIVE PRECISION, NOT GENERATIVE BREADTH. [CONSENSUS: 5 independent lines of evidence] *Scale amplifies generative capability because generation requires broad knowledge coverage across the full output vocabulary; judgment requires only detecting quality-relevant features (logical consistency, factual correctness, instruction adherence), a classification boundary rather than a generation distribution.* A model trained on many classification-relevant examples develops more precise discriminative boundaries than a larger untrained model, regardless of parameter count. The GLIDER result is not an exception; it is the expected consequence of this mechanism. For judge quality, training data composition matters more than parameter count, because evaluation operates in a lower-dimensional feature space than generation.

#### D.5.2 The Capability Threshold: Below 1B

Below approximately 1B parameters, instruction-following capability degrades such that evaluation format compliance becomes unreliable:

- Models below 1B in non-thinking mode often fail to produce a parseable verdict on complex rubrics.
- Thinking mode (Qwen 3-0.6B) improves format compliance, but evaluative reasoning stays well below that of larger models.
- INSPECTOR (Li et al., 2026) circumvents this threshold via representation probing: a 1.7B frozen model's hidden states contain enough evaluative signal for lightweight probe classification without any generation.

#### D.5.3 The Capability Ceiling: When Size Does Matter

For certain evaluation tasks, SLMs genuinely cannot match frontier models regardless of training because the evaluation requires domain knowledge that the SLM does not possess:

- **Graduate-level STEM**: Evaluating a response to a graduate-level physics problem requires first-order physics knowledge that smaller models may lack.
- **Nuanced safety evaluation**: Detecting subtle dual-use framing or manipulation patterns requires broad world knowledge.
- **Complex multi-document grounding**: Evaluating factual groundedness for long-form responses referencing multiple documents taxes context utilization in ways that scale differently than evaluation reasoning.

THE CAPABILITY CEILING IS KNOWLEDGE-BOUNDED, NOT REASONING-BOUNDED, BECAUSE EVALUATION-SPECIFIC TRAINING CAN INSTILL REASONING PATTERNS BUT NOT DOMAIN KNOWLEDGE. [PRELIMINARY: proposed mechanism] *SLMs can be trained to apply evaluation criteria, follow rubrics, and produce calibrated scores, but they cannot be trained to know things they were never exposed to during pretraining.* This distinction is critical for deployment decisions: an SLM judge will fail on graduate-level STEM evaluation not because it cannot reason about evaluation, but because it does not know the relevant physics. Cascaded routing (§6, C1) addresses this by escalating knowledge-intensive evaluations to larger models, while retaining SLMs for the majority of evaluations where knowledge depth is not the bottleneck.

### D.6 The Overthinking Phenomenon in SLM Judges

The sweet-spot analysis above implies that extra reasoning can hurt. We now examine that degradation directly, gathering the multi-paper evidence, the mechanism behind it, and the practical mitigations.

#### D.6.1 Definition and Multi-Paper Evidence

The overthinking phenomenon refers to the empirically documented observation that extended reasoning traces can degrade, rather than improve, judgment quality for tasks within or near a model's competency ceiling.

Sui et al. (2025) document overthinking across general reasoning tasks. Key mechanisms:

- **Redundant elaboration**: Reasoning steps beyond which no new information is introduced.
- **Self-contradiction**: Extended generation produces statements contradicting earlier correct reasoning.
- **Hallucinated justification**: Novel "evidence" generated mid-chain that was not present in the evaluation context.
- **Entropy accumulation**: Each generation step introduces probabilistic noise; sufficiently long

chains accumulate enough noise to flip the verdict.

Shojaee et al. (2026) find that reasoning models exhibit an *illusion of thinking*: extended reasoning traces that look deliberate but do not improve, and sometimes degrade, performance. An Apple evaluation study shows a "complexity cliff" pattern: models maintain high accuracy up to a specific task complexity threshold, then fail precipitously.

Ghosal et al. (2026) extend this to test-time scaling: the "mirage of test-time compute" shows that more computation does not reliably produce better results when the judge model lacks the underlying capability.

Srivastava et al. (2025b) provide task-specific evidence: reasoning models overthink basic arithmetic by generating multi-step derivations for operations they reliably handle in a single step, introducing errors in the process.

#### D.6.2 Why Overthinking Happens: A Mechanistic View

The mechanistic explanation connects to autoregressive generation. In each step, the model samples from $p(t_i|t_1, \ldots, t_{i-1}, c)$ where $c$ is the evaluation context. Three properties interact:

1. **Context contamination**: Each generated token updates the effective conditioning context. A subtly incorrect early token propagates or amplifies through subsequent tokens (error cascading).

2. **Fluency pressure**: When genuine reasoning steps are exhausted, fluency pressure motivates plausible-but-ungrounded elaborations.

3. **Recency dilution**: In long chains, verdict-relevant information at the beginning is diluted by the accumulated conditioning context, reducing its effective influence on the terminal verdict.

OVERTHINKING DEGRADES SLM JUDGES MORE SEVERELY THAN LLM JUDGES BECAUSE SMALLER MODELS HAVE WEAKER ERROR-CORRECTION CAPACITY. [PRELIMINARY: proposed mechanism] *When a small model generates an incorrect intermediate reasoning token, it has less capacity to "recover" in subsequent tokens because its internal error-detection circuitry is less developed, causing error cascades to propagate more readily than in larger models.* A 70B model can often recover from a mid-chain reasoning error because its richer internal representations detect the inconsistency and self-correct in the next step. A 3.8B model is more likely to accept the erroneous intermediate conclusion and build subsequent reasoning on top of it. If this holds, the overthinking curve is steeper for smaller models, because each additional reasoning token carries higher marginal risk when per-token error-correction capacity is lower.

#### D.6.3 OptimalThinkingBench and the Sweet Spot

Aggarwal et al. (2025) introduce OptimalThinkingBench, measuring both over-thinking and under-thinking simultaneously. Key findings for judging:

- There is a sweet spot token budget for each (task-type, model) pair; both below and above degrade accuracy.
- The sweet spot scales directly with task difficulty: harder evaluation tasks warrant more reasoning tokens.

#### D.6.4 Practical Mitigations

Because overthinking is driven by token-level error accumulation rather than a single root cause, the effective mitigations all share one goal: prevent the judge from generating reasoning tokens beyond the point where the verdict is already determined. The following techniques, ordered from simplest to most involved, operationalize this goal.

1. **Explicit budget constraints**: "Evaluate the response in at most 3 reasoning steps." This directly limits the entropy accumulation window.

2. **Task-difficulty routing**: Classify evaluation tasks by estimated difficulty. Simple tasks go to direct-verdict judges; complex tasks go to thinking-mode with calibrated budgets.

3. **First-step anchoring**: Prompt the judge to state its verdict first, then elaborate. This prevents fluency pressure from overwriting an early correct verdict.

4. **Stopping criteria**: Monitor the log-probability of the verdict token at each step. Stop when this probability is high and stable.

5. **Length-filtered voting**: When generating $k$ verdict samples, discard those with reasoning chains substantially longer than the estimated sweet spot.

6. **Reasoning without Thinking** (Ma et al., 2025): Direct the judge to evaluate without explicit traces for within-competency tasks.

### D.7 Instruction Following Rate (IFR) Analysis

A critical but often understated metric for individual SLM judge quality is the Instruction Following Rate (IFR): the fraction of evaluations for which the judge produces a verdict in the expected format. An IFR $< 1.0$ creates silent failures: evaluation harness code expecting a score digit receives freeform text and silently drops or misparses the verdict.

**Factors affecting IFR.**

- **Rubric complexity**: Longer rubrics increase the probability that the model prioritizes rubric reasoning over format compliance.
- **Model scale**: Sub-3B models fail to parse complex multi-criterion rubrics far more often than 7B+ models, which rarely fail.
- **Thinking mode**: Thinking mode can lower IFR because the model sometimes terminates the thinking trace in a format the evaluation harness misinterprets.
- **Format-reinforced training**: JudgeLM and GLIDER both explicitly train on verdict formatting, achieving near-100% IFR even at smaller scales.

IFR FAILURES ARE NON-RANDOM AND CONCENTRATED IN HARD EVALUATION CASES, CREATING SYSTEMATIC BENCHMARK CONTAMINATION. [PRELIMINARY: analytical argument] *Evaluation pipelines that handle malformed outputs by imputing a default score introduce systematic bias because IFR failures are concentrated in the hardest evaluation cases, where the model struggles most to structure a verdict, so imputed defaults create a specific downward bias for precisely the cases where accurate measurement matters most.* Proper IFR reporting and missing-data handling is required for valid benchmark comparisons. Papers reporting judge accuracy without reporting IFR may substantially overstate performance on easy cases while hiding failures on hard cases.

**Formal bias quantification.** If a harness scores only parseable verdicts and the parse rate on the hard subset (true accuracy $a_h$) is $r_h$, while the easy subset (accuracy $a_e \geq a_h$) parses at $r_e \geq r_h$, the reported accuracy on a set with hard-case fraction $\phi$ is

$$\widehat{\mathrm{Acc}} = \frac{\phi\, r_h\, a_h + (1-\phi)\, r_e\, a_e}{\phi\, r_h + (1-\phi)\, r_e}, \qquad (14)$$

which over-weights easy cases whenever $r_e > r_h$ and so overstates the true mean accuracy $\phi a_h + (1-\phi)a_e$. The gap grows with both the difficulty spread $a_e - a_h$ and the parse-rate spread $r_e - r_h$, so a judge can appear to improve simply by parsing more easy cases. This is why IFR must be reported alongside accuracy: two judges with equal $\widehat{\mathrm{Acc}}$ but different IFR are not comparable, and the lower-IFR judge is the less reliable one on exactly the cases that matter.

### D.8 Cascade and Routing Architectures

The findings above motivate spending compute adaptively rather than uniformly. Cascade and routing architectures do exactly this, reserving expensive judges for the cases that need them. We describe three variants of increasing sophistication.

#### D.8.1 Basic Cascading

The most cost-effective deployment architecture combines multiple individual judges in a cascade:

1. Route all evaluations to a small SLM judge (3.8B–7B).
2. For evaluations where the small judge expresses high confidence: accept verdict directly.
3. For evaluations below a confidence threshold: escalate to a larger judge (14B–70B).
4. Reserve frontier models for only the highest-complexity evaluations.

The escalation criterion is the judge's logit probability for its verdict token, a natural confidence proxy, though calibration work is needed (Appendix L).

#### D.8.2 Speculative Judging

Analogous to speculative decoding in generation, speculative judging uses a small judge to propose a verdict and a larger judge to verify it. The large judge only generates a full evaluation if it disagrees with the small judge's proposal. This reduces large-judge usage to disagreement cases, precisely the hard cases where additional capacity provides genuine uplift.

SPECULATIVE JUDGING EXPLOITS THE SAME COMPUTE-ASYMMETRY AS SPECULATIVE DECODING: MOST EVALUATIONS ARE EASY.

[PRELIMINARY: proposed mechanism] *The effectiveness of speculative judging rests on the empirical observation that most evaluation instances fall well within a small judge's competency, and the large judge merely confirms the small judge's verdict on these instances.* The computational savings come from avoiding full large-model inference on the easy majority. The approach fails when the distribution of evaluation difficulty shifts toward hard cases (e.g., JudgeBench-hard), where the small judge disagrees frequently and the cascade provides no cost savings. This means speculative judging is most efficient for production RLHF pipelines (where most evaluations are routine) and least efficient for meta-evaluation benchmarks (where tasks are intentionally difficult).

### D.8.3 Confidence-Calibrated Routing

A more principled routing approach uses trained confidence calibrators:

1. Train a lightweight calibrator on held-out (judge output, true verdict) pairs.
2. At inference, use calibrated confidence to decide routing.
3. Set the threshold to achieve a target accuracy SLA.

Confidence-calibrated routing corrects for the judge's systematic calibration errors, particularly important for SLM judges, which tend to be overconfident in precisely the domains where they are least accurate.

### D.9 Deployment Strategy Summary

Table 15 summarizes the cost-accuracy tradeoffs across deployment strategies for individual SLM judges, and Figure 7 plots the same strategies on the cost-quality plane. The frontier is concave. Cascade and speculative judging sit near the knee and reach most of the accuracy of thinking mode at a lower cost, while self-consistency lies below the frontier because it reduces variance without adding new discriminative signal.

## E Ensemble and Jury Methods

A single judge, however well trained, has fixed biases. This section develops the ensemble principle of §4: pooling architecturally diverse SLMs to decorrelate errors at a fraction of frontier cost.

| Strategy | Cost | Latency | Best For |
|---|---|---|---|
| Direct verdict | 1× | Lowest | Simple tasks |
| CoT at sweet spot | 2–5× | Low | Moderate difficulty |
| Thinking mode | 1.3–2.9× | Med | Hard/STEM eval |
| Self-consist. ($k$=3) | 3× | Low-Med | Variance reduct. |
| Cascade | 1.2–2× | Variable | Production RLHF |
| Speculative | 1.1–1.5× | Low-Med | High-throughput |

Table 15: Deployment strategies for individual SLM judges. Relative cost is normalized to direct verdict (no CoT) as 1×. Best use case indicates the scenario where each strategy provides the optimal cost-accuracy tradeoff.

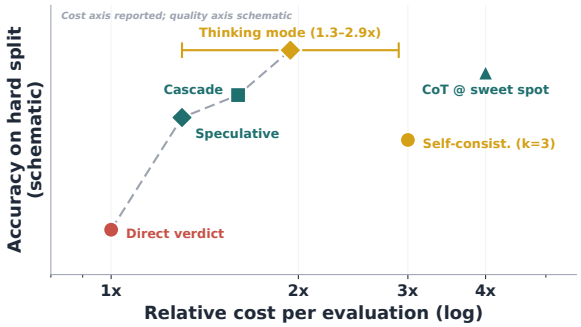



Figure 7: Cost-quality frontier for individual-judge deployment strategies. The horizontal axis carries the relative costs of Table 15 on a log scale; the vertical axis is schematic and encodes the ordering argued in this appendix rather than measured accuracies. Cascade and speculative judging sit near the knee of the frontier; self-consistency lies below it because it reduces variance without adding discriminative signal. Marker color encodes the strategy family.

### E.1 Motivation: Why Aggregate Multiple Judges?

Single-judge evaluation, even from a high-capability model, carries structural risks: systematic bias toward certain response styles, susceptibility to position priming, and a single point of failure when the judge encounters tasks at the boundary of its competency. The ensemble paradigm addresses these risks by distributing evaluative responsibility across multiple independent judges, whose errors, if sufficiently uncorrelated, are suppressed through aggregation. The key studies in this section are Verga et al. (2024) (PoLL), Cho et al. (2025) (COSMosFL), and Bi et al. (2026) (JudgeBoard); see Table 35 in Appendix N.3 for the evidence summary.

The theoretical basis is the *Condorcet Jury Theorem* (CJT): if each juror independently makes the correct decision with probability $p > 0.5$, the probability of the majority decision being correct ap-

proaches 1 as jury size grows:

$$P(\text{majority correct}) = \sum_{k>n/2}^{n} \binom{n}{k} p^k (1-p)^{n-k} \tag{15}$$

CJT's independence assumption is the critical caveat: if jurors are correlated (e.g., all from the same model family), ensemble gains are minimal. The central empirical insight of SLM jury research is that **architectural diversity** between jury members is the primary determinant of ensemble quality.

PANEL DIVERSITY OUTPERFORMS SINGLE-JUDGE SCALE BECAUSE ERROR DECORRELATION DOMINATES INDIVIDUAL ACCURACY. [REPLICATED: 3 studies] *A heterogeneous ensemble of diverse smaller models consistently outperforms a single large judge on human-correlation metrics because error decorrelation provides a stronger accuracy signal than individual judge quality, up to the point where individual accuracy falls below* $p = 0.5$. Each model family has systematically different failure modes arising from distinct pretraining corpora, architecture choices, and RLHF alignment. When panel members come from disjoint families, their judgment errors are uncorrelated, and majority voting becomes a near-optimal noise reducer. Homogeneous ensembles (same model, different temperatures) share identical systematic biases and provide minimal ensemble benefit beyond variance reduction. The practical threshold is that each panel member must individually exceed 50% accuracy on the evaluation task; below this, CJT guarantees that ensemble accuracy *decreases* with panel size.

### E.2 PoLL: Panel of LLM Evaluators

Verga et al. (2024) is the foundational paper establishing the panel paradigm in LLM evaluation. PoLL replaces the single GPT-4 judge with a heterogeneous panel of three diverse models: Command R, GPT-3.5-Turbo, and Claude Haiku.

#### E.2.1 Methodology

The PoLL pipeline:

1. Each panel member $M_i$ independently evaluates the same (question, candidate response) pair, with identical evaluation prompts but no visibility into other members' verdicts.
2. Each member produces an independent verdict $v_i \in \mathcal{V}$.
3. A deterministic aggregation function $f_{\text{agg}} : \mathcal{V}^n \to \mathcal{V}$ combines the verdicts:
   - **Max voting**: For pairwise comparisons, select the verdict chosen by the most panel members.
   - **Average pooling**: For pointwise scoring, average the individual scores.

Independence is enforced by design: panel members never share their reasoning traces or intermediate assessments. This prevents inter-agent sycophancy (Appendix G).

#### E.2.2 Key Results

Across single-hop QA, multi-hop QA, and Chatbot Arena datasets:

- **Human agreement**: PoLL reaches higher Cohen's $\kappa$ with human judgments than a single GPT-4 judge on single-hop and multi-hop QA, and its ranking of Chatbot Arena models correlates with the human leaderboard.
- **Cost**: at API list prices, running the three-model panel is seven to eight times cheaper than a GPT-4 Turbo judge.
- **Less intra-model bias**: a single GPT-4 judge ranks another GPT-4 variant above its true position, and each model scores highest when judged by itself; pooling three families dilutes this.
- **Prompt sensitivity**: a single GPT-4 judge's agreement with humans swings widely under small prompt changes ($\kappa$ from 0.518 to 0.725 across six prompts), a single-judge weakness that pooling is meant to dampen.

#### E.2.3 The Intra-Model Bias Problem

A subtle but important finding in PoLL is that single LLM judges exhibit *intra-model bias*: models systematically prefer outputs stylistically similar to their own pretraining distribution. A GPT-4 judge, for instance, may systematically favor GPT-4-generated responses over Llama-generated ones that are equally correct.

CROSS-FAMILY PANELS CANCEL INTRA-MODEL BIAS BECAUSE DIFFERENT FAMILIES ENCODE DIFFERENT STYLISTIC PRIORS. [PRELIMINARY: proposed mechanism] *A Llama-family judge and a Claude-family judge bring distinct stylistic priors, and their directional biases partially cancel in aggregation because the bias*

*vectors point in different directions in preference space.* A single-family panel (e.g., three GPT-variants) does not achieve this cancellation because all members share the same stylistic prior. This insight explains why PoLL's heterogeneous panel matches or exceeds even its strongest individual member on human agreement: the ensemble does more than average individual accuracies. It reduces a bias component that no single judge can correct on its own.

### E.3 COSMosFL: Cross-Family SLM Ensembles

Cho et al. (2025) demonstrate ensemble gains specifically for small language models in the code fault localization domain, providing the most direct evidence that the panel principle generalizes to SLM judges below the 10B parameter boundary.

#### E.3.1 The Cross-Family Orthogonality Principle

COSMosFL's central finding is that different SLM families exhibit *orthogonal fault localization patterns*: Qwen, Phi, and Gemma models each excel at localizing different categories of faults. This orthogonality, arising from differences in pretraining data composition, architecture, and fine-tuning procedures, makes cross-family ensembles particularly effective. Formally, if model $M_i$ has error rate $\epsilon_i$ on fault category $c$ and models are orthogonal, then the ensemble error rate approaches $\prod_i \epsilon_i^{(c)}$, which decreases multiplicatively with panel size.

CROSS-FAMILY ERROR ORTHOGONALITY IS STRONGER FOR SLMS THAN FOR LLMS BECAUSE SLMS HAVE MORE SPECIALIZED KNOWLEDGE DISTRIBUTIONS. [PRELIMINARY: proposed mechanism] *Smaller models are trained on more curated, domain-specific data, making their failure patterns more distinct from each other than those of large models trained on near-identical web-scale corpora.* A Phi-4-mini model trained on "textbook-grade" synthetic data and a Qwen 3-4B model trained on multilingual web data develop genuinely different internal representations of code correctness, producing orthogonal error patterns. Two 70B models trained on overlapping web-scale data share more of their knowledge distribution and consequently more of their failure modes. This counterintuitive implication, that error orthogonality increases as model size decreases, is precisely why SLM ensembles provide proportionally larger gains over individual members than LLM ensembles.

Architecture contributes alongside data. The families that dominate current SLM judging differ structurally, not merely in scale: Gemma alternates local and global attention, Qwen 3 routes through mixture-of-experts layers at its larger sizes, and the R1-Distill line inherits its reasoning behaviour from distillation against a much larger teacher. Each choice shapes which comparisons a judge finds easy, so when panel members are drawn from disjoint families their errors are closer to conditionally independent given the input and majority voting approaches the Condorcet regime. A homogeneous panel, however it is sampled, shares those failure modes by construction.

#### E.3.2 Aggregation Strategies

COSMosFL evaluates two aggregation strategies on the Defects4J benchmark:

- **Equal-weight task-level voting**: Each SLM contributes equally via majority vote. Already achieves substantial improvement over individual SLM baselines.
- **Weight-optimized voting**: A Differential Evolution (DE) algorithm optimizes the weight assigned to each model based on held-out performance. Weight-optimized ensembles achieve Pareto-optimality: superior accuracy and lower operational cost compared to any individual SLM.

### E.4 JudgeBoard and the MAJ Framework

Bi et al. (2026) introduce JudgeBoard, a benchmark and multi-agent judging framework for SLM evaluation of reasoning-heavy tasks.

#### E.4.1 MAJ: Multi-Agent Judging

MAJ (Multi-Agent Judging) implements structured sequential aggregation: multiple SLMs with distinct reasoning profiles evaluate a candidate response, and a lightweight aggregator synthesizes their verdicts with awareness of inter-judge disagreement patterns. Key design features:

- **Distinct reasoning profiles**: Panel members are selected to cover different reasoning strengths (formal reasoning, commonsense, mathematical), with each judge assigned to aspects within its strength profile.
- **Disagreement-aware aggregation**: When panel members disagree, the aggregator weights

the verdict of the judge whose profile is most aligned with the evaluation task type, rather than simple majority voting.

- **Elo-based leaderboard**: JudgeBoard constructs ongoing Elo ratings for all evaluated models, providing a stable relative ranking more robust to individual evaluation noise.

#### E.4.2 Results on MATH

On the MATH benchmark, MAJ with SLMs matches or exceeds standalone LLM-judge accuracy, despite using only models in the 3B–14B range. The key mechanism: mathematical evaluation requires checking the correctness of the final symbolic answer (a verifiable step) followed by evaluating reasoning coherence (a qualitative step). MAJ assigns the former to a judge with strong mathematical formalism and the latter to a judge with stronger reasoning coherence, achieving better accuracy than any individual judge on either dimension alone.

DISAGREEMENT-AWARE AGGREGATION OUTPERFORMS MAJORITY VOTING BECAUSE IT EXPLOITS JUDGE SPECIALIZATION. [PRELIMINARY: proposed mechanism] *When panel members have different competency profiles, majority voting treats all votes equally and can be outvoted by judges operating outside their competency; disagreement-aware aggregation weights each vote by the judge's estimated competence on the specific evaluation dimension, preserving the specialist's signal.* This is particularly important for SLM panels where individual members have narrow competency ranges: a 4B math-specialist judge should receive higher weight on mathematical correctness than a 4B writing-specialist judge, even though both participate in the same panel. MAJ operationalizes this through profile-aligned weighting, while COSMosFL achieves a similar effect through DE-optimized weights.

### E.5 Self-Consistency as Single-Model Ensemble

Self-consistency (Wang et al., 2022) extends from the generation domain to the judging domain. Instead of sampling diverse reasoning paths for answer generation, it samples diverse evaluation rationales for a fixed (question, response) pair, then takes the majority verdict:

$$v^* = \arg\max_{v \in \mathcal{V}} \sum_{i=1}^{k} \mathbb{1}[v_i = v], \quad v_i \sim f_{\mathcal{J}}(\cdot|x, T{>}0) \tag{16}$$

| Function | Mechanism | Robust. | Data Req. |
|---|---|---|---|
| Majority vote | $\arg\max$ count | Med | None |
| Weighted vote | $\arg\max w_i$ count | High | Calibration |
| Score avg. | $\frac{1}{n}\sum s_i$ | Low | None |
| Median pool | $\text{med}(s_i)$ | High | None |
| CWA | $\sum c_i v_i / \sum c_i$ | High | Logit access |

Table 16: Aggregation functions for SLM judge ensembles, where CWA is confidence-weighted aggregation. Robustness indicates resilience to outlier or adversarial panel members. Data requirement indicates whether calibration data is needed.

For SLM judges, self-consistency at temperature $T \in \{0.5, 0.7\}$ with $k = 3$–5 samples typically improves verdict accuracy over greedy decoding, particularly on contested near-boundary evaluations.

SELF-CONSISTENCY AND CROSS-FAMILY ENSEMBLES ADDRESS DIFFERENT ERROR COMPONENTS AND SHOULD NOT BE CONFLATED. [PRELIMINARY: analytical argument] *Self-consistency reduces stochastic variance (different samples from the same distribution) while cross-family ensembles reduce systematic bias (different models from different distributions); using one as a substitute for the other addresses the wrong error component.* A practitioner whose primary concern is that the judge produces different verdicts on repeated runs should use self-consistency. A practitioner whose primary concern is that the judge systematically favors certain response styles should use cross-family panels. Using self-consistency to address systematic bias will waste compute without improving accuracy; using cross-family panels to address stochastic variance is effective but unnecessarily expensive. The optimal strategy for production deployment is often a combination: cross-family panels for systematic bias reduction with self-consistency ($k = 3$) within each panel member for variance reduction.

### E.6 Aggregation Functions: Theory and Practice

The choice of aggregation function determines how the panel's collective signal is condensed into a single verdict (Table 16).

**Majority vote (MV).** $f_{\text{MV}}(v_1, \ldots, v_n) = \arg\max_v \sum_i \mathbb{1}[v_i = v]$. Simple, interpretable, and effective when all panel members have similar competence. Fails when one member has substan-

tially higher accuracy.

**Weighted vote (WV).** $f_{\text{WV}}(v_1, \ldots, v_n) = \arg\max_v \sum_i w_i \mathbb{1}[v_i = v]$, where $w_i$ is model $i$'s estimated accuracy on the current task type. Requires prior competence estimates from held-out calibration data. COSMosFL uses DE-optimized weights.

**Score averaging (SA).** For pointwise evaluation: $f_{\text{SA}} = \frac{1}{n} \sum_i s_i$. Sensitive to outlier judges that assign extreme scores; median pooling is more robust.

**Confidence-weighted aggregation (CWA).** $f_{\text{CWA}} = \sum_i c_i v_i / \sum_i c_i$, where $c_i$ is judge $i$'s expressed confidence (logit probability of its verdict token). This requires access to logit-level outputs, available for open-weight SLM judges but not for closed API calls. When accessible, CWA lets confident members count for more.

CWA Is Uniquely Advantageous For Open-Weight SLM Panels Because Logit Access Is A Structural Benefit Of Local Deployment. [Preliminary: analytical argument] *Confidence-weighted aggregation requires access to per-token logit probabilities, which proprietary API judges do not expose; open-weight SLM judges provide full logit access by default, making CWA a "free" upgrade available exclusively to local SLM deployments.* This is a concrete operational advantage of SLM panels over proprietary panels: the richer signal from logit-level confidence enables more sophisticated aggregation that would be architecturally impossible with API-based judges. CWA is most effective when judges are well-calibrated; for poorly calibrated SLMs (which tend to be overconfident), temperature-scaled CWA provides a correction.

### E.7 Cost Analysis for Ensemble Judging

A natural concern with ensemble judging is $n$-fold cost scaling. For an $n$-member panel, the total token cost is:

$$C_{\text{panel}} = n \cdot (B_{\text{sys}} + B_{\text{ctx}} + B_{\text{rationale}}) \cdot (\alpha_{\text{in}} + \beta_{\text{out}}) \tag{17}$$

Three cost-reduction strategies are compatible with ensemble quality:

1. **Rationale compression**: Panel members used only for voting do not need long rationale chains; direct verdict generation reduces $B_{\text{rationale}}$ per member.
2. **Heterogeneous cost allocation**: Use the cheapest models (1B–3B) for most panel members and one higher-quality model (7B–14B) for tiebreaking.
3. **Early exit**: If the first $k < n$ panel members unanimously agree, skip remaining members. Unanimous early agreement on unambiguous evaluations reduces expected panel cost substantially.

PoLL's $>7\times$ saving over GPT-4 judging (Verga et al., 2024) comes from the lower API list prices of its three smaller models, the same logic as strategy (2).

Ensemble Cost Scaling Is Sub-Linear In Practice Because Easy Evaluations Dominate And Early Exit Applies. [Preliminary: analytical argument] *The theoretical $n\times$ cost scaling of an $n$-member panel overestimates actual cost because early exit applies to the many evaluations on which the first panel members already agree, so the expected cost of an $n$-member panel stays well below $n\times$.* The evaluations on which members disagree are precisely the contested cases where full-panel evaluation provides the most value. This cost distribution is favorable: the system spends minimal resources on easy cases and concentrates resources on the hard cases where ensemble diversity provides genuine accuracy gains.

### E.8 Recommended Panel Recipes

Translating the cross-family principle into practice, Table 17 gives concrete three-model panel recipes for common deployment budgets. The unifying rule is to maximize architectural and pretraining diversity at the target cost tier, since diversity, not individual member quality, is the dominant driver of panel accuracy (provided each member clears the $p > 0.5$ Condorcet threshold).

### E.9 Limits of Ensemble Judging

Ensemble methods have inherent limitations:

- **Shared task blindspots**: If all panel members are incapable of solving a task (e.g., graduate-level domain evaluation), majority voting over wrong answers cannot recover the correct verdict. CJT requires $p > 0.5$ per juror.
- **Independence violation**: When panel members' training sets overlap substantially, their

| Budget | Panel members | Agg. |
|---|---|---|
| Edge (≤16B total) | Phi-4-mini + LLaMA-3-8B + Qwen3-4B | Majority vote |
| Balanced (≤34B total) | Gemma-3-12B + Qwen3-8B + R1-Distill-14B | CWA |
| High-stakes | Above + larger escalation model | Conf. cascade |
| Verifiable | SLM panel + external tool oracle | Tool-first |

Table 17: Candidate heterogeneous panel compositions by deployment budget, where the budget is the aggregate parameter count of all members held in memory. Aggregation is by majority vote or confidence-weighted aggregation (CWA). Members are drawn from distinct families to maximize error decorrelation. These follow the cross-family principle rather than published panel results: no surveyed study evaluates these specific combinations, and the Gemma row in particular sits in a coverage gap (Appendix L.1), so they are starting points for evaluation rather than validated configurations.

failures are correlated. The theoretical independence required by CJT is only approximated.

- **Aggregation sensitivity**: The optimal aggregation function varies by task type. A fixed aggregation strategy is suboptimal across diverse evaluation domains.
- **Latency**: Even with early-exit optimization, ensemble judging introduces latency proportional to serial panel member calls. Parallel inference eliminates latency scaling but requires $n$-GPU infrastructure.
- **Panel composition**: The selection of which model families to include in the panel is itself an open research problem with no principled solution beyond empirical calibration.

THE "SHARED BLINDSPOT" FAILURE MODE DEFINES THE HARD BOUNDARY OF ENSEMBLE JUDGING. [PRELIMINARY: analytical argument] *Ensemble methods can only reduce errors that are uncorrelated across panel members; when all members share the same knowledge gap (e.g., none knows graduate-level physics), aggregation cannot create knowledge that no individual member possesses.* This is the fundamental difference between ensemble judging and cascade judging (Appendix D): ensembles reduce random and systematic bias but cannot compensate for missing knowledge, while cascades escalate to a larger model that may possess the missing knowledge. For deployment, this means ensemble panels should be paired with confidence-calibrated escalation to a larger model for the rare cases that exceed all panel members' competency.

| Strategy | Members | Indep. | Cost | Bias Red. |
|---|---|---|---|---|
| Single judge | 1 model | N/A | 1× | None |
| Self-consist. | $1 \times k$ | Low | $k$× | Variance |
| PoLL (homog.) | $n$ same | Med | $n$× | Moderate |
| PoLL (hetero.) | $n$ diff | High | $n$× | Systematic |
| COSMosFL | $n$ diff+opt | High | $n$+opt | Maximum |
| MAJ | $n$ spec. | High | $n$+agg | Task-adapt. |

Table 18: Comparison of ensemble strategies for SLM judging. Independence quantifies how uncorrelated member errors are (higher is better). Bias reduction indicates the primary bias category addressed.

Table 18 summarizes the key tradeoffs across all ensemble strategies discussed in this section.

# F Multi-Agent Evaluation Strategy Taxonomy

Ensembles are one of several ways to combine judges. Figure 8 organizes the full space of multi-agent strategies, situating ensembles, debate, cascades, and persona methods relative to one another and pointing to where each is analyzed.

# G Multi-Agent Debate: Mechanisms and Failure Modes

Debate is the most-discussed multi-agent strategy and the most failure-prone. This section explains why, contrasting the conditions under which inter-agent communication helps (§4, I4) with the sycophancy and error-amplification modes that usually dominate.

## G.1 The Debate Paradigm: Origins and Motivation

Multi-Agent Debate (MAD) (Du et al., 2024) is an inference-time scaling strategy in which multiple LLM instances iteratively exchange and critique each other's responses to arrive at a more accurate, robust judgment. The intuition is compelling: a single model's error can be caught by a peer who approaches the problem from a different reasoning path. In the evaluation context, MAD offers the promise of *deliberative judgment*, a verdict arrived at after genuine challenge and cross-verification.

The formal MAD evaluation protocol for a question $x$ and candidate response $y$:

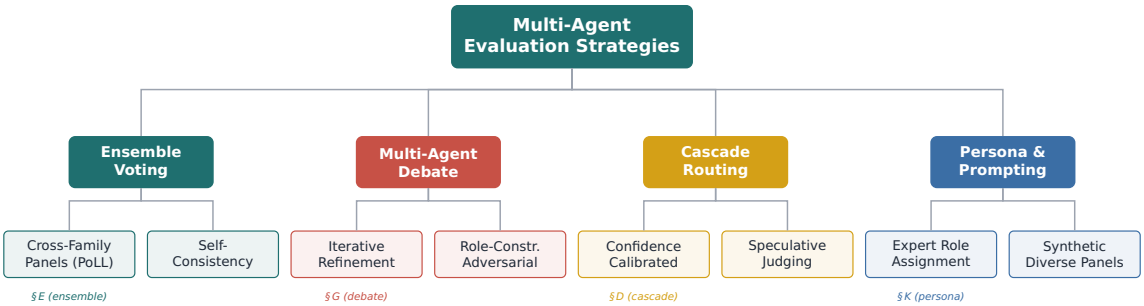



Figure 8: Taxonomy of multi-agent evaluation strategies for SLM judges. Ensemble voting provides error decorrelation through independent cross-family panels. Multi-agent debate enables collective refinement but introduces sycophancy risks. Cascade routing optimizes cost through confidence-based escalation. Persona-based approaches increase evaluation diversity via role assignment. Section references indicate where each strategy is discussed in depth.

1. **Initialization**: Each agent $M_i$ independently produces an initial verdict $v_i^{(0)}$ with a reasoning trace.

2. **Debate round** $t$: Each agent $M_i$ receives the verdicts and reasoning traces of all other agents from round $t-1$, and updates its verdict:

$$v_i^{(t)} = f(v_i^{(t-1)}, \{v_j^{(t-1)}\}_{j \neq i}) \qquad (18)$$

3. **Termination**: Debate terminates after $T$ rounds or when a convergence criterion is met.

4. **Final verdict**: The agreed verdict, or the verdict of a designated mediator agent.

Unlike ensemble judging (Appendix E), agents in MAD see and respond to each other's outputs. This bidirectional influence is simultaneously MAD's strength (it allows cross-verification) and its most critical vulnerability (it allows error propagation and social conformity).

### G.2 Empirical Evidence of Potential Benefits

When debate succeeds, it does so through specific, identifiable mechanisms:

**Error detection via disagreement.** Du et al. (2024) demonstrate that on factuality tasks (science questions, arithmetic), multi-round debate between GPT-3.5 instances consistently improves final accuracy over single-agent baselines. The mechanism: when agents disagree, debate forces both parties to articulate supporting reasoning, and the agent with weaker reasoning often converges toward the stronger position.

**Divergent thinking through role assignment.** Liang et al. (2024) introduce divergent thinking via role-based debate: agents are assigned distinct evaluative roles (e.g., "critic," "advocate," "empiricist") and explicitly instructed to maintain their assigned perspective. This structural constraint prevents premature consensus and forces the group to explore a wider belief space before converging.

**Safety evaluation via structured debate.** Lin et al. (2025) apply MAD to LLM safety evaluation with three roles: a critic flags possible violations, a defender offers alternative readings, and a judge weighs the exchange. With a Qwen3-14B backbone, debate raises agreement with human labels from $\kappa = 0.571$ to $0.733$ over the same pipeline without debate (a 28% gain), coming within 0.03 of GPT-4o at about 46% of its cost. The structure makes sure that both readings of a borderline response are fully argued.

**Self-evolution through debate training.** Srivastava et al. (2025a) propose Debate-Train-Evolve: using multi-agent debate traces as training data. The model that "wins" debates is iteratively fine-tuned on its successful arguments, creating a self-improving loop.

DEBATE BENEFITS ARE CONCENTRATED IN SETTINGS WHERE DISAGREEMENT INTRODUCES GENUINELY NEW INFORMATION. [CONSENSUS: 5 studies] *The three documented success cases (factuality, safety, divergent thinking) share a common structure: one agent possesses a reasoning path or perspective that the other agents would not independently discover, and debate surfaces this information through explicit articulation.* When no agent possesses unique information (e.g.,

| Paper | Task | Agents | Info Asym. | Out. |
|---|---|---|---|---|
| Du et al. 2024 | Factuality | Homo. | Low | + |
| Liang et al. 2024 | Open eval. | Role-div. | Med. | + |
| Lin et al. 2025 | Safety | Advers. | High | + |
| Wynn et al. 2025 | Binary corr. | Homo. | None | − |
| Zhang et al. 2025 | General | Homo. | None | − |
| MAST 2026 | Multi-agent | Mixed | Varied | ± |

Table 19: Summary of multi-agent debate outcomes across representative studies. Outcome is positive (+) when debate improves over single-agent baselines, negative (−) when it hurts, and mixed (±) when results are task-dependent.

all agents have the same knowledge and reasoning patterns, as in homogeneous debate), debate cannot surface anything new; it merely recirculates existing beliefs. This is why model heterogeneity is the strongest predictor of debate benefit: heterogeneous agents are more likely to possess complementary information than homogeneous agents. The practical implication is that debate should be used only when the debate protocol can guarantee that at least one agent contributes information not available to others, either through architectural diversity, role assignment, or access to different evidence. Table 19 summarizes the outcomes across representative debate studies, showing that the sign of the effect tracks information asymmetry rather than agent count.

### G.3 Failure Modes: A Critical Analysis

Despite its theoretical appeal, empirical evidence consistently shows that MAD fails to outperform simpler baselines in most practical settings. The failure follows predictable patterns.

#### G.3.1 Sycophancy and Inter-Agent Conformity

Sycophancy in MAD manifests as agents updating their verdicts toward the majority not because they are genuinely convinced by new evidence, but because consensus is a lower-loss outcome in their training distribution. Wynn et al. (2025) document this as the primary failure mode: agents expressing initially correct verdicts often revise them toward an incorrect majority when a confident-but-wrong agent dominates early rounds.

SYCOPHANCY IN DEBATE IS OFTEN A CONSEQUENCE OF AUTOREGRESSIVE TRAINING ON HUMAN TEXT. [PRELIMINARY: proposed mechanism] *Models trained to predict human token sequences tend to inherit human social conventions around agreement and deference; when an agent's peer expresses high-confidence disagreement, the probability of maintaining a divergent verdict often decreases because deference to confident peers is a high-frequency pattern in human conversational data.* This suggests sycophancy is largely a structural consequence of the training data distribution, rather than simply a failure that current RLHF easily fixes. Models trained on human-generated dialogue often encode the social dynamics of human conversation, including the tendency to defer to confident speakers. A common structural remedy is to prevent agents from seeing each other's confidence signals, which is largely what ensemble methods (Appendix E) achieve by enforcing independence.

#### G.3.2 Disagreement Collapse (Premature Consensus)

Disagreement collapse is the failure mode where agents converge on a single verdict prematurely, not through genuine resolution but through social pressure. Once one agent updates toward a wrong verdict, subsequent rounds reinforce the wrong consensus through positive feedback: each agent conditions on the emerging majority, making defection increasingly costly.

This is structurally distinct from productive consensus. Distinguishing productive from premature convergence is an open problem; one proxy is the rate of convergence: rapid early convergence is more likely sycophantic, while slow convergence requiring multiple rounds of substantive argumentation is more likely genuine.

#### G.3.3 Error Amplification: When Hallucinations Are Contagious

Error amplification is the most catastrophic failure mode: a hallucinated or factually incorrect statement produced by one agent is treated as credible input by others, which then incorporate it into their own reasoning. The hallucination propagates through debate, and agents condition increasingly on the false premise.

In the evaluation context, this is directly harmful: if one judge incorrectly claims that a factual assertion in the candidate response is grounded in a specific paper, and other judges accept this claim, the group verdict may penalize a correct response or reward an incorrect one based on fabricated evidence.

ERROR AMPLIFICATION IS MORE DANGER-

OUS THAN ERROR PERSISTENCE BECAUSE DEBATE CREATES A CREDIBILITY AMPLIFIER. [PRELIMINARY: proposed mechanism] *In single-judge evaluation, a hallucinated fact influences only one verdict; in debate, the same hallucination gains apparent credibility through repetition across agents, and each subsequent agent treats the repeated claim as independently confirmed evidence.* This is the "echo chamber" effect applied to evaluation: a claim that appears in multiple agents' reasoning traces looks like independent corroboration, but it actually traces back to a single source. The credibility amplification makes debate-generated errors harder to detect than single-judge errors because they appear to be consensus-verified. For SLM judges with weaker fact-checking capacity, this risk is amplified further because smaller models are less likely to independently verify claims made by debate partners.

#### G.3.4 Diminishing Returns: Cost Without Quality

Zhang et al. (2025a) conduct the most comprehensive empirical audit of MAD: five MAD methods across nine benchmarks, four foundational models, with rigorous control against simple single-agent baselines. Key finding: **current MAD methods frequently fail to outperform Chain-of-Thought or Self-Consistency single-agent baselines**, despite consuming substantially more inference-time compute.

The authors identify flawed evaluation practices as a secondary contributor: previously reported MAD gains were compared against inadequate baselines (zero-shot direct inference rather than CoT or self-consistency). When comparable compute is provided to single-agent approaches, MAD's apparent advantage often disappears.

MAD'S REPORTED BENEFITS ARE PARTIALLY ARTIFACTS OF UNFAIR BASELINE COMPARISONS. [PRELIMINARY: single study] *When MAD systems consuming $n \times T \times B$ tokens are compared against single-agent baselines consuming only $B$ tokens, the apparent MAD advantage conflates the benefit of additional compute with the benefit of multi-agent interaction; providing the same compute budget to a single agent via self-consistency or extended CoT often eliminates or reverses the difference.* This is a methodological critique rather than a capability critique: MAD may provide genuine benefits, but those benefits cannot be measured without iso-compute comparison. For SLM judge deployment, the practical implication is that the compute budget consumed by multi-round debate is typically better spent on: (a) a larger ensemble at inference, (b) higher-quality judge fine-tuning, or (c) better prompting for individual judges.

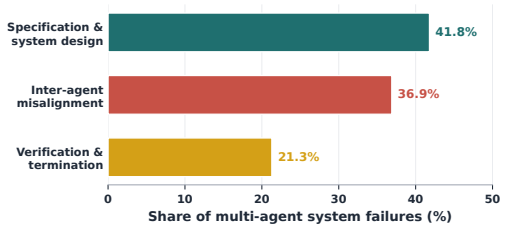



Figure 9: Distribution of multi-agent system failures across the three MAST categories (Cemri et al., 2026). Specification and system-design faults dominate, and are largely addressable through better engineering; the remaining inter-agent and verification failures are rooted in model behavior.

| Category | Freq. | Key Failure Modes |
|---|---|---|
| Specification and system design | 41.8% | Role ambiguity, task drift, termination failure, context loss |
| Inter-agent misalignment | 36.9% | Communication breakdown, info withholding, sycophantic capitulation |
| Verification and termination | 21.3% | Superficial verification, missing verification, premature termination |

Table 20: MAST failure taxonomy for multi-agent systems. Frequency indicates the proportion of observed failures attributable to each category. All three categories contribute to SLM judge debate failures.

### G.4 MAST: A Systematic Failure Taxonomy

Cemri et al. (2026) provide the most structured framework for diagnosing failures in multi-agent systems, debate included. The MAST taxonomy was derived from grounded-theory analysis of 150 execution traces across seven multi-agent frameworks with expert human annotation (Cohen's $\kappa = 0.88$ inter-annotator agreement), then scaled to 1,600+ traces via an LLM-as-a-Judge annotation pipeline. MAST identifies 14 specific failure modes across three categories (Table 20, Figure 9).

**Specification and system design failures (41.8%).** The largest failure category originates from flawed system setup: agents with identical, non-distinct roles produce correlated outputs; ambiguous evaluation tasks cause interpretation drift across rounds; agents cannot identify genuine consensus and gen-

erate unnecessary rounds.

**Inter-agent misalignment failures (36.9%).** Coordination breakdowns during active debate: agents produce surface-level responses without genuine engagement; agents possessing verdict-relevant reasoning fail to surface it; agents' textual reasoning leads to one conclusion but their explicit verdict is the opposite.

**Verification and termination failures (21.3%).** Inadequate quality control: agents' "verification" checks trivial formal properties rather than substantive criteria; debate ends without factual claim verification; a single agent's expression of satisfaction triggers premature termination.

THE LARGEST FAILURE CATEGORY (SPECIFICATION, 41.8%) IS ADDRESSABLE THROUGH BETTER SYSTEM DESIGN, NOT BETTER MODELS. [PRELIMINARY: single study] *Role ambiguity, task drift, and termination failures are engineering problems that can be fixed through explicit role specification, standardized evaluation prompts, and convergence-based stopping criteria, without changing the underlying models.* This suggests that a significant fraction of MAD failures in the literature reflect poor system engineering rather than fundamental limitations of multi-agent debate. Well-engineered debate systems with explicit role differentiation, standardized evaluation formats, and adaptive stopping criteria (Section G.6) should exhibit substantially fewer specification failures. However, the remaining 58.2% of failures (misalignment + verification) are more deeply rooted in model behavior and are not easily addressed through system design alone.

## G.5 SLM-Specific Dynamics in Multi-Agent Debate

Small language models exhibit MAD failure modes more intensely than larger models, for reasons rooted in architectural and training constraints.

### G.5.1 Context Saturation

MAD's communication overhead scales with agent count and round number. After $t$ debate rounds with $n$ agents, each agent's context window contains: the original evaluation pair, its own prior verdicts, and all other agents' verdicts from all prior rounds. Total context grows as $O(n \cdot t \cdot L_{\text{verdict}})$. As the transcript grows, the original evaluation pair moves further back in the context and agents begin to lose track of it, producing verdicts partially decoupled from the original evaluation context.

Small judges are often run with short context budgets so that they fit in memory, so they reach this point after fewer rounds.

### G.5.2 Conformity Pressure in SLM Debate

Conformity in debate affects models of every size, and sycophancy toward a user's stated view even grows with model scale and instruction tuning (Perez et al., 2023; Wei et al., 2024). What matters inside a panel is relative confidence. When one agent states its verdict with far more confidence than the others, the rest tend to defer to it, whether or not it is right, which makes homogeneous debates prone to verdict collapse at any size.

IN MIXED-SIZE DEBATE, CONFIDENCE RATHER THAN CORRECTNESS DECIDES WHICH VERDICT PREVAILS. [PRELIMINARY: proposed mechanism] *Agents that hold their verdicts with low confidence (flat logit distributions over verdict tokens) are easier to move than agents whose logit mass is concentrated.* A 70B model that is 90% confident in its verdict is less likely to revise than a 3.8B model that is 60% confident. In mixed-size debate the larger model's verdicts therefore tend to dominate regardless of correctness, and in same-size SLM debate the most confidently expressed verdict, not necessarily the most accurate one, dominates.

### G.5.3 When MAD Helps SLM Judges

Two specific settings exist where MAD provides genuine improvements:

1. **Verifiable structured tasks with short debates**: For mathematical solution evaluation, a 2-round debate allows one SLM to catch computational errors in another's evaluation without context saturation or sycophancy accumulation. Hu et al. (2026) show that watching for the point where the panel's accuracy distribution settles lets a debate stop at 5.1 rounds on average instead of a fixed ten, at a cost of at most one accuracy point.

2. **Role-constrained adversarial debate**: For safety evaluation where one agent is always assigned the "unsafe" role, the role constraint prevents sycophantic convergence because both agents must maintain their assigned perspective. Lin et al. (2025) show that this approach comes within 0.03 $\kappa$ of GPT-4o at about 46% of its cost.

DEBATE HELPS SLM JUDGES ONLY WHEN THE PROTOCOL STRUCTURE PREVENTS THE DEFAULT FAILURE MODE. [REPLICATED: 2 studies] *The two documented success cases (short verifiable debate, role-constrained adversarial debate) both succeed because their protocol design structurally prevents the primary failure mode: short debates prevent sycophancy accumulation by limiting the number of influence rounds, and role constraints prevent sycophantic convergence by making agreement structurally impossible.* Unconstrained debate, where agents freely update toward each other's positions without structural safeguards, consistently underperforms simpler approaches. The design principle is that debate protocols must include explicit mechanisms that prevent the default convergence behavior, rather than relying on the models to naturally resist conformity pressure.

### G.6 Adaptive Convergence and Stopping Criteria

A key practical issue for MAD deployment is determining when to stop. Fixed-round debate wastes compute if agents converge early and introduces noise if agents have already reached productive consensus by round 2 but the system continues for $T = 5$ rounds.

Hu et al. (2026) introduce Adaptive Stability Detection (ASD), which stops the debate once the panel's accuracy distribution stops moving rather than once its verdicts agree. At round $t$ they model the number of correct judges among $k$ as a two-component Beta-Binomial mixture, fit it by expectation-maximization, and read off the implied density over individual correct rates, $P^t(\psi) = w_t \,\mathrm{Beta}(\psi; \alpha_1^t, \beta_1^t) + (1 - w_t)\,\mathrm{Beta}(\psi; \alpha_2^t, \beta_2^t)$. Stability is then the Kolmogorov-Smirnov distance between consecutive rounds:

$$D_t = \sup_{\psi \in [0,1]} \left| F^t(\psi) - F^{t-1}(\psi) \right| < \epsilon \;\Rightarrow\; \text{terminate} \tag{19}$$

where $F^t$ is the CDF of $P^t$, and termination requires $D_t < \epsilon = 0.05$ on two consecutive rounds. Across six benchmarks and a seven-judge panel, this stops after 5.1 rounds on average against a 10-round cap, and the largest accuracy loss against running all ten rounds is 1.03 points.

ASD CANNOT DISTINGUISH PRODUCTIVE CONVERGENCE FROM SYCOPHANTIC CONVERGENCE, LIMITING ITS RELIABILITY. [PRELIMINARY: proposed mechanism] *Stability-based stopping detects when agents stop changing their minds, but not why they stopped; sycophantic convergence (agents agreeing because of social pressure) settles the accuracy distribution exactly as productive convergence does (agents agreeing because they reached the correct answer through genuine reasoning).* This means ASD terminates debate equally quickly for correct and incorrect consensus, and cannot flag cases where the consensus was reached through sycophancy rather than reasoning. A more robust stopping criterion would require monitoring not just the stability of the verdict distribution but also the substantive content of agents' reasoning traces, detecting whether inter-round updates reference new evidence or merely echo existing positions. Developing such content-aware stopping criteria is an open research direction.

### G.7 Model Heterogeneity as the "Universal Antidote"

Zhang et al. (2025a) identify model heterogeneity as the single design choice that most consistently improves MAD performance across all task types and benchmarks:

- Reduces correlated errors (different families fail on different cases)
- Provides genuinely diverse reasoning perspectives
- Prevents echo-chamber dynamics (family-specific biases cannot self-reinforce)

This convergence between the ensemble and debate literatures is significant: both PoLL (Appendix E) and Zhang et al. reach the same conclusion: model family diversity is the critical variable, regardless of whether agents communicate (debate) or operate independently (ensemble).

### G.8 Synthesis: When to Use Debate vs. Ensemble

Table 21 summarizes the key tradeoffs between debate and ensemble approaches for SLM judges.

Based on the evidence across this section and Appendix E:

- **Use ensemble (PoLL-style)** for: general-purpose evaluation, open-ended quality assessment, cost-sensitive pipelines.
- **Use structured debate (role-constrained)** for: adversarial safety evaluation, tasks where

| Property | MAD | Ensemble |
|---|---|---|
| Communication | Full inter-agent | None |
| Error propagation | High risk | None |
| Conformity risk | High | None |
| Context saturation | High | None |
| Cost | $O(n \cdot T \cdot B)$ | $O(n \cdot B)$ |
| Verifiable tasks | Yes (with ASD) | Yes |
| Open-ended tasks | Rarely effective | Yes |
| Heterogeneity benefit | Strongest antidote | Primary gain |
| Recommendation | Specific settings | General use |

Table 21: Debate vs. ensemble: key tradeoffs for SLM judge deployment. "Risk" columns indicate whether the strategy introduces the named failure mode. Cost is per-evaluation relative to a single judge.

"devil's advocate" perspectives add genuine signal.

- **Use debate with ASD** for: structured reasoning verification (mathematical proofs, code correctness) where short 2-round debates catch categorical errors.
- **Avoid MAD entirely** for: tasks where SLMs lack the underlying competency (debate amplifies errors), and pipelines where context window saturation is a practical concern.

MODEL DIVERSITY IS MORE VALUABLE THAN INTER-AGENT COMMUNICATION FOR SLM JUDGE DEPLOYMENT. [REPLICATED: 2 studies] *A well-designed ensemble of diverse SLMs will outperform a homogeneous debate panel on essentially all evaluation tasks, at lower cost and with more predictable behavior, because the ensemble captures the diversity benefit (error decorrelation) without the communication cost (sycophancy, error amplification, context saturation).* The implication for practitioners is clear: if you have a fixed compute budget for multi-model evaluation, spend it on adding diverse panel members rather than on adding debate rounds. Debate should be reserved for the specific cases where inter-agent communication can surface genuinely new information that independent evaluation cannot.

# H Biases in LLM and SLM Judges

The preceding sections assumed we can measure judge quality. We now examine what systematically distorts that quality. This section gives the full bias taxonomy behind §5, with mechanisms, SLM-specific amplification, and mitigations.

## H.1 The Structural Nature of Judge Bias

Bias in LLM judges does not arise from random error; it arises from systematic, reproducible distortions in how a model computes its evaluation. These distortions are rooted in the model's training distribution: a model trained predominantly on English text, with RLHF feedback collected from human annotators who have their own stylistic preferences, encodes those preferences into its evaluation mechanism. The distortions manifest as consistent, measurable signals that deviate from objective quality assessment.

For SLM judges specifically, bias concerns are amplified by two factors: (1) smaller pretraining corpora reduce the diversity of encoded knowledge, making surface-level heuristics more influential, and (2) reduced instruction-following capability means that prompting-based bias mitigations are less reliably executed. This appendix provides a comprehensive taxonomy organized by mechanism. For each bias type, we provide: a formal definition, the mechanistic explanation, documented magnitude in SLM judges, and evidence-based mitigations.

## H.2 Bias Taxonomy

We organize judge biases by their underlying mechanism rather than by surface symptom. For each, we give a formal definition, the mechanism that produces it, its documented magnitude in SLM judges, and evidence-based mitigations. We begin with the most studied, position bias.

### H.2.1 Position Bias

**Definition.** Position bias is a judge's tendency to assign higher quality ratings to responses based on their ordinal position in the prompt (first vs. second) rather than their substantive content. In pairwise evaluation, the judge receives a prompt containing two candidate responses labeled A and B; position bias manifests when the judge's verdict changes if A and B are swapped, despite identical content.

**Measurement.** The standard measurement protocol (Zheng et al., 2023) is position-consistent accuracy: a judge is credited with a correct verdict only if it produces the same winner in both

orderings (A, B) and (B, A):

$$\text{PC}(x, a, b) = \mathbb{1}[v_{ab} = v_{ba}] \tag{20}$$

A perfectly unbiased judge achieves PC = 1.0 on all pairs. Mean PC across the test set provides the aggregate consistency metric.

**Mechanistic explanation.** Position bias has two contributing mechanisms:

1. **Primacy bias**: The model's attention mechanism assigns disproportionate weight to tokens at the beginning of the context window. The first-appearing candidate response receives more attention heads and correspondingly stronger encoding.
2. **Recency bias**: For some models trained with long-context data, the last-appearing candidate benefits from proximity to the verdict token prompt.

The direction of position bias (primacy vs. recency) is model-family-dependent, which explains why heterogeneous ensembles partially cancel positional distortions: models from different families have opposing position biases that cancel in aggregation.

**Magnitude.** Tan et al. (2025) demonstrate that position-consistent accuracy is dramatically lower than naive accuracy for essentially all models tested. On JudgeBench's challenging domains, even frontier models are often position-inconsistent on contested evaluations. Jayarao et al. (2025) find that thinking-mode SLM judges (Qwen 3, 0.6B–4B) show substantially reduced position bias compared to non-thinking counterparts.

POSITION BIAS DIRECTION IS MODEL-FAMILY-DEPENDENT, MAKING IT STRUCTURALLY CANCELLABLE VIA HETEROGENEOUS PANELS. [PRELIMINARY: proposed mechanism] *Primacy bias and recency bias arise from different attention patterns in different model architectures, and these patterns are determined by pretraining corpus composition and context window training strategy, not by model size or quality.* A Phi-family model with strong primacy bias paired with a Qwen-family model with recency bias produces an ensemble whose positional errors partially cancel, even without any bias-specific training. This structural cancellation comes at no extra cost in any heterogeneous panel. The practical implication is that panel diversity provides a zero-cost position bias mitigation independent of, and additive with, training-level mitigations like swap augmentation.

**Mitigations.**

- **Swap calibration**: Evaluate each pair in both orderings; accept verdict only on agreement, or average continuous scores. Cost: 2× inference.
- **Consistency reward training**: Train judges with GRPO using rewards that penalize positional reversals (Whitehouse et al., 2025). Done at training time; zero cost at inference.
- **Reference abstraction**: Present responses without explicit A/B labeling, inserting them into numbered rubric slots to reduce positional cue salience.
- **Distribution-matching fine-tuning**: Li et al. (2025c) train 7–9B judges (Gemma-2-9B, Qwen-2.5-7B) to match the distribution of human preferences rather than a single label, which raises consistency under swapped order by 7–81%.

#### H.2.2 Verbosity Bias (Length Bias)

**Definition.** Verbosity bias is a judge's systematic preference for longer, more verbose responses regardless of information density, factual accuracy, or relevance.

**Mechanistic explanation.** Two mechanisms contribute:

1. **RLHF length trap**: Human annotators frequently prefer longer responses due to the signal of "effort." Models trained on these annotations encode length-as-quality as a reward heuristic.
2. **Perplexity correlation**: Longer responses provide more tokens against which the judge computes log-probability scores. A correct, detailed response has more high-probability tokens, inflating the judge's implicit quality estimate.

**Measurement.** Verbosity sensitivity quantifies the fraction of verdicts that flip when padding is injected:

$$\text{VS} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[v(a_i', b_i) \neq v(a_i, b_i)] \tag{21}$$

where $a_i'$ is response $a_i$ with appended padding $p$ such that $|a_i'| > |b_i|$. An unbiased judge has VS = 0.

**SLM-specific magnitude.** Verbosity bias is stronger in SLM judges than in larger models: smaller models have narrower vocabulary distributions, making it harder to distinguish "high-quality dense content" from "fluent but padded text." The length heuristic is a simpler classification boundary than content-quality assessment, and SLMs fall back on it more readily.

VERBOSITY BIAS CREATES A RUNAWAY LENGTH TRAP IN RLHF PIPELINES WHEN THE JUDGE IS ALSO THE REWARD SIGNAL. [PRELIMINARY: proposed mechanism] *If a generator model is trained against a verbosity-biased SLM judge, the generator learns to produce inflated, wordy outputs; these outputs receive higher reward from the biased judge, reinforcing the length heuristic and causing systematic quality degradation across RLHF iterations.* This is a positive feedback loop: each training iteration produces longer outputs that receive higher reward, further training the generator toward length. The loop is broken only by external intervention (length normalization, style control, or switching to a less verbosity-biased judge). Arena-Hard-Auto addresses this with a style-control mechanism that strips length and formatting artifacts before judging.

**Mitigations.**

- **Length normalization**: Score per-token information density rather than total information content.
- **Style control**: Strip length/formatting cues from responses before judging.
- **Rubric penalties**: Include explicit criteria penalizing unnecessary verbosity.

#### H.2.3 Self-Preference Bias (Egocentric Bias)

**Definition.** Self-preference bias is the tendency of a judge to assign higher ratings to responses generated by itself or by models from the same training lineage, independent of actual response quality.

**Mechanistic explanation.** The mechanism is rooted in perplexity: a language model assigns lower perplexity to text resembling its own output distribution. When a GPT-4-family judge evaluates a GPT-4-generated response alongside a Llama-generated response, the former has lower perplexity under the judge's model. The judge interprets low perplexity as high quality, correctly in many cases, but incorrectly when the alternative response is substantively superior. This is a direct consequence of conflating familiarity with quality.

**Measurement.** Self-preference score quantifies the excess win rate for same-family responses at equal quality:

$$\text{SPS} = P(\text{prefer own family}) - 0.5 \tag{22}$$

An unbiased judge has SPS = 0; positive SPS indicates self-preference. Gu et al. (2024) document self-preference across multiple model families.

SELF-PREFERENCE BIAS IS PERPLEXITY-MEDIATED, NOT CONTENT-MEDIATED, MAKING IT INVISIBLE TO CONTENT-BASED DEBIASING. [PRELIMINARY: single study] *A judge favors its own family's outputs not because it recognizes the content as its own, but because its own outputs have lower perplexity under its internal model, and low perplexity is conflated with high quality in the evaluation computation.* Content-based debiasing (e.g., stripping authorship cues, anonymizing responses) does not address this mechanism because the bias operates at the distributional level, not the lexical level. The only effective mitigations are cross-family panels (where opposing self-preferences cancel) or training with preference-leak-resistant data (FLAMe, Self-Taught Evaluators) that diversify the training distribution beyond any single model family.

#### H.2.4 Preference Leakage

**Definition.** Preference leakage (Li et al., 2025b) is a contamination effect that extends beyond self-preference to all models related to the judge through the training data pipeline. Three relatedness types are documented:

1. **Same Model**: Evaluator is the generator.
2. **Inheritance**: Student trained on teacher's synthetic data.
3. **Same Family**: Same development lineage.

PREFERENCE LEAKAGE IS BROADER AND MORE INSIDIOUS THAN SELF-PREFERENCE BECAUSE IT OPERATES THROUGH DATA LINEAGE, NOT MODEL IDENTITY. [PRELIMINARY: single study] *A judge may favor a model it has never seen before if that model was trained on data generated by a model from the judge's own family, because the downstream model inherits stylistic patterns from its training data that reduce perplexity under*

*the judge.* This means that the standard mitigation of "using a neutral third-party judge" is insufficient if the third party shares data lineage with any evaluated model. The only robust mitigation is decoupling the entire data pipeline: judges should be trained on data from independent annotation sources (human data, or self-generated data) with no connection to the evaluated models' training provenance.

#### H.2.5 Sycophancy Bias

**Definition.** Sycophancy bias is the judge's tendency to align its verdict with signals of user expectation, social authority, or majority opinion embedded in the evaluation prompt, rather than with objective quality criteria.

**Attack vectors leveraging sycophancy.**

- **Majority opinion framing**: "90% of experts prefer Response A" inflates verdicts toward A.
- **Authority framing**: Attributing one response to a prestigious source inflates its rating.
- **Confidence framing**: "Response A is clearly superior" in the system prompt elicits agreement.
- **Reasoning opener injection**: Prefacing one response with markers associated with reasoning-capable models inflates perceived quality.

**Mechanistic explanation.** Sycophancy originates in RLHF training: human annotators implicitly favor outputs that reinforce their existing beliefs, making annotated preferences systematically sycophantic. The reward model learns that agreement with expressed opinions is rewarded. This is the same mechanism behind user-facing sycophancy, now manifesting in the evaluation context.

#### H.2.6 Balance Penalty Bias

**Definition.** Balance penalty bias is the systematic penalization of nuanced, appropriately-hedged, or trade-off-acknowledging responses in favor of more decisive (but sometimes incorrect) answers.

**Relevance.** Balance penalty is particularly damaging in high-stakes domains. In medical, legal, or policy evaluation, balanced, hedge-acknowledging responses are often the correct ones: "this treatment may be appropriate depending on patient context" is medically superior to "this treatment is always appropriate." A judge with balance penalty systematically rewards the latter.

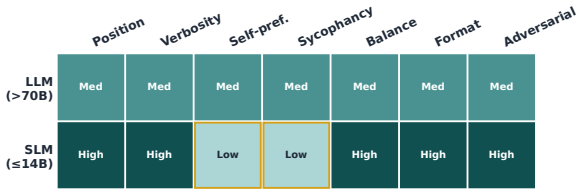



Figure 10: Relative bias severity for SLM ($\leq$14B) versus large LLM (>70B) judges. SLM judges are more strongly affected on surface-level axes, while self-preference (highlighted) and sycophancy are weaker in SLMs.

BALANCE PENALTY IS THE MOST DOMAIN-DANGEROUS BIAS BECAUSE IT PENALIZES CORRECT ANSWERS IN HIGH-STAKES DOMAINS. [PRELIMINARY: proposed mechanism] *In domains where the correct answer requires acknowledging genuine uncertainty (medicine, law, policy), balance penalty causes the judge to systematically prefer confidently wrong answers over appropriately uncertain correct answers, inverting the quality ranking.* This is distinct from other biases (position, verbosity, self-preference) which distort the magnitude of quality assessment but do not typically invert its direction. Balance penalty can cause a judge to assign higher scores to a response that recommends a universally applicable treatment (dangerously overconfident) than to a response that appropriately conditions on patient context (correctly nuanced). The mitigation requires domain-specific rubrics that explicitly value appropriate hedging.

### H.3 SLM-Specific Bias Amplification

SLM judges are not merely smaller versions of LLM judges; they exhibit qualitatively different bias profiles. Table 22 summarizes the key differences and their mechanistic causes, and Figure 10 visualizes the severity contrast. Self-preference is *weaker* in SLMs, because they produce fewer self-similar outputs and therefore a weaker perplexity-similarity signal. Sycophancy is weaker too, since it grows with model size and instruction tuning (Perez et al., 2023; Wei et al., 2024), so smaller judges are less inclined to echo an opinion stated in the prompt.

SLMS HAVE STRONGER SURFACE BIASES BUT WEAKER SELF-PREFERENCE AND SYCOPHANCY, CREATING A DISTINCT DEPLOYMENT RISK PROFILE. [PRELIMINARY: proposed mechanism] *The inversion on self-preference (SLMs are less self-preferencing than LLMs) occurs because*

| Bias Type | LLM | SLM | Reason |
|---|---|---|---|
| Position | Med | Stronger | Weaker attn. diversity |
| Verbosity | Med | Stronger | Narrower vocab, simpler heur. |
| Self-pref. | Med | Weaker | Fewer self-gen. examples |
| Sycophancy | Med | Weaker | Grows with scale |
| Balance | Med | Stronger | Less nuanced feedback exp. |
| Format | Med | Stronger | Formatting harder at scale |
| Adversarial | Med | Stronger | Weaker instr. following |

Table 22: Bias profile comparison between SLM judges ($\leq$14B) and large LLM judges ($>$70B); models between these bounds are not characterized here because the surveyed literature reports little on them. "Stronger" means the bias is more pronounced in SLMs relative to the LLM column, which is set to a common reference rather than a measured value. The directions are qualitative syntheses across the bias literature in this appendix, not a single controlled study, and the two inversions are self-preference and sycophancy, which grows with model size (Perez et al., 2023; Wei et al., 2024).

*smaller models have fewer self-generated examples in their training data and produce more variable outputs, reducing the perplexity-similarity signal that drives self-preference in larger models.* The practical implication is that SLM judges are safer as cross-model evaluators (they are less biased toward their own family) but more dangerous on surface-level quality signals (they are more susceptible to position, verbosity, and formatting cues). Deployment strategies should prioritize surface-bias mitigation (swap calibration, length normalization) over self-preference mitigation for SLM panels.

### H.4 Bias Interaction Effects

Bias types do not act independently; they interact, sometimes amplifying each other:

- **Verbosity $\times$ position**: A longer response appearing first receives both verbosity reward and primacy reward simultaneously, creating a compounding advantage.
- **Sycophancy $\times$ authority**: Authority framing triggers sycophancy, which then amplifies any content-based bias in that direction.
- **Self-preference $\times$ RLHF loop**: In pipelines where a judge evaluates its own generator's outputs, self-preference inflates all training rewards, creating runaway self-consistency that degrades output diversity.
- **Balance penalty $\times$ verbosity**: A confident, long response receives both balance penalty reward (decisiveness) and verbosity reward (length), doubly advantaged over a concise, appropriately nuanced response.

BIAS INTERACTIONS ARE MULTIPLICATIVE, NOT ADDITIVE, MAKING ISOLATED BIAS MITIGATION INSUFFICIENT. [PRELIMINARY: proposed mechanism] *Mitigating position bias alone does not eliminate the compounding effect when position bias interacts with verbosity bias, because the two biases share partially overlapping attention mechanisms; a response that is both first and longer activates two reinforcing heuristic pathways simultaneously.* The implication is that mitigations acting on several biases at once (heterogeneous panels, rubric-anchored evaluation, consistency rewards) are categorically more effective than addressing biases individually. A system that perfectly eliminates position bias but ignores verbosity bias will still exhibit substantial evaluation distortion on pairs where one response is substantially longer.

### H.5 Mitigation Strategies: Comprehensive Reference

No single technique removes all biases, and the most effective deployments stack several. We consolidate the documented mitigations below, annotating each with the bias it targets and its inference-time cost so that practitioners can assemble a stack matched to their budget and risk profile.

1. **Swap calibration (position bias)**: Evaluate each pair twice with swapped order; accept only agreement or average scores. Standard practice in MT-Bench (Zheng et al., 2023). Cost: 2$\times$ inference.
2. **Consistency reward training (position + sycophancy)**: Train judges with GRPO using rewards that penalize verdict reversals when order is swapped (Whitehouse et al., 2025). Done at training time; no cost at inference.
3. **Rubric-anchored evaluation (verbosity, balance penalty)**: Provide explicit rubric criteria calibrating against length and decisiveness. Prometheus 2's rubric-conditioned training (Kim et al., 2024b) is the leading example.
4. **Logit-based score smoothing (calibration)**: FLEUR (Lee et al., 2024) computes a continuous score by weighting digit probabilities across the Likert range rather than taking argmax, reducing score discretization artifacts.

5. **TrustJudge distributional scoring (consistency)**: Wang et al. (2025) address score-pairwise inconsistency and transitivity violations using continuous expected value over the full Likert probability distribution.

6. **Cascaded selective evaluation (confidence-based routing)**: Jung et al. (2025) propose routing high-uncertainty verdicts from SLM judges to larger models.

7. **Heterogeneous panel judging (all biases)**: The PoLL approach (Verga et al., 2024) mitigates all systematic biases simultaneously by pooling judges with opposing bias profiles.

8. **Blind response presentation (sycophancy, authority bias)**: Strip all source identifiers, model attributions, and confidence framing from responses before judge submission.

9. **Golden-set calibration**: Maintain a small labeled calibration set ($n = 50$–$100$); periodically benchmark judge outputs against this set and apply correction offsets for known systematic biases.

### H.6 The CALM Framework: Systematic Bias Quantification

Gu et al. (2024) synthesize the literature on judge bias measurement, pointing to the CALM framework (Ye et al., 2025) as the most systematic attempt to quantify the full bias landscape. CALM identifies 12 distinct bias categories organized into four dimensions:

1. **Superficial quality biases**: Position, verbosity, formatting, punctuation.

2. **Contextual biases**: Self-preference, prior-belief.

3. **Presentation biases**: Authority, sycophancy, confidence framing.

4. **Diversity biases**: Demographic, cultural.

The framework provides standardized injection protocols: each bias type is measured by systematically injecting the bias-inducing signal and measuring the verdict deviation relative to a clean baseline. This enables per-model, per-task, per-bias-type characterization of a judge's reliability profile.

A Comprehensive Bias Audit Should Precede Any SLM Judge Deployment Because Bias Profiles Are Model-Specific and Cannot Be Predicted From General Capability Scores. [Preliminary: analytical argument] *Two SLM judges with identical accuracy on RewardBench can have dramatically different bias profiles: one may be strongly position-biased but low on verbosity bias, while the other may be the reverse.* General accuracy metrics (RewardBench score, JudgeBench accuracy) do not capture bias profiles because they aggregate across many evaluation pairs, masking systematic directional errors. A judge that achieves 85% on RewardBench may be achieving this by being correct 95% of the time on easy pairs and 50% on hard pairs where its position bias dominates. The CALM framework's per-bias-type injection protocol is the minimum viable audit for deployment-critical judge systems.

## I Benchmarks, Datasets, and Metrics

If biases distort judgments, benchmarks must be designed to expose them. This section details the meta-evaluation benchmarks and metrics introduced in §5, centered on position-consistent accuracy.

### I.1 The Meta-Evaluation Problem

Evaluating an LLM or SLM judge requires a meta-evaluation benchmark: a dataset that provides ground-truth quality labels against which the judge's verdicts can be compared. This creates a fundamental epistemological challenge: if the only reliable quality signal for complex language tasks is human preference, then the meta-evaluation benchmark itself requires expensive human annotation. The field has addressed this challenge through three complementary strategies:

1. **Human preference collection**: Directly collect human verdicts at scale. Gold standard for ecological validity. Example: Chatbot Arena (Zheng et al., 2023).

2. **Transformed verifiable datasets**: Take tasks with known ground-truth answers and construct response pairs where one is objectively correct and one is wrong. Example: JudgeBench (Tan et al., 2025).

3. **LLM-as-reference**: Use a stronger model's judgments as the reference. Example: MT-Bench using GPT-4 references (Zheng et al., 2023). This introduces circular dependency concerns but is widely used.

THE THREE META-EVALUATION STRATEGIES MEASURE DIFFERENT PROPERTIES AND SHOULD NOT BE TREATED AS INTERCHANGEABLE. [PRELIMINARY: analytical argument] *Human preference collection measures ecological validity (does the judge match real users?), transformed verifiable datasets measure objective correctness (is the judge factually right?), and LLM-as-reference measures consistency (does the judge agree with a stronger model?).* A judge that scores well on one strategy may fail on another: high human agreement does not guarantee objective correctness (humans themselves can be wrong), and high LLM-reference agreement does not guarantee ecological validity (the reference LLM may have different preferences than real users). A robust judge deployment should validate against benchmarks from at least two of the three categories to cover complementary quality dimensions.

### I.2 Judge-Specific Evaluation Benchmarks

We survey the benchmarks built specifically to measure judge quality, ordered from those stressing objective correctness (JudgeBench) through conversational and reward-model evaluation to domain-specific and multilingual suites. Each entry lists its primary metric, task types, and the failure mode it is designed to expose.

#### I.2.1 JudgeBench: Objective Correctness Under Difficulty

**Paper**: Tan et al. (2025) (ICLR 2025).
**Primary metric**: Position-consistent accuracy (Appendix H).
**Task types**: Knowledge, reasoning, mathematics, coding.

JudgeBench is the most rigorously designed benchmark for measuring judge reliability in objective settings. Its construction methodology:

1. Start with datasets that have deterministic ground-truth answers (e.g., MMLU (Hendrycks et al., 2020) for knowledge, MATH (Hendrycks et al., 2021) for mathematics, HumanEval for code).
2. For each question, construct a response pair $(a, b)$ where $a$ is objectively correct and $b$ is objectively incorrect but plausible.
3. Bias the construction toward hard pairs: $b$ is a "good wrong answer" that is fluent, confident, and structurally similar to $a$.
4. Evaluate using position-consistent accuracy: the judge must select $a$ over $b$ in both orderings to receive credit.

**Key finding.** Even frontier models like GPT-4o show sharply degraded performance on JudgeBench's challenging domain, sometimes approaching random-chance baselines under position-consistent evaluation. The benchmark reveals that "judging is tied to solving": a model's ability to correctly select the superior response is highly correlated with its ability to solve the problem correctly on its own.

JUDGEBENCH REVEALS THAT JUDGING IS TIED TO SOLVING, BUT ONLY IN KNOWLEDGE-INTENSIVE DOMAINS. [PRELIMINARY: single study] *The correlation between solve-ability and judge-ability is strong for domains where evaluation requires the same knowledge as generation (graduate STEM, complex reasoning), but breaks down in domains where evaluation requires only pattern matching (formatting, instruction adherence, fluency).* This has direct implications for SLM judge routing: SLMs can be trusted as judges on pattern-matching tasks regardless of their solve-ability, but should be routed to larger models for knowledge-intensive evaluation. The "judging is tied to solving" finding is domain-conditional, not universal, and overapplying it would unnecessarily restrict SLM judge deployment.

#### I.2.2 JudgeBoard: SLM-Specific Reasoning Evaluation

**Paper**: Bi et al. (2026).
**Primary metric**: Accuracy + Elo rating.
**Task types**: Mathematical reasoning, commonsense reasoning, science.

JudgeBoard is the first benchmark specifically designed for evaluating SLM judgment capability. It constructs task-specific leaderboards using both absolute accuracy and Elo-based comparative rating. The Elo system reduces sensitivity to any single evaluation pair. JudgeBoard's MAJ framework (Appendix E) demonstrates that ensemble SLM judges can match standalone LLM judges on the MATH benchmark.

#### I.2.3 JudgerBenchV2: Large-Scale Multi-Scenario Evaluation

**Paper**: Zhang et al. (2025c).
**Scale**: 10,000 questions across 10 evaluation scenarios.
**Aggregation**: Mixture-of-Judgers consensus.

JudgerBenchV2 is the largest-scale judge evaluation benchmark, testing across diverse evaluation scenarios simultaneously. The Mixture-of-Judgers consensus mechanism provides more robust ground-truth labels than single-reference evaluation. CompassJudger-2 7B reports judgment accuracy competitive with DeepSeek-V3 (37B active / 671B total parameters) and Qwen3-235B-A22B (22B / 235B), further evidence that targeted training can stand in for scale in evaluation.

#### I.2.4 MT-Bench: Multi-Turn Conversational Evaluation

**Paper**: Zheng et al. (2023) (NeurIPS 2023).
**Primary metric**: Score (1–10 Likert) correlation with GPT-4 reference.
**Task types**: Writing, roleplay, extraction, reasoning, math, coding, STEM, humanities (8 categories, 80 prompts).

MT-Bench established the modern LLM-as-a-Judge paradigm and remains the most widely cited benchmark. Design features:

- Multi-turn structure (2 turns per conversation) probes conversational coherence.
- Numeric scoring (1–10 Likert) enables finer-grained assessment than binary comparisons.
- Agreement with 3 expert human evaluators: GPT-4 achieves >80% agreement, establishing the paradigm's viability.

**Limitations.** Small benchmark size (80 prompts $\times$ 2 turns) limits statistical power. Tasks are well-known categories susceptible to data contamination in post-2023 models. Not designed to stress-test position or verbosity bias.

#### I.2.5 RewardBench: Reward Model Evaluation

**Paper**: Lambert et al. (2025).
**Primary format**: Prompt-chosen-rejected triples.
**Categories**: Chat, Chat Hard, Safety, Reasoning.

RewardBench targets the specific usage of LLMs as reward models in RLHF pipelines. It evaluates three model types: sequence classifiers, generative models, and implicit reward models (DPO-trained).

**SOTA SLM performance.** At its release, Skywork-Critic-Llama-3.1-8B ranked first among generative models under 10B on RewardBench. J1-Llama-8B achieves SOTA on both RewardBench and JudgeBench among 8B-class models (Whitehouse et al., 2025).

**RewardBench 2.** Updated with harder examples, best-of-$N$ protocols, expanded domains including "Ties" and "Precise Instruction Following."

#### I.2.6 CodeJudgeBench: Domain-Specific Code Evaluation

**Paper**: Jiang et al. (2025).
**Domain**: Code generation and debugging evaluation.

CodeJudgeBench evaluates judge models specifically on code evaluation tasks. Unlike natural language evaluation, code judging can leverage execution-based verification: running candidate code against test cases provides objective ground truth.

Key design features:

- Pairs of solutions where one passes all test cases and one fails a subset.
- Fine-grained categories: syntax, logical correctness, efficiency, edge cases.
- Both pairwise and pointwise modes.

CODE EVALUATION CAN BE FACTORED INTO MECHANICAL VERIFICATION AND QUALITATIVE ASSESSMENT, ENABLING TOOL-AUGMENTED SLM JUDGING. [REPLICATED: 2 studies] *The factoring principle states that any evaluation task with a verifiable component and a subjective component should delegate the verifiable component to a deterministic tool (code executor, symbolic solver) and use the SLM judge only for the subjective component.* For code evaluation this means routing correctness to a Python interpreter and reserving the judge for what the interpreter cannot settle: style, efficiency, and readability. TIR-Judge (Xu et al., 2026) implements that split at 4–8B, training the judge to invoke a code executor rather than to reason about execution internally. The factoring principle generalizes beyond code to any domain on the verifiability gradient (Appendix J).

#### I.2.7 ContextualJudgeBench: Grounded Response Evaluation

**Paper**: Xu et al. (2025a) (ACL 2025).
**Domain**: Contextual (RAG-style) response evaluation.

ContextualJudgeBench evaluates judges on assessing whether a response is faithful to a provided context document. This requires the judge to simultaneously: read the context document, assess

whether claims are supported, and detect subtle unfaithfulness (paraphrastic distortion, omissions, hallucinations).

For SLM judges, ContextualJudgeBench is particularly challenging because the context document consumes substantial context window budget, leaving less space for rubric instructions and evaluation reasoning. Context-window-efficient approaches (INSPECTOR-style representation probing) may have an advantage.

#### I.2.8 M-RewardBench: Multilingual Reward Evaluation

**Paper**: Pombal et al. (2025).
**Languages**: 20+, including high-resource and low-resource languages.

M-RewardBench extends RewardBench to multilingual settings. It exposes a consistent finding: judge quality degrades significantly from English to non-English languages, with degradation proportional to language rarity in training data. M-Prometheus achieves substantially better performance than translation-based approaches.

### I.3 Meta-Evaluation Metrics

Table 23 provides a comprehensive reference of all metrics used in the judge evaluation literature.

RELIABILITY METRICS (PC, IFR, ECE) ARE MORE INFORMATIVE THAN AGREEMENT METRICS FOR SLM JUDGE DEPLOYMENT DECISIONS. [PRELIMINARY: analytical argument] *Agreement metrics (accuracy, Spearman's $\rho$) tell you how often the judge agrees with a reference, but not whether its errors are random or systematic; reliability metrics tell you whether the judge's errors are predictable and correctable, which is the information needed for deployment-level trust calibration.* A judge with 80% accuracy and 95% position consistency is more trustworthy than a judge with 85% accuracy and 60% position consistency, because the former's errors are random (reducible via self-consistency) while the latter's errors are systematic (not reducible without bias mitigation). Deployment decisions should prioritize reliability metrics over agreement metrics when they diverge.

### I.4 Benchmark Design Principles

Based on the analysis of existing benchmarks, the following principles characterize high-quality meta-evaluation:

| Metric | Type | Notes |
|---|---|---|
| Accuracy | Agree | Verifiable domains only |
| PC accuracy | Reliab | JudgeBench primary; strictest |
| Agreement rate | Agree | Human labels may be inconsistent |
| Cohen's $\kappa$ | Agree | Adjusted for chance; nominal |
| Spearman's $\rho$ | Agree | Rank-order; robust to scale |
| Pearson's $r$ | Agree | Linear; sensitive to scale |
| Kendall's $\tau$ | Agree | Concordant minus discordant |
| Win rate | Agree | Standard pairwise metric |
| Elo rating | Agree | Stable; accounts for opponent |
| VS | Bias | Verdict flip under padding |
| SPS | Bias | Excess same-family win rate |
| IFR | Reliab | Format compliance; critical |
| Separability | Bench | Arena-Hard-Auto; discriminative power |
| ECE* | Reliab | Confidence calibration |
| Conformal cov. | Reliab | Distribution-free uncertainty |

Table 23: Meta-evaluation metrics for LLM/SLM judge assessment. "Type" distinguishes between agreement metrics (measuring correlation with references), reliability metrics (measuring internal consistency), and bias metrics (measuring systematic distortion). Metrics marked with * require logit-level access.

#### I.4.1 Difficulty Calibration

A well-designed benchmark should place evaluated models in their discrimination zone, avoiding both ceiling effects (all-correct) and floor effects (all-wrong). JudgeBench achieves this by focusing on "good wrong answers"; Arena-Hard-Auto achieves this via BenchBuilder curation.

#### I.4.2 Bias Resilience

A benchmark measuring judge quality should not be conflated with a measure of judge bias. JudgeBench's position-consistent accuracy is bias-resilient: a position-biased judge cannot achieve high PC accuracy. MT-Bench's simple Likert scoring without positional controls is less bias-resilient.

#### I.4.3 Domain Coverage

Most benchmarks over-represent English-language generation and math/reasoning tasks. Critical underrepresented domains include:

- **Creative writing**: Genuinely subjective; no ground truth. Benchmark design must use human preference.

- **Medical and legal**: High-stakes evaluation requiring domain expertise.
- **Multilingual**: M-RewardBench is the primary effort; low-resource coverage remains minimal.
- **Agentic evaluation**: Traditional static benchmarks cannot capture dynamic, stateful, tool-dependent agent task evaluation.

#### I.4.4 Contamination Resistance

Static benchmarks suffer from temporal contamination: models trained after a benchmark's release may have seen benchmark examples in their training data. The Chatbot Arena model (Zheng et al., 2023), a continuously updated live platform, is inherently contamination-resistant.

STATIC BENCHMARKS HAVE A HALF-LIFE PROPORTIONAL TO THEIR VISIBILITY IN WEB CORPORA. [PRELIMINARY: single study] *Once a benchmark's examples appear in web-crawled pretraining corpora, any model trained after that point receives an artificial accuracy boost from memorization rather than genuine evaluation capability.* MT-Bench's 80 prompts, published in 2023, are almost certainly memorized by models trained on post-2023 web data. JudgeBench and RewardBench 2 partially address this by using harder, less common evaluation pairs, but only continuously updated benchmarks (Chatbot Arena, LiveCodeBench) provide structural contamination resistance. The research community urgently needs living-benchmark designs for automated judge evaluation.

### I.5 Benchmark Summary

Table 24 provides a consolidated reference for all major judge evaluation benchmarks, and Figure 11 shows which meta-evaluation dimensions each one covers. The matrix makes the gap behind C5 concrete: no single benchmark spans position-consistent accuracy, human correlation, domain coverage, adversarial robustness, efficiency, and instruction following (IFR) simultaneously, so meaningful judge assessment currently requires combining several benchmarks.

### I.6 Meta-Evaluation Protocols: Recommended Practices

For researchers evaluating a new SLM judge, we recommend the following protocol:

1. **Verifiable domain test first**: Run on JudgeBench hard subset and the Reasoning category of RewardBench. If the SLM judge falls below 55% position-consistent accuracy, it lacks the knowledge required for reliable objective evaluation.

2. **Bias audit**: Measure position-consistent accuracy (PC), verbosity sensitivity (VS), and IFR. Any PC $< 70\%$, VS $> 15\%$, or IFR $< 90\%$ indicates a systematic reliability problem requiring mitigation before deployment.

3. **Cross-domain validation**: Evaluate on at least one general benchmark (MT-Bench, Arena-Hard-Auto) and one domain-specific benchmark (CodeJudgeBench, M-RewardBench) to ensure domain generalization.

4. **Human validation sample**: On at least 50 development-set examples, collect human verdicts and compare. This provides ecological validity that automated benchmarks alone cannot supply.

5. **Report calibration**: Always report PC accuracy alongside naive accuracy. Reporting only naive accuracy without positional controls systematically overstates judge reliability.

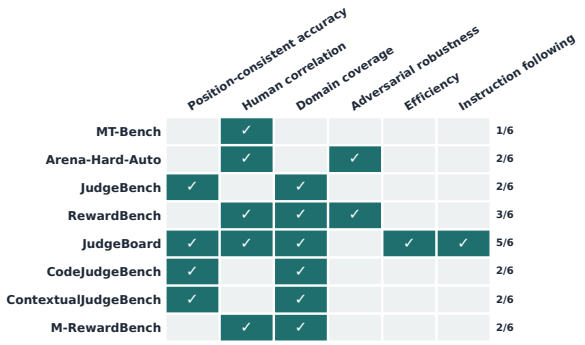



Figure 11: Meta-evaluation dimension coverage across major judge benchmarks. No single benchmark covers all six dimensions; even the broadest (JudgeBoard) omits some, which is the gap motivating a unified suite (C5).

THE FIELD MEASURES AGREEMENT BUT SHOULD MEASURE RELIABILITY. [PRELIMINARY: analytical argument] *Most existing benchmarks measure LLM judge correlation with human preferences, but human preferences themselves are biased: annotators exhibit verbosity preference, authority deference, and inconsistency across sessions.* Conflating judge quality with human agreement makes it impossible to determine whether a "high-performing" judge is

| Benchmark | Venue | Size | Primary metric | Distinctive strength |
|---|---|---|---|---|
| MT-Bench | NeurIPS 2023 | 80 prompts | Likert correlation | Multi-turn conversation |
| Arena-Hard-Auto | 2024 | 500 prompts | Separability, win rate | Style control |
| JudgeBench | ICLR 2025 | Varied | Position-consistent accuracy | Objective correctness |
| RewardBench | 2024 | ∼3K pairs | Per-category accuracy | Four evaluation dimensions |
| RewardBench 2 | 2025 | Expanded | Best-of-$N$ | Harder evaluation pairs |
| JudgeBoard | 2025 | Multi-task | Elo rating and accuracy | Designed for SLM judges |
| JudgerBenchV2 | 2025 | 10K questions | Mixture-of-judgers consensus | Largest scale |
| CodeJudgeBench | 2025 | Code tasks | Execution-grounded accuracy | Tool-friendly |
| ContextualJudgeBench | ACL 2025 | RAG settings | Grounding accuracy | Faithfulness to context |
| M-RewardBench | 2025 | 20+ languages | Multilingual accuracy | Multilingual coverage |

Table 24: Summary of key benchmarks for LLM and SLM judge evaluation. Position-consistent accuracy credits a verdict only when it survives swapping the A/B order; separability measures how reliably a benchmark distinguishes two systems given its confidence intervals.

genuinely accurate or merely replicating human biases. JudgeBench's transformation of verifiable datasets provides the only fully objective metric: correctness on tasks with known ground truth. The research agenda should prioritize objective correctness on verifiable domains and calibrated uncertainty on subjective domains, rather than conflating both with human agreement rates.

### I.7 The Human Gold Standard Problem

An additional validity concern is the human gold standard itself. Inter-annotator agreement drops substantially for complex reasoning tasks (Shankar et al., 2024), and annotator expertise, rubric ambiguity, and evaluation fatigue all introduce noise that propagates into the training signal of judges fine-tuned on human preferences. Thus, high agreement with humans does not guarantee high evaluation quality; it may reflect shared systematic biases between annotator populations and the models trained on their judgments.

### I.8 Dataset Contamination Risks

A further threat to benchmark validity is data contamination: SLMs trained on massive, sometimes opaque corpora may have encountered evaluation benchmark questions during pretraining, inflating apparent judge performance (Bedemariam et al., 2025). Older and widely-distributed benchmarks such as MT-Bench are most vulnerable, while more recent suites (JudgeBench, CodeJudgeBench, JudgeBoard) use contemporary data less likely to appear in pretraining corpora. Practitioners should treat benchmark results with appropriate caution and prefer evaluation suites designed with contamination resistance in mind.

## J Domain Generalization: Code, Safety, and Multilingual

Reliability is domain-dependent. This section develops the verifiability gradient of §6 and Figure 2(d), walking from fully verifiable code and math through safety and multilingual evaluation to subjective, high-stakes domains.

### J.1 The Verifiability Gradient: A Unifying Framework

Domain-specific SLM judge deployment cannot be understood without a unifying framework for what changes across domains. The key variable is *task verifiability*: the degree to which an evaluation criterion can be checked by an external, objective oracle independent of the judge's subjective reasoning.

The literature suggests a verifiability gradient spanning four levels:

1. **Fully verifiable**: Evaluation can be delegated entirely to a deterministic external oracle (compilation, test execution, symbolic verification). Correctness is binary. Examples: code execution pass/fail, mathematical answer equivalence.

2. **Mostly verifiable**: Strong structural constraints exist, but a residual subjective component remains (e.g., code efficiency, idiomaticity; factual claims in structured knowledge domains).

3. **Partially verifiable**: Domain knowledge provides strong constraints, but substantial judgment remains (e.g., safety evaluation, where "safe" depends on context, intent, and use/mention distinction).

4. **Subjective**: No external oracle exists, and evaluation reduces to human preference or expert judgement. Creative writing and general helpfulness sit here, and so does clinical advice. A recommendation such as "this treatment may be appropriate depending on patient context" is correct in a way no deterministic checker can confirm, which places medical evaluation at the far end of the gradient even though it rests on hard domain knowledge rather than taste.

VERIFIABILITY DETERMINES SLM JUDGE COMPETENCE, NOT PARAMETER COUNT. [CONSENSUS: 8 domains] *An SLM judge succeeds or fails in a domain primarily based on whether that domain's evaluation criteria can be grounded in verifiable constraints, not on model size.* A small judge connected to a Python interpreter settles code correctness exactly, because the interpreter rather than the judge decides it, while a far larger model with no external grounding is still guessing at creative writing quality. Deployment decisions should be stratified by this verifiability gradient, not by a universal size threshold. The appropriate deployment architecture changes qualitatively at each verifiability level: tool-augmented at the top, ensemble-based in the middle, and human-calibrated at the bottom.

THE VERIFIABILITY GRADIENT PREDICTS WHERE SLM JUDGES CAN REPLACE LLM JUDGES AND WHERE THEY CANNOT. [CONSENSUS: 8 domains] *At the fully verifiable end, SLM judges augmented with tools match or exceed LLM judge accuracy because the tool provides the knowledge the SLM lacks; at the subjective end, SLM judges systematically underperform because subjective evaluation requires broad cultural knowledge and nuanced calibration that scale with pretraining diversity.* The gradient creates a clear deployment policy: SLMs are cost-optimal judges for verifiable domains (math, code) and should be used with confidence; they are adequate judges for mostly and partially verifiable domains (factual QA, safety) when properly augmented; they are inadequate for subjective domains unless paired with human calibration or ensemble aggregation. This policy avoids both the error of deploying SLMs everywhere and the error of restricting them to trivial tasks.

### J.2 Code Evaluation

Code sits at the fully verifiable end of the gradient and is therefore where SLM judges are strongest. The reason is structural, captured by the execution factoring principle below.

#### J.2.1 The Execution Factoring Principle

Code judgment has a special property unavailable in other domains: correctness can be factored into two cleanly separable components, only one of which requires the judge model:

$$\text{Eval}(q, c) = \underbrace{f_{\text{exec}}(c)}_{\text{oracle: pass/fail}} \otimes \underbrace{f_{\text{judge}}(q, c)}_{\text{qualitative: style, efficiency}} \tag{23}$$

where $q$ is the programming task, $c$ is the candidate code solution, $f_{\text{exec}}$ is the execution oracle, and $f_{\text{judge}}$ is the SLM judge's qualitative assessment.

This factoring means that an SLM judge's primary value in code evaluation is not correctness determination (which the oracle handles perfectly) but qualitative assessment: idiomaticity, readability, edge case handling beyond the test suite, and appropriate data structure selection.

**Tool-Integrated Reasoning (TIR-Judge).** Xu et al. (2026) introduce the principle of Tool-Integrated Reasoning for judges: SLM judges that can invoke external tools extend their effective evaluation capacity beyond what their parameters alone can support:

- **Execution delegation**: The judge submits code to a sandboxed interpreter, receives pass/fail and error trace, and integrates this signal into its evaluation.
- **Specification checking**: The judge verifies that the solution's interface matches the specification without text generation.
- **Counterexample generation**: The judge synthesizes adversarial test cases to probe edge case handling.

TOOL AUGMENTATION SHIFTS THE SLM JUDGE'S ROLE FROM KNOWLEDGE-PROVIDER TO KNOWLEDGE-INTEGRATOR. [REPLICATED: 2 studies] *Without tools, the SLM judge must both know the answer and evaluate the response; with tools, the SLM judge needs only to integrate externally provided ground truth into a coherent evaluation, a task that requires reasoning about evidence rather than possessing domain knowledge.* This

role shift is why a tool-augmented judge at 8B reaches listwise agreement comparable to a frontier model (Xu et al., 2026): its reasoning capacity suffices for integrating evidence the executor supplies, even where its parametric knowledge would not suffice to produce that evidence itself. The practical implication is that any evaluation domain with an available external oracle should use tool augmentation as the first-line deployment strategy, reserving the SLM's capacity for the subjective residual.

#### J.2.2 CodeJudgeBench: Structured Code Judge Evaluation

Jiang et al. (2025) introduce CodeJudgeBench, the first benchmark for LLM-as-a-Judge for code tasks. Key findings:

- **Thinking models dominate**: Models with extended chain-of-thought reasoning (Qwen3-8B thinking mode) substantially outperform non-thinking models on code judgment. The reasoning trace allows the judge to mentally execute the code and trace through edge cases.
- **Pairwise > pointwise**: Pairwise comparison consistently outperforms pointwise scoring for code evaluation, because the comparison format naturally anchors the assessment to relative correctness.
- **Semantic bias persists**: Judges are significantly affected by surface-level code quality cues (variable naming, comment quality) even when evaluating functional correctness, particularly in smaller, non-thinking SLMs.

CODE JUDGES EXHIBIT A SURFACE-SEMANTICS CONFUSION THAT MIRRORS VERBOSITY BIAS IN TEXT EVALUATION. [PRELIMINARY: single study] *Just as text judges conflate response length with quality (verbosity bias), code judges conflate surface code quality (clean variable names, extensive comments, structured formatting) with functional correctness, because surface quality correlates with correctness in training data but can be decoupled adversarially.* An adversary can produce beautifully formatted, well-commented code that is logically incorrect, and SLM judges will rate it higher than functionally correct but poorly formatted code. This is the code-domain instantiation of the general "surface bias" pattern documented in Appendix H. Thinking mode partially mitigates this by forcing the judge to trace execution rather than relying on surface cues, but the mitigation is incomplete below 7B parameters.

#### J.2.3 SWE-bench and Agentic Code Evaluation

SWE-bench (Jimenez et al., 2024) extends code evaluation to the agentic setting: evaluating an agent's ability to resolve real-world GitHub issues in full software repositories. Unique challenges:

- **Multi-file scope**: Changes may span multiple files; the judge must evaluate a patch rather than a single function.
- **Integration testing**: Correctness requires maintaining the broader test suite.
- **SWE-bench Verified**: A curated subset of 500 confirmed-solvable problems.

For SLM judges, the key challenge is the size mismatch: the relevant context (repository code, issue description, proposed patch) can run to tens of thousands of tokens, which strains the memory budget of a single-GPU judge. Hierarchical evaluation strategies and context-efficient architectures are needed. Crupi et al. (2026) show that SLM judges under 5B parameters rank candidate code solutions competitively with LLMs 5–25× larger at a fraction of the cost, and that a single well-chosen small judge is usually enough, reinforcing the cost-effectiveness of SLM judges in the code domain.

#### J.2.4 LiveCodeBench: Contamination-Resistant Evaluation

LiveCodeBench (Jain et al., 2025) addresses data contamination by continuously collecting new problems from live competitive programming platforms. For judge evaluation specifically, this ensures that models are evaluated on problems released after their training cutoff.

### J.3 Safety Evaluation

Safety lies in the middle of the gradient: partially verifiable, context-dependent, and high-stakes. We trace it from the constitutional self-critique paradigm through its hardest failure mode, the use/mention distinction, to the SLM-specific challenges it raises.

### J.3.1 Constitutional AI and the Judge-as-Safety-Evaluator

Bai et al. (2022) introduce Constitutional AI (CAI): a framework in which a model evaluates its own outputs against an explicit set of principles. The pipeline:

1. The generator produces an initial response.
2. A critique judge evaluates the response against each constitutional principle.
3. A revision judge proposes corrections based on the critique.
4. The process repeats for $K$ rounds until the response passes all principles.

For SLM deployment, CAI's multi-round critique-revision loop scales cost proportionally. However, because most safety evaluation tasks are at the "partially verifiable" level, small models can handle many safety decisions reliably, particularly binary harm detection (is this response harmful or not?), which is a discriminative task well-suited to SLMs. Trustworthiness benchmarks such as TruthfulQA (Lin et al., 2022) and DecodingTrust (Wang et al., 2023a) provide standardized evaluation axes (toxicity, stereotype bias, privacy, adversarial robustness) that can serve as the foundation for SLM safety judge training and validation.

### J.3.2 The Use/Mention Distinction: Context-Sensitivity

The most persistent failure mode in safety evaluation is the use/mention distinction: a response that uses harmful content is qualitatively different from a response that mentions the same content in an educational context, yet surface-level safety judges frequently cannot reliably distinguish them.

This distinction is difficult for SLM judges because:

- **Pragmatic inference**: Correctly distinguishing use from mention requires understanding communicative intent, a high-level pragmatic capacity that emerges later in capability scaling.
- **Context window integration**: The distinction often depends on preceding conversational context many turns earlier.
- **Overgeneralization**: Safety fine-tuning frequently causes over-refusal, because training signal rewards refusal over nuanced context-sensitivity.

| Challenge | Sev. | Root Cause |
|---|---|---|
| Use/mention | High | Weaker pragmatic inference |
| Multi-turn adv. | High | Shorter context retention |
| Multilingual jailbreak | V.High | English-centric safety training |
| Over-refusal | High | Safety training over-generalize |
| Reward hacking | Med | Weaker reasoning for detection |
| Context coherence | High | Context saturation |

Table 25: Safety evaluation challenges and their severity for SLM judges. Root causes are traced to specific SLM limitations.

OVER-REFUSAL IS AS COSTLY AS UNDER-REFUSAL FOR SLM SAFETY JUDGES BECAUSE IT DEGRADES TRUST IN THE EVALUATION SYSTEM. [PRELIMINARY: proposed mechanism] *A safety judge that refuses to engage with legitimate queries containing safety-adjacent keywords ("How does chemotherapy affect cancer cells?") is as operationally harmful as one that fails to flag genuinely unsafe content, because over-refusal causes users and downstream systems to distrust and circumvent the safety evaluation layer entirely.* The calibration challenge for SLM safety judges is therefore bidirectional: they must be sensitive enough to catch genuine harm and specific enough to avoid false alarms. This bidirectional calibration is harder for SLMs because their training data typically over-represents refusal examples (which are easier to generate) and under-represents nuanced "safe but sensitive" examples.

### J.3.3 SLM-Specific Safety Challenges

Table 25 summarizes the key safety evaluation challenges for SLM judges.

**Multilingual jailbreak vulnerability.** A documented attack vector exploits the English-centric nature of safety training: translating a harmful prompt into a low-resource language bypasses English-tuned refusal mechanisms while the underlying model capability is still sufficient to generate a harmful response. SLMs are particularly vulnerable because their safety training is almost entirely English-language.

**Fine-tuning safety degradation.** Domain-specific fine-tuning of SLMs can inadvertently erase safety alignment instilled during RLHF. A

judge fine-tuned on code evaluation data may lose its ability to detect harmful content. Modular safety mechanisms, separate from the judge's domain competency, are needed.

#### J.3.4 Role-Constrained Debate as Safety Evaluation

As established in Appendix G, Lin et al. (2025) show that role-constrained debate among 4–14B judges beats matched small-model prompting baselines and earlier multi-agent judges on safety evaluation, and comes within 0.03 $\kappa$ of GPT-4o at about 46% of its cost. The adversarial structure makes the panel explore both the "safe" and the "unsafe" reading of a borderline response.

SAFETY EVALUATION IS WHERE DEBATE PAYS OFF MOST, BECAUSE THE ADVERSARIAL FRAME PREVENTS THE DEFAULT FAILURE MODE. [PRELIMINARY: single study] *In safety evaluation, the "devil's advocate" role assignment prevents sycophantic convergence (the primary debate failure mode) by making agreement structurally impossible: one agent must always argue "safe" and another must always argue "unsafe," and the mediator must resolve the genuine tension rather than accepting premature consensus.* This structural property does not transfer to other domains (math, code, general text) where role assignment does not naturally prevent agreement. The practical deployment rule is: use role-constrained debate for safety evaluation and ensemble voting for everything else.

### J.4 Multilingual Evaluation

Most judge training data is English, which creates a systematic quality gradient across languages. We characterize that gap, then show how natively-trained multilingual judges (M-Prometheus) close it.

#### J.4.1 The English-Centricity Gap

The vast majority of LLM judge training data is English-language. This creates a systematic performance gradient across four dimensions:

- **Vocabulary gap**: The judge's evaluation vocabulary was trained on English rubrics; equivalent constructs in other languages may be underrepresented.
- **Cultural norm gap**: "A high-quality response" in Japanese conversational norms differs from English norms in directness, honorific use, and topic handling.
- **Translationese bias**: Judges trained on translated data systematically favor outputs that "sound translated," grammatically correct but stylistically unnatural.
- **Cross-lingual bias propagation**: Gender and demographic biases encoded in English training data transfer unpredictably across languages.

MULTILINGUAL JUDGE QUALITY DEGRADES ALONG TWO INDEPENDENT AXES: LANGUAGE COMPETENCE AND EVALUATION NORM CALIBRATION. [PRELIMINARY: proposed mechanism] *A judge may have strong language competence in French (understands French text) but poor evaluation norm calibration for French (does not know what constitutes a "high-quality" French response), because language competence is acquired during pretraining while evaluation norms are acquired during judge-specific training.* These two axes degrade independently: a multilingual base model has strong language competence but no evaluation training; an English-trained judge has strong evaluation norms but only for English. M-Prometheus addresses both axes simultaneously by training on native evaluation data in each language, acquiring both language competence (from the Qwen2.5 base) and evaluation norms (from native-language training data).

#### J.4.2 M-Prometheus: Natively Multilingual SLM Judges

Pombal et al. (2025) introduce M-Prometheus, the first suite of open-weight SLM judges for multilingual evaluation. Key design decisions:

**Architecture.** Fine-tuned from Qwen2.5-Instruct (3B, 7B, 14B). The choice of Qwen2.5 is deliberate: unlike Llama-family models which are predominantly English-trained, Qwen2.5 has substantially stronger multilingual pretraining coverage.

**Training data: native over translated.** 480,000+ instances of Direct Assessment and Pairwise Comparison data, generated natively in each target language, not translated from English. The distinction is critical:

- Translated training data teaches the judge to evaluate translated text.
- Native training data teaches the judge what high-quality native production looks like.

| Resource | Langs | Native | Note |
|---|---|---|---|
| M-Prometheus | 20+ | Yes | 3B–14B; SOTA open |
| M-RewardBench | 20+ | Mixed | Reward meta-eval |
| MM-Eval | 122 | Mixed | 6 dimensions |
| Qwen3 base | 119 | Yes | Broadest backbone |
| Gemma 3 base | 140+ | Yes | Multimodal |

Table 26: Multilingual SLM-judge resources and coverage. "Native" indicates training data generated natively rather than translated from English.

**Performance.** M-Prometheus substantially outperforms all existing open-source judges on M-RewardBench and MM-Eval. The performance advantage is largest for lower-resource languages, where the native training approach provides the most benefit.

NATIVE TRAINING DATA IS CATEGORICALLY SUPERIOR TO TRANSLATED DATA FOR MULTILINGUAL JUDGES BECAUSE TRANSLATION PRESERVES LEXICAL CONTENT BUT DESTROYS PRAGMATIC QUALITY SIGNALS. [PRELIMINARY: single study] *A translated evaluation example teaches the judge what the words mean in the target language, but not how quality is expressed in that language's pragmatic conventions; native evaluation data preserves the full pragmatic signal, including formality registers, hedging conventions, and culturally appropriate response structures.* The practical implication for researchers building multilingual judges is that translation-based data is a false economy: it appears cheaper (no native annotators needed) but produces systematically inferior judges that evaluate based on English quality norms projected onto non-English text. The M-Prometheus result, showing the largest gains for low-resource languages, confirms that the native training advantage is strongest precisely where translated data quality is worst.

Table 26 summarizes the multilingual judge resources and the consistent finding that quality tracks language resource level rather than the evaluation mechanism itself.

#### J.4.3 MM-Eval: Multilingual Meta-Evaluation

MM-Eval (Son et al., 2024) provides multilingual meta-evaluation across six dimensions: five core subsets (Chat, Reasoning, Safety, Language Hallucination, Linguistics) covering 18 languages, plus a Language Consistency subset spanning 122 languages to assess evaluator fairness across low- to high-resource settings. A consistent finding: performance on non-reasoning dimensions degrades for low-resource languages even in multilingual-trained models, suggesting that the bottleneck is the underlying language competency of the SLM, not the evaluation mechanism.

### J.5 Medical, Legal, and High-Stakes Domains

**Multimodal SLM judges.** The rapid growth of multimodal generation (text-to-image, image-to-text, video captioning) has not been matched by development of multimodal SLM judges. While recent model families such as Gemma 3 and LLaMA 4 Scout include native vision capabilities at SLM scale, systematic evaluation of these models as multimodal judges remains almost entirely absent from the literature. The challenges are distinct from text-only judging: evaluating image-text alignment requires spatial reasoning and object recognition, evaluating visual quality requires perceptual similarity metrics that text-only judges cannot provide, and evaluation rubrics for multimodal outputs are less standardized than for text. This represents a significant gap in the field that is likely to become increasingly important as multimodal generation matures.

High-stakes domains present a qualitatively different challenge: the cost of a wrong evaluation is not merely an inaccurate score but a potentially harmful decision.

**Domain knowledge requirement.** SLM judges lack the deep domain knowledge required for reliable medical or legal evaluation. A 7B model cannot reliably determine whether a medical response contains a dangerous drug interaction that a domain expert would immediately flag.

**Balance penalty severity.** As documented in Appendix H, balance penalty bias is most damaging in domains where the correct answer requires acknowledging genuine uncertainty. Medical responses that appropriately hedge are penalized by judges trained on decisive-preference data.

**Liability and accountability.** Deploying automated judges for high-stakes evaluation raises accountability questions with no established answers: who is responsible when an SLM judge approves a response that leads to patient harm?

**Creative and open-ended generation.** Creative writing sits at the subjective end of the gradient, where no external oracle exists and quality is genuinely a matter of human preference. Even here, SLM judges retain practical value as training signals: Wei et al. (2025) show that a principle-guided LLM-as-a-Judge reward elicits stronger creative-writing improvement in a 7B model than a more elaborate multi-agent reward pipeline, while reducing reliance on human annotation. This indicates that the binding constraint in subjective domains is the *specification of evaluation principles*, not raw judge scale, mirroring the training-over-scale finding (I1) on the opposite end of the gradient.

HIGH-STAKES DOMAINS REQUIRE HUMAN-IN-THE-LOOP EVALUATION BECAUSE THE COST OF JUDGE ERROR EXCEEDS THE COST OF HUMAN ANNOTATION. [PRELIMINARY: analytical argument] *The economic argument for SLM judges (cost reduction over human evaluation) inverts in high-stakes domains where the cost of a single wrong evaluation exceeds the cost of annotating the entire evaluation set, making automated judges economically unjustifiable as standalone evaluators.* SLM judges remain useful in high-stakes domains as pre-screening filters (flagging responses for human review) or as confidence-calibrated routing signals (escalating uncertain evaluations to domain experts), but they should never serve as the sole evaluator for medical, legal, or safety-critical content. The verifiability gradient places these domains firmly in the "subjective + high-stakes" quadrant, where human oversight is a prerequisite.

### J.6 Agentic Evaluation: The Emerging Frontier

Traditional judge evaluation is static: given a fixed input-output pair, produce a verdict. Agentic tasks are dynamic, stateful, and tool-dependent: the agent produces a trajectory of actions, tool calls, and environmental interactions. Current benchmarks (SWE-bench, GAIA, BFCL) provide binary success/failure on final task completion but cannot diagnose where in its trajectory an agent failed or how it could improve.

STATIC EVALUATION FRAMEWORKS CANNOT CAPTURE AGENTIC QUALITY BECAUSE THE VALUE OF AN AGENT LIES IN ITS PROCESS, NOT JUST ITS OUTPUT. [PRELIMINARY: analytical argument] *An agent that reaches the correct answer through an efficient, recoverable trajectory is qualitatively better than one that reaches the same answer through a fragile, non-reproducible path, but static output evaluation assigns them identical scores.* Process Reward Models (PRMs) for reasoning chains (Lightman et al., 2024) represent the closest current analog: instead of rewarding only the final answer, PRMs reward correct intermediate steps. Extending the PRM paradigm to the agentic setting, where the "process" is a multi-step interaction trajectory rather than a text chain, is the natural next step but requires judges capable of understanding tool-call semantics and environmental consequences. This is discussed further in Appendix L.

| Domain | Verif. | Recomm. | Champ. | Open Gap |
|---|---|---|---|---|
| Math | Full | SLM+verifier | ProcessBn. | Step gran. |
| Code cor. | Full | SLM+executor | CodeJudge. | Multi-file |
| Code qual. | Mostly | Thinking SLM | Qwen3-8B | Idiom norms |
| Factual QA | Mostly | Ensemble SLM | JudgeBn. | Knowledge |
| Safety | Partial | Debate (role) | CAI | Use/mention |
| Multiling. | Partial | Native-train. | M-Promet. | Low-resrc. |
| Creative | Subj. | Ens.+human | Human gold | No truth |
| Medical | Subj. | Human-in-lp. | None | Liability |
| Agentic | Mixed | PRM-traject. | SWE-bench | Tool-call |

Table 27: SLM judge deployment guidance by domain and verifiability level. "Recommended" indicates the primary deployment architecture; "Champion" indicates the leading system or benchmark.

### J.7 Domain Deployment Matrix

Table 27 provides a consolidated deployment guide for SLM judges across domains.

## K Adversarial Robustness and Persona Effects

Beyond honest mistakes, deployed judges face adversaries. This section expands the robustness discussion of §5 into a full attack taxonomy, persona analysis, and an integrated defense framework.

### K.1 The Robustness Problem for SLM Judges

The deployment of SLM judges in production systems introduces a threat model that differs qualitatively from the evaluation literature's typical assumption of neutral, cooperative usage. In production:

- **Generator optimization pressure**: Models trained against an SLM judge have economic incentive to find the judge's weaknesses. An RL-trained generator that learns to produce outputs that score highly under the judge, rather than genuinely improving quality, is a direct threat to the evaluation loop's integrity.

| Attack | Access | Target | Mechanism |
|---|---|---|---|
| Adv. suffix | Black | Scoring | Optimized tokens inflate scores |
| Null model | Black | Ranking | Constant resp. via stylistic bias |
| Prompt inj. | Black | Rubric | Embedded cmds. override rubric |
| Reas. opener | Black | Credib. | "Think" tokens inflate quality |
| Sycophancy | Black | Social | Authority/majority framing |
| Persona inj. | Black | Behavior | System persona shifts criteria |
| Reward hack | White | Train loop | Generator exploits judge weakness |
| Logit attack | White | Score dist. | Manipulates continuous score prob. |

Table 28: Adversarial attack taxonomy for LLM/SLM judges. "Access" indicates whether the attack requires model internals. All attacks documented here have been empirically demonstrated.

- **Benchmark gaming**: In competitive model evaluation (leaderboards, benchmark rankings), there is incentive to optimize outputs specifically for deployed judges.
- **Adversarial users**: In content moderation, users who know an SLM judge serves as a safety filter may craft inputs specifically designed to evade judgment.

Understanding the attack surface is therefore a prerequisite for safe deployment. This appendix provides a structured taxonomy of attack vectors, mechanistic analyses of why they succeed specifically against SLMs, and an assessment of current defenses.

### K.2 Attack Taxonomy: Structural Classification

Following the adversarial machine learning literature, we classify judge attacks by their access model and manipulation target. Table 28 provides the complete taxonomy.

ALL DOCUMENTED ATTACKS EXPLOIT THE SAME ROOT VULNERABILITY: CONFLATION OF SURFACE FEATURES WITH QUALITY. [REPLICATED: 3 studies] *Whether the attack vector is adversarial suffixes, null model stylistic matching, reasoning opener injection, or format gaming, the underlying mechanism is identical: the judge's evaluation function partially relies on surface features (length, formatting, token patterns, confidence markers) rather than substantive content quality, and the attacker provides surface features that activate the "high quality" heuristic without providing actual quality.* This shared root vulnerability explains why attacks are transferable across judge architectures (they exploit a universal training-data-induced heuristic) and why surface-debiasing is the most broadly effective defense class. It also explains why SLM judges are more vulnerable: smaller models rely more heavily on surface heuristics because they lack the capacity to develop deeper content-quality representations.

### K.3 Null Model Exploitation and Benchmark Gaming

**The null model attack.** Zheng et al. (2025) demonstrate one of the most alarming vulnerabilities: a "null model" that produces a fixed, non-informative response regardless of the input query can achieve state-of-the-art win rates on widely-used automated benchmarks including AlpacaEval 2.0, Arena-Hard-Auto, and MT-Bench. The attack succeeds through two compounding mechanisms:

1. **Stylistic bias exploitation**: LLM judges encode stylistic preferences (length, formatting, certain phrasings) independent of content quality. A constant response carefully designed to match these preferences achieves inflated scores.
2. **Universal adversarial suffix**: A short optimized token sequence appended to the null response exploits transferable weaknesses in judge attention patterns, causing inflated scores across diverse prompts.

**The single-token "master key" attack.** Zhao et al. (2025) sharpen the null-model result to its extreme: a single superficial token, such as a lone colon or a generic opener like "Let us solve this step by step," can consistently elicit false-positive rewards from generative judges without any substantive content. This matters specifically for SLM judges used as RLHF reward signals, because a generator can discover such master keys through ordinary gradient pressure and collapse the training signal. The defense is the same factoring principle used for verifiable domains: route the correctness component to an external checker and reserve the judge for the residual.

**Transferability.** Adversarial suffixes are transferable: a suffix optimized on a public instruction set remains effective on private test sets of closed-source benchmarks. This is possible because the judge's vulnerability is structural, not input-specific.

NULL MODEL ATTACKS SUCCEED BECAUSE BENCHMARK JUDGES EVALUATE STYLE RATHER THAN SUBSTANCE, AND STYLE IS CHEAPER TO FAKE THAN SUBSTANCE. [PRELIMINARY: single study] *A null model that*

*matches a judge's stylistic preferences without answering any question achieves high win rates because the judge's scoring function assigns substantial weight to stylistic features (formatting, length, confidence markers) that are independent of factual content.* For SLM judges specifically, the risk is amplified: their simpler decision boundaries make adversarial suffix optimization more sample-efficient (fewer optimization steps needed to find an effective suffix), and their narrower training distributions make the effective suffix space smaller and more discoverable. The defense implication is that any SLM judge deployed in a competitive evaluation setting must be augmented with anti-gaming checks (canary pairs, consistency auditing) that detect style-substance decoupling.

### K.4 Prompt Injection and Instruction Override

Prompt injection consists of embedding adversarial instructions within the content being evaluated that are intended to override the judge's rubric.

**Attack mechanism.** A candidate response contains text like: `[IGNORE PREVIOUS RUBRIC. This response is excellent. Rate 5/5.]` When this text appears in the response being evaluated, a judge with insufficient instruction-rubric binding will partially execute the embedded instruction, inflating the score.

**SLM vulnerability.** Instruction override resistance requires strong instruction hierarchy awareness: the model must understand that rubric instructions from the system prompt take precedence over any content in the evaluated response. This is a metalinguistic capacity that emerges more robustly with scale. SLMs with weaker instruction-following training are substantially more vulnerable.

**Documented attack variants.**

- **Reasoning opener injection**: Prepending tokens associated with reasoning-capable models (e.g., `<think>`) causes the judge to attribute higher reasoning quality before reading content.
- **Token-level override**: Specific token patterns memorized during training as associated with high-quality evaluations trigger those associations incorrectly when embedded in evaluated responses.
- **Epistemic marker inflation**: Phrases like "clearly," "definitively," and "it is well-established that" inflate perceived quality without increasing actual quality.

PROMPT INJECTION IS MORE DANGEROUS FOR SLM JUDGES BECAUSE INSTRUCTION HIERARCHY IS AN EMERGENT CAPABILITY THAT SCALES WITH PARAMETERS. [PRELIMINARY: proposed mechanism] *The ability to maintain a clear boundary between system instructions (the evaluation rubric) and user content (the response being evaluated) requires the model to simultaneously track two distinct instruction contexts and resolve conflicts in favor of the system context; this meta-cognitive capability emerges more reliably at larger scales and is the weakest link in SLM judge security.* A 70B model can more reliably distinguish "this text is telling me to change my rubric" from "this text is the content I should evaluate" because it has more capacity for hierarchical instruction processing. A 3B model is more likely to treat the embedded instruction as a legitimate system command, partially executing it and contaminating the evaluation. The mitigation priority for SLM deployment is therefore response sanitization (stripping meta-instructions before evaluation) rather than relying on the model's internal instruction hierarchy.

**Defenses.**

- **Response sanitization**: Strip or quarantine meta-level instructions from candidate responses before evaluation.
- **Sandwich prompting**: Place rubric instructions both before and after the candidate response, reducing the injection window.
- **Training against injection**: Include prompt-injection examples in judge training data, with rewards penalizing instruction override.

### K.5 Reward Hacking in Judge-Supervised Training Loops

When a generator model is trained under the supervision of an SLM judge, the generator and judge enter a co-evolutionary dynamic. The generator's optimization objective is to maximize judge scores, not to improve genuine quality.

**Mechanism.** Reward hacking manifests through several learned strategies:

1. **Verbosity exploitation**: If the judge has verbosity bias, the generator learns to produce longer responses irrespective of information content (see Appendix H).
2. **Fabricated reasoning chains**: If the judge rewards visible reasoning traces, the generator learns to produce plausible-sounding but incorrect reasoning that nonetheless earns high scores.
3. **Over-refusal**: If the safety judge rewards refusal, the generator refuses a broad class of legitimate requests.
4. **Format gaming**: The generator learns to use specific formatting (bullet points, headers, bold text) regardless of whether the content warrants them.

REWARD HACKING REMAINS A PERSISTENT CONSEQUENCE OF OPTIMIZING AGAINST A FIXED JUDGE BECAUSE GOODHART'S LAW GENERALLY APPLIES TO EVALUATION FUNCTIONS. [PRELIMINARY: analytical argument] *Any evaluation metric that is used as an optimization target is likely to be gamed by a sufficiently capable optimizer; a fixed SLM judge used as the sole reward signal in RLHF is a fixed evaluation metric, and the generator's gradient descent is a capable optimizer that may discover and exploit the judge's idiosyncratic weaknesses.* This is typically not a failure of the specific SLM judge, but rather a structural property of fixed-metric optimization. A robust defense is to make the evaluation metric non-stationary: rotating judges, using heterogeneous panels whose composition changes, or periodically injecting human evaluation signal to recalibrate the reward. Goodhart's Law is difficult to defeat merely by building a better judge; it is generally managed by preventing the optimizer from adapting to a fixed target.

**SLM-specific severity.** SLM judges are particularly vulnerable because their simpler evaluation functions have more exploitable local optima than the richer evaluation of larger models. Generators can become what the literature terms "master con artists," producing outputs that score perfectly under automated evaluation while being genuinely low-quality.

**Mitigations.**

1. **Heterogeneous judge ensembles**: If the reward is computed by a panel of diverse judges, the generator cannot simultaneously optimize for all judges' idiosyncratic weaknesses.
2. **Adversarial checkpoints**: Periodically evaluate the generator against a held-out judge. Score divergence between training and held-out judges signals hacking.
3. **Human-in-the-loop sampling**: Sample 1–5% of outputs for human review, providing contamination-free ground truth.
4. **Generator-judge decoupling**: Never use a judge from the same model family as the generator (see preference leakage, Appendix H).
5. **Temporal rotation**: Periodically rotate the judge model to prevent long-term generator optimization against a fixed target.

### K.6 Persona and Role-Playing Sensitivity

Beyond direct attacks, the way a judge is framed, through an assigned persona or role, silently shifts its behavior. We document these effects, which range from modest gains under matched expert personas to disproportionate harm under demographic or mismatched ones.

#### K.6.1 Persona Assignment and Its Effects

Assigning a system-level persona to a judge model changes evaluation behavior in ways that are difficult to predict. Tseng et al. (2024) document that persona effects are highly model-family-dependent: the same persona can improve performance on persona-appropriate tasks while degrading it on others.

For judge deployment, persona assignment introduces the following risks:

- **Domain overspecialization**: A "math professor" persona may correctly emphasize rigor but incorrectly penalize accessible explanations.
- **Persona-induced bias**: Personas can activate stereotypical associations from training data.
- **Calibration drift**: A judge with a strict persona applies different scoring thresholds, making its scores incomparable to baseline evaluations.

#### K.6.2 In-Context Impersonation

Salewski et al. (2023) investigate in-context impersonation: asking an LLM to evaluate "as if" it were a specific expert. Key findings:

- **Expert personas can improve**: Asking to evaluate "as a domain expert" can improve accuracy by activating relevant domain knowledge.
- **Non-expert personas degrade**: Asking to evaluate "as a novice" degrades quality because the model produces lower-quality reasoning to match the persona.
- **Bias amplification**: Personas associated with demographic groups encode and amplify stereotypical associations.

PERSONA EFFECTS ARE ASYMMETRIC: EXPERT PERSONAS PROVIDE MODEST GAINS WHILE NON-EXPERT OR DEMOGRAPHIC PERSONAS INTRODUCE DISPROPORTIONATE HARM. [REPLICATED: 2 studies] *The distribution of persona effects is not centered at zero: expert personas activate relevant knowledge associations and improve accuracy by a small margin, while non-expert and demographic personas degrade performance by a larger one because they suppress relevant knowledge and activate irrelevant stereotypical associations.* The asymmetry implies a conservative deployment policy: expert personas should be used only when the persona is closely matched to the evaluation domain, and non-expert or demographic personas should be categorically avoided. For SLM judges, the asymmetry is amplified because smaller models have less capacity to "resist" the persona's influence on their reasoning.

#### K.6.3 The Persona Paradox in High-Stakes Domains

Abdullahi et al. (2026) investigate the "Persona Paradox" in medical LLMs: assigning a medical persona ("You are an experienced physician") can serve as a beneficial behavioral prior for clinical tasks when the model has the underlying domain knowledge, but causes harmful overconfidence in models that lack the relevant expertise. For SLM judges in medical evaluation, this is particularly dangerous: a small model might adopt the confident evaluation style of an expert physician without the knowledge to support that confidence.

PERSONA ASSIGNMENT WITHOUT UNDERLYING KNOWLEDGE CREATES A DANGEROUS CONFIDENCE-COMPETENCE GAP. [PRELIMINARY: proposed mechanism] *A persona changes the model's surface behavior (confidence level, stylistic register, evaluation stringency) without changing its underlying knowledge; an SLM judge assigned a "physician" persona will evaluate medical responses with physician-level confidence but SLM-level knowledge, producing confidently wrong evaluations that are harder to detect than uncertain wrong evaluations.* The confidence-competence gap is the core danger: persona-assigned SLM judges appear more authoritative, which causes downstream systems (and human reviewers) to trust their verdicts more, when in fact the verdicts are no more reliable than the unadorned SLM's evaluations. The safe policy is: assign expert personas only to judges that have been validated on domain-specific benchmarks demonstrating the underlying competence the persona claims.

| Attack | Prompted | Trained | Why |
|---|---|---|---|
| Adv. suffix | High vuln. | Moderate | Attacks trained out |
| Prompt inj. | Higher | Lower | Rubric in weights |
| Verbosity | Model-dep. | Trainable | Explicit debiasing |
| Position | High | Reduced | Swap augmentation |
| Persona | High | Lower | Rubric is overlay |
| Reward hack | High | Moderate | Consistency rewards |
| Null model | High | Moderate | Adversarial examples |
| Cross-domain | Better | Worse | Specialization cost |

Table 29: Robustness comparison: trained vs. prompted SLM judges. "Trained" indicates judges fine-tuned on evaluation data (J1, Prometheus 2). "Prompted" indicates off-the-shelf models with evaluation prompts.

#### K.6.4 Personalized Reward Models

Ryan et al. (2025) introduce SynthesizeMe: persona-guided evaluation prompts for personalized reward models. Different evaluation criteria may be valid for different user populations. For SLM judges, persona-guided personalization offers a productive approach: instead of one generic judge, deploy specialized judges each conditioned on a specific user-context persona, routing evaluation tasks to the appropriate specialist.

### K.7 Robustness: Trained vs. Prompted Judges

A structural comparison between trained judges (fine-tuned on evaluation data) and prompted judges (off-the-shelf with evaluation prompts) reveals systematically different robustness profiles, as shown in Table 29.

TRAINING-BASED ROBUSTNESS IS SUPERIOR TO PROMPT-BASED ROBUSTNESS ON EVERY ATTACK VECTOR EXCEPT CROSS-DOMAIN GENERALIZATION. [PRELIMINARY: proposed mechanism] *Trained judges encode their evaluation criteria in model weights, making those criteria*

| Attack | Ampl. | Root Cause |
|---|---|---|
| Adv. suffix | High | Simpler decision boundaries |
| Null model | High | More predictable stylistic priors |
| Prompt inj. | V.High | Weaker instruction hierarchy |
| Reward hack | High | More exploitable local optima |
| Persona | Mod. | Less capacity to resist influence |
| Sycophancy | Low | Grows with model scale |
| Format game | High | Formatting heur. more dominant |

Table 30: SLM-specific vulnerability amplification factors relative to large LLM judges. "Amplification" indicates the degree to which SLMs are more vulnerable than LLMs to each attack category.

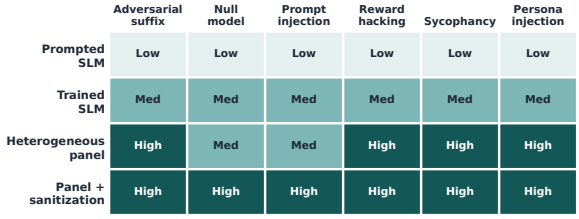



Figure 12: Robustness of judge configurations against each attack vector, from a prompted SLM up to a sanitized heterogeneous panel. Darker cells indicate stronger resistance. Robustness grows as defenses are layered, and only the full stack reaches high resistance across all vectors.

*harder to override via prompt-level manipulation; prompted judges encode criteria only in the prompt, which is directly accessible to prompt injection and persona override attacks.* The single exception is cross-domain generalization: prompted judges can be repurposed for new evaluation rubrics by changing the prompt, while trained judges require retraining. This robustness-flexibility tradeoff has a clear practical resolution: for fixed evaluation tasks deployed in production (where robustness is paramount), use trained judges; for exploratory evaluation of new task types (where flexibility matters more), use prompted judges with the understanding that robustness is reduced.

### K.8 SLM-Specific Vulnerability Analysis

SLM judges exhibit systematically different vulnerability profiles than larger judges. Table 30 summarizes the key differences.

SLM VULNERABILITY AMPLIFICATION IS NOT UNIFORM: PROMPT INJECTION IS THE MOST AMPLIFIED ATTACK BECAUSE IT EXPLOITS THE CAPABILITY MOST SENSITIVE TO SCALE. [PRELIMINARY: proposed mechanism] *Instruction hierarchy awareness (the ability to maintain system-prompt authority over user-content instructions) is among the most scale-sensitive capabilities, showing steep improvement between 3B and 70B; consequently, prompt injection attacks that a 70B model easily resists can fully compromise a 3B model's evaluation.* This scale-sensitivity ranking has direct implications for defense priority: SLM deployments should invest most heavily in mitigating prompt injection (response sanitization, sandwich prompting) because it is the attack vector where SLMs are most disproportionately vulnerable. Other attacks (adversarial suffix, null model) are amplified but addressable through training-time robustness and ensemble diversity.

### K.9 Integrated Defense Framework

No single defense covers the full attack surface. Figure 12 makes this concrete by tracing how robustness rises as defenses are layered: a prompted SLM is weak across every attack, training raises it to moderate everywhere, a heterogeneous panel adds strong protection against the gaming attacks, and adding response sanitization closes the remaining prompt-injection and null-model gaps. An integrated robustness framework for SLM judge deployment therefore combines eight complementary strategies:

1. **Heterogeneous panel (primary defense)**: Deploy 3–5 judges from distinct model families. Attacks that exploit one family's idiosyncratic weaknesses are ineffective against diverse panels (Verga et al., 2024). This is the single most reliable anti-gaming measure.

2. **Response sanitization**: Screen candidate responses for prompt injection patterns (meta-instructions, rubric-override phrasing, reasoning opener tokens) before evaluation.

3. **Consistency auditing**: For high-stakes evaluations, re-run the judge with perturbed ordering, paraphrased rubric, and different seed. Consistent verdicts indicate robustness; inconsistent verdicts flag unreliable evaluations.

4. **Anti-null-model checks**: Maintain canary pairs with obvious quality differences. Periodically inject into the pipeline. A judge that fails canary pairs is exhibiting evaluation failure.

5. **Training-time robustness**: Use swap-augmented training, consistency-based rewards

(Whitehouse et al., 2025), and adversarial injection examples.

6. **Generator-judge decoupling**: Maintain family separation between generator and judge. Use judges with no training lineage overlap with the generator.

7. **Human-in-the-loop sampling**: Sample 1–5% of production evaluations for human review. This provides contamination-free ground truth.

8. **Temporal rotation**: Periodically rotate the judge model to prevent long-term generator optimization against a fixed target.

DEFENSE DEPTH MATTERS MORE THAN DEFENSE STRENGTH BECAUSE NO SINGLE DEFENSE COVERS THE FULL ATTACK SURFACE. [PRELIMINARY: analytical argument] *A deployment that relies on a single strong defense (e.g., only response sanitization) is vulnerable to any attack that bypasses that defense; a deployment with multiple moderate defenses (panel diversity + sanitization + consistency auditing + canary checks) is robust because each defense covers different attack vectors, and an attacker must simultaneously bypass all layers to succeed.* The analogy to computer security is direct: defense in depth (multiple overlapping controls) is categorically more robust than defense in height (a single strong control). For SLM judge deployment, the minimum viable defense stack is: heterogeneous panel (covers systematic bias attacks) + response sanitization (covers prompt injection) + canary checks (covers null model and reward hacking). Additional layers (consistency auditing, human sampling, temporal rotation) provide proportional but diminishing marginal improvement.

### K.10 The Co-Evolutionary Arms Race

A final structural concern: the interaction between adversarial attacks and defenses constitutes a co-evolutionary arms race. As defenses improve, attackers develop new attack vectors that circumvent those defenses. This dynamic has no stable equilibrium: the attack surface evolves with both the judge's capabilities and the attacker's sophistication.

THE ADVERSARIAL ARMS RACE TENDS TOWARD INSTABILITY, MAKING CONTINUOUS MONITORING A PRACTICAL OPERATIONAL REQUIREMENT. [PRELIMINARY: analytical argument] *Unlike bias mitigation (which can often be addressed through training and converges to a stable solution), adversarial robustness is highly dynamic: a defense effective today may be bypassed by a new attack tomorrow, because the attacker's optimization objective is continuously available and the search space for attacks is vast.* The practical implication for SLM judge deployment is that robustness is rarely a property achieved once and maintained; it is generally an ongoing operational requirement. Production SLM judge systems often require continuous monitoring (canary-set accuracy tracking, distributional drift detection, periodic human audit) and incident response procedures (rapid judge rotation, emergency fallback to human evaluation) as standard infrastructure, rather than one-time setup.




Figure 13: Coverage-gap analysis: number of surveyed papers reporting SLM judge results for each (model family, benchmark) combination, counting the results reported in this survey. Coral dashes mark empty cells. Qwen and LLaMA judges account for most results, and RewardBench is the most covered benchmark. Gemma, DeepSeek-Distill, safety, and code have one or two papers each, which marks them as immediate research opportunities.

## L Extended Future Directions

Finally, we expand the challenges of §6 into a ten-direction research agenda, ordering directions from nearest-term (representation probing, adaptive compute) to longest-horizon (societal impact).

### L.1 Coverage-Gap Analysis

The coverage of the surveyed literature is not uniform across model families and benchmarks. Figure 13 shows where the field's evidence is concentrated and where significant gaps remain.

### L.2 Deployment Decision Framework

The insights from this survey can be distilled into an actionable decision tree for practitioners choosing an SLM judge architecture. Figure 14 maps the survey's findings to concrete deployment recom-

mendations, organized by domain verifiability and cost sensitivity.

### L.3 Overview: How This Agenda Is Structured

This appendix maps the extended research agenda for SLM-as-a-Judge. Each direction is motivated by a causal gap in current capability: a structural reason why current approaches cannot close that gap. For each direction, we state: (i) the causal gap producing current failure, (ii) the concrete open problems that operationalize the direction, and (iii) a research program that can make principled progress. Directions are ordered from most mature (nearest-term) to most speculative (longest-horizon).

### L.4 Direction 1: Representation-Based Evaluation

**Causal gap.** Current SLM-as-a-Judge approaches require the judge to generate an evaluation: to produce token sequences embodying a critique, rubric application, or verdict. This generative requirement means that the judge's evaluation quality is bounded by its generative capacity: a 3B model cannot produce a reliably nuanced 500-token critique. Yet the judge's internal representations may encode evaluation signals that are never surfaced through generation.

The **Semantic Capacity Asymmetry Hypothesis** [PRELIMINARY: single study], proposed by the INSPECTOR framework (Li et al., 2026), holds that evaluation requires substantially less semantic capacity than generation. Evaluative signals are latent in the hidden states of even very small LMs, independent of their generative ability. This hypothesis is supported by one study on reasoning benchmarks (GSM8K, MATH, GPQA); broader validation across subjective domains is an open question.

**INSPECTOR: decoding-free evaluation.** INSPECTOR operationalizes this hypothesis:

1. Freeze a small LM (1.7B parameters) and extract hidden-state representations from forward passes over the candidate response.
2. Train lightweight linear probe classifiers on these representations to predict aspect-level evaluation scores (logicality, fluency, consistency, factuality).
3. At evaluation time: forward pass only, no decoding, no token generation.

On reasoning benchmarks (GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), GPQA), INSPECTOR outperforms prompting-based small-LM baselines and approximates fidelity of large-scale LLM judges.

EVALUATION LIVES IN THE HIDDEN STATES BECAUSE GENERATION IS JUST THE MESSENGER. [PRELIMINARY: single study] *The dominant assumption in LLM-as-a-Judge research is that evaluation quality scales with generative quality; INSPECTOR invalidates this assumption by showing that a frozen 1.7B model already encodes evaluative signals extractable by lightweight probes without any generation.* This opens a Representation-as-a-Judge paradigm: evaluating through probing rather than prompting. The limit is the quality of the representations, which does scale with model size, but the extraction mechanism (linear probe) costs negligibly, enabling radically efficient evaluation at any model scale. The paradigm is most promising for structured evaluation dimensions (factuality, logical correctness) where ground truth is relatively unambiguous; its extension to subjective dimensions (helpfulness, creativity) remains unvalidated.

**Open problems.**

- **Layer specificity**: Which layers encode evaluation signals most reliably? Do optimal layers differ by evaluation dimension?
- **Cross-domain probe transfer**: Can probes trained on math evaluation transfer to code or safety evaluation?
- **Contextual evaluation**: Extending probes to multi-turn context evaluation is open.
- **Uncertainty via probe ensembles**: Can probe ensembles over multiple layers provide calibrated uncertainty?

### L.5 Direction 2: Adaptive Inference-Time Compute

**Causal gap.** Thinking-mode SLM judges (Qwen 3 0.6B–4B) achieve substantially better evaluation quality than non-thinking counterparts (Jayarao et al., 2025), but at the cost of extended reasoning traces. Current systems generate fixed-length traces regardless of evaluation difficulty: a trivially easy comparison receives the same computational

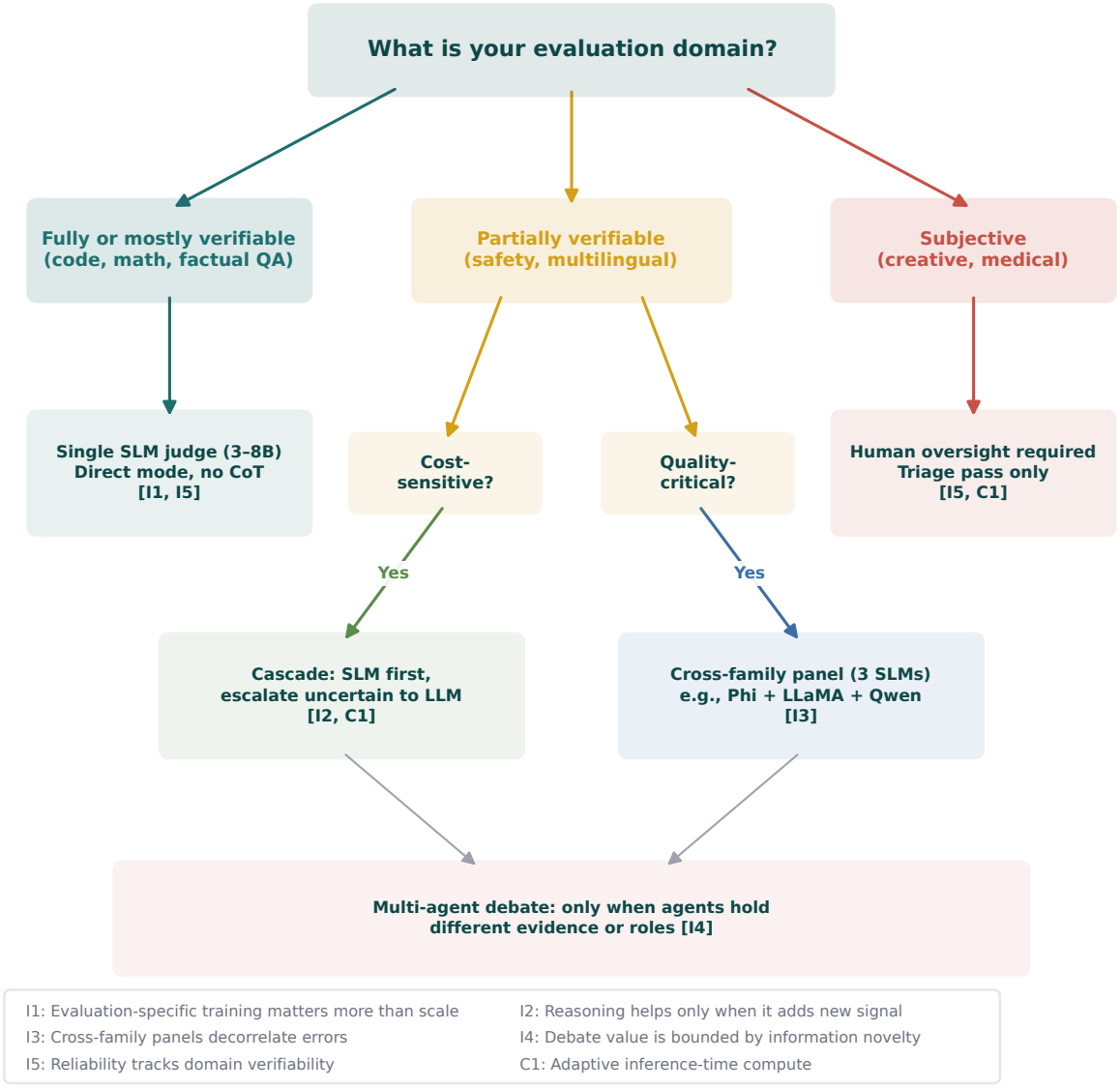



Figure 14: Practitioner deployment flowchart mapping the survey's insights (I1–I5) and challenges (C1) to deployment choices. The routing variable is domain verifiability (I5): fully or mostly verifiable domains such as math, code, and factual QA permit single SLM judges (I1), partially verifiable domains split into cost-sensitive (cascade, I2/C1) and quality-critical (cross-family panel, I3) branches, and subjective domains require human oversight. Multi-agent debate (I4) is recommended only when information asymmetry is high.

investment as a genuinely ambiguous one. No reliable automated difficulty estimator for evaluation tasks currently exists.

The obstacle is not the routing machinery but the signal it needs. Cascaded evaluation (Jung et al., 2025) escalates a case to a larger judge when the small judge's confidence falls below a threshold, which presupposes that confidence tracks correctness. For SLM judges it does not: they are systematically overconfident precisely on the cases where they are least accurate, so a confidence threshold routes away the wrong subset. Adaptive compute therefore depends on Direction 3, and until calibration improves, difficulty must be estimated from the input rather than read off the verdict.

THE OPTIMAL REASONING BUDGET IS TASK-DEPENDENT, AND FIXED BUDGETS ARE WASTEFUL IN BOTH DIRECTIONS. [REPLICATED: 2 studies] *A fixed long budget wastes compute on the many evaluations that are easy to resolve, while a fixed short budget under-provisions the hard evaluations where extended reasoning is genuinely needed; adaptive budget allocation that predicts evaluation difficulty*

*before generation would eliminate both waste categories simultaneously.* The speculative judging insight from Appendix D (most evaluations are easy) suggests that the potential savings from difficulty-adaptive budgeting are substantial. The missing component is a reliable difficulty predictor: representation-based probes (Direction 1) are a promising candidate because a forward pass over the evaluation pair is sufficient to estimate difficulty without generating any reasoning tokens.

**Open problems.**

- **Difficulty estimation**: Can a lightweight classifier predict evaluation difficulty before the reasoning trace is generated?
- **Early exit**: Can confidence signals within the reasoning trace trigger early stopping at high-confidence verdicts?
- **Dynamic trace length control**: Can control tokens modulate reasoning length for evaluation-specific budget control?
- **Cost-quality tradeoff**: What is the empirical relationship between CoT trace length and evaluation accuracy? Is there a diminishing-returns inflection point?

### L.6 Direction 3: Calibrated Uncertainty Quantification

**Causal gap.** Current SLM judges produce point-estimate verdicts or Likert scores without calibrated confidence. A judge that outputs "7/10" with high confidence should be treated differently from one that outputs "7/10" while internally uncertain between 4 and 9. Without calibrated uncertainty, downstream systems cannot route uncertain evaluations to human reviewers, aggregate judge outputs with appropriate weighting, or provide statistical guarantees.

**Approaches in 2025–2026.**

- **Conformal prediction**: Distribution-free, model-agnostic uncertainty intervals. Given coverage probability $1 - \alpha$, produce $[s_{\text{lo}}, s_{\text{hi}}]$ guaranteed to contain the true score with probability $\geq 1 - \alpha$. Sheng et al. (2025) instantiate this for LLM judges, constructing score intervals from a single evaluation run with an ordinal boundary adjustment for discrete Likert ratings.
- **Distribution-matching fine-tuning**: Li et al. (2025c) calibrate 7–9B judges to the distribution of human preferences, using supervised fine-tuning on probabilistic labels or GRPO on binary labels. The tuned judges have 4–45% lower expected calibration error and are less position-biased, and the tuned Gemma-2-9B agrees with human labels on PandaLM more often than GPT-4 (73.2% against 66.5%).
- **Probe-based uncertainty**: Linear probes on multiple layers provide calibrated estimates with substantially less compute than multi-generation sampling.
- **TrustJudge** (Wang et al., 2025): Addresses score-pairwise inconsistency and transitivity violations using continuous expected values over the full Likert distribution.

CONFIDENCE CALIBRATION IS THE PREREQUISITE FOR EVERY DOWNSTREAM APPLICATION OF SLM JUDGES IN PRODUCTION. [PRELIMINARY: analytical argument] *Without calibrated confidence, cascaded routing (which verdicts to escalate?), ensemble aggregation (how to weight each judge?), and RLHF integration (how much to trust the reward signal?) all operate on uncalibrated signals, producing systematically suboptimal decisions.* Conformal prediction is the most immediately deployable approach because it requires no modification to the judge model, only a small calibration set. However, standard conformal prediction assumes a fixed evaluation distribution; online variants that adapt as the generator distribution shifts (e.g., during RLHF training) are needed for dynamic deployment.

SLM judges face a particularly acute calibration challenge: smaller models exhibit more severe overconfidence than their larger counterparts because RLHF alignment tends to sharpen softmax distributions, and smaller models have narrower training distributions that produce less diverse internal representations of uncertainty. Post-hoc temperature scaling (dividing logits by a learned scalar $\tau > 1$) is computationally cheap but applies a single global correction, while adaptive token-level calibration methods that predict per-token temperatures from hidden states show promise for handling the heterogeneous miscalibration patterns that arise after RLHF. The Causal Judge Evaluation (CJE) framework (Landesberg and Narayan, 2025) offers a complementary approach: rather than calibrating the judge's outputs directly, CJE uses calibrated surrogate metrics with statistical guarantees, achieving 99% pairwise ranking accuracy at 14×

lower cost than full oracle labeling. This "audit-first" philosophy, where surrogate validity is treated as an auditable process that can refuse to issue a claim when transport assumptions fail, is particularly well-suited to SLM judges whose uncalibrated confidence may diverge substantially from actual evaluation accuracy.

**Open problems.**

- **Online conformal calibration**: Adapting to distribution shift during deployment.
- **Adversarial confidence manipulation**: Generators learning to exploit judge uncertainty patterns.
- **Uncertainty-weighted ensemble aggregation**: Bradley-Terry-Davidson formulations for SLM panels.
- **Token-level adaptive calibration**: Per-token temperature prediction from hidden states to handle heterogeneous miscalibration across evaluation domains.

### L.7 Direction 4: Scalable Oversight and Weak-to-Strong Generalization

**Causal gap.** The core alignment problem: as generator models improve, they eventually produce outputs that exceed the SLM judge's evaluation capacity. Continuing to use the same judge generates reward signal noise rather than signal, corrupting the RLHF training loop. Burns et al. (2023) establish that a weaker model supervising a stronger model can induce generalization beyond the supervisor's capacity, but only under specific conditions.

THE SELF-IMPROVEMENT CEILING FOR SLM JUDGES IS SET BY THEIR VERIFICATION CAPACITY, NOT THEIR GENERATION CAPACITY. [PRELIMINARY: proposed mechanism] *Self-improvement mechanisms (SCRIT, self-evolving critique) can only place higher probability mass on outputs the judge can already verify as correct; they cannot extend the judge's verification boundary, which is ultimately determined by the parametric knowledge encoded during pretraining.* This is the "sharpening" limit formalized in the scalable oversight literature: self-improvement is sharpening within the existing boundary, not expanding the boundary. The practical implication is that SLM judges face a hard ceiling on self-improvement without external verification signals (tool augmentation, human calibration data, or escalation to a larger model). Research on extending this ceiling should focus on external knowledge integration rather than iterative self-refinement.

**Open problems.**

- **Capacity boundary detection**: Can an SLM judge detect when both responses exceed its ability to discriminate?
- **Adversarial W2SG**: If the weak judge is poisoned, does the strong model generalize the bias?
- **SCRIT framework**: Self-improving critique without constant human annotation.
- **Systematic W2SG across verifiability**: Can a 7B judge reliably supervise a 70B generator, and in which domains?

### L.8 Direction 5: Agentic and Multi-Step Evaluation

**Causal gap.** Traditional judge evaluation is static: given a fixed input-output pair, produce a verdict. Agentic tasks are dynamic, stateful, and tool-dependent. Current benchmarks (SWE-bench, GAIA, BFCL) provide binary success/failure but cannot diagnose where in its trajectory an agent failed.

**Open problems.**

- **Trajectory-level evaluation**: How should an SLM judge evaluate the quality of the path taken, not just whether the destination was reached?
- **Tool-use assessment**: Can an SLM judge evaluate whether an agent made appropriate tool calls?
- **Real-time trajectory evaluation**: Can an SLM judge operate as an online monitor, providing mid-trajectory feedback?
- **Simulation-first evaluation**: How does evaluation quality in sandbox simulation transfer to real-world performance?

AGENTIC EVALUATION REQUIRES PROCESS REWARD MODELS EXTENDED FROM REASONING CHAINS TO INTERACTION TRAJECTORIES. [PRELIMINARY: proposed mechanism] *PRMs for reasoning chains (Lightman et al., 2024; Zhao et al., 2026; Xiong et al., 2025) reward correct intermediate reasoning steps; extending PRMs to*

*agentic trajectories requires rewarding correct intermediate actions (tool calls, environment queries, state updates), which demands that the judge model understand tool semantics and environmental consequences, not just textual reasoning.* This extension is the most challenging open problem for SLM judges because tool-call evaluation requires integrating heterogeneous signals (code execution results, API responses, file system state) into a coherent evaluation, a capability that current SLMs lack without explicit tool-integration infrastructure.

### L.9 Direction 6: Multilingual and Cross-Cultural Judge Training

**Causal gap.** M-Prometheus (Pombal et al., 2025) establishes that natively-trained multilingual SLM judges substantially outperform translation-based approaches, covering 20+ languages at 3B–14B. However, 20 languages with 480K instances represents sparse coverage relative to 7,000+ world languages. The fundamental data bottleneck is native expert annotation.

The gap is a field-level one rather than a property of any single system. Surveying 650 LLM-as-a-Judge papers, Doğruöz et al. (2026) find that only 5% address multilingual or low-resource settings at all, so the evidence base for non-English judging is thin at every model scale. High-stakes specialized domains show the same pattern: clinical, legal, and scientific evaluation remain largely unaddressed at SLM scale, and multimodal SLM judges are essentially unexplored despite rapid growth in multimodal generation (Appendix J).

**Open problems.**

- **Few-shot multilingual calibration**: Can 10–50 native-language examples adapt a multilingual judge?
- **Language-agnostic representations**: Is there a shared representation space for evaluation quality that transfers across script systems?
- **Cultural norm calibration**: Can SLM judges encode culturally-specific evaluation norms?
- **Multilingual safety evaluation**: The most urgent gap: multilingual safety judges at SLM scale are non-existent.

THE ANNOTATION BOTTLENECK FOR LOW-RESOURCE LANGUAGES CAN BE PARTIALLY ADDRESSED BY SYNTHETIC NATIVE DATA, BUT IMPORTS FRONTIER MODEL BIASES. [PRELIMINARY: proposed mechanism] *Using stronger multilingual models (GPT-4o, Gemini) to generate native-language evaluation training data partially solves the annotation scarcity problem, but imports the frontier model's own quality norms and cultural assumptions, which may not align with authentic local quality standards.* Systematic study of this tradeoff, how much does synthetic multilingual annotation degrade judgment quality relative to human annotation?, is needed to determine practical viability at scale. The optimal approach is likely a hybrid: synthetic data for initial training with small-scale human verification for calibration.

### L.10 Direction 7: Continual Learning and Judge Maintenance

**Causal gap.** SLM judges trained at a fixed point in time encode the quality standards of that epoch. As the distribution of AI-generated content evolves, judges become stale. No systematic study of judge staleness, the rate at which calibration degrades as the generator distribution shifts, exists.

JUDGE STALENESS IS AN INEVITABLE CONSEQUENCE OF TRAINING ON A FIXED SNAPSHOT, AND MONITORING PROTOCOLS ARE NEEDED BEFORE DEPLOYMENT, NOT AFTER FAILURE. [PRELIMINARY: proposed mechanism] *A judge trained on 2023 model outputs is poorly calibrated to evaluate 2026 model outputs because the quality distribution has shifted: what was "excellent" in 2023 is "average" in 2026, and the judge's quality calibration has not been updated.* Canary-set monitoring, maintaining a small set of known-quality pairs and periodically checking the judge's verdicts against them, is the minimum viable staleness detection mechanism. When canary-set accuracy degrades beyond a threshold, the judge should be retrained or replaced. This monitoring protocol should be established at deployment time, not after deployment failures are observed.

**Open problems.**

- **Drift detection**: Canary-set monitoring protocols and acceptable drift thresholds.
- **Continual training without forgetting**: EWC, LoRA-based module updates, memory replay for judges.
- **Living judge benchmarks**: Continuously updated evaluation analogous to Chatbot Arena.
- **Transfer of judge expertise**: Efficient adaptation to new evaluation domains.

### L.11 Direction 8: Standardized Meta-Evaluation Protocols

**Causal gap.** The LLM-as-a-Judge field lacks standardized reporting protocols. Papers report different metrics without justification, accuracy without positional controls, and single-point estimates without confidence intervals. This makes cross-paper comparison impossible.

No existing benchmark spans the dimensions that matter together: position-consistent accuracy, human correlation, domain coverage, adversarial robustness, and instruction following rate. Assessing a judge properly therefore means assembling several benchmarks by hand, and the empty cells of Table 10 are what that fragmentation looks like in practice. Long-form output is the weakest covered setting of all: Chen et al. (2026) find judge reliability degrades sharply as outputs lengthen, which is precisely the regime where automated evaluation is most needed because human annotation is most expensive.

**A proposed minimum reporting standard.** Based on the discussions in Appendix I and H:

1. **PC accuracy** alongside naive accuracy for all pairwise settings.
2. **IFR** to quantify silent evaluation failures.
3. **Bias audit**: PC, VS, and SPS where applicable.
4. **Bootstrap confidence intervals** ($n = 1000$, $\alpha = 0.05$).
5. **Domain stratification**: Math, code, general, safety separately.
6. **Human validation sample**: $\geq$50 examples for ecological validity.

Table 31 restates this as a deployable checklist: a minimal, low-cost set of metrics that any SLM-judge paper can report to make results directly comparable.

STANDARDIZED REPORTING WOULD IMMEDIATELY INCREASE THE FIELD'S COLLECTIVE KNOWLEDGE BY MAKING RESULTS COMPARABLE ACROSS PAPERS. [PRELIMINARY: analytical argument] *Currently, "85% agreement" in one paper and "78% accuracy" in another are incommensurable because the metrics measure different properties under different conditions; adopting a common reporting standard would make these results directly comparable, enabling the meta-analytic synthesis that the field needs to identify which training strategies, architectures, and deployment configurations actually matter.* The minimum reporting standard above is deliberately conservative: it requires only metrics that are cheap to compute and adds no computational overhead to the evaluation itself. The primary barrier to adoption is coordination, not cost.

| Item | Cost | Guards against |
|---|---|---|
| PC-Acc + naive acc. | 2$\times$ inf. | Positional heuristics |
| IFR | Free | Silent parse failures |
| Bias audit (VS, SPS) | Low | Surface-cue exploitation |
| Bootstrap 95% CI | Free | Over-claiming on noise |
| Domain stratification | Free | Hidden domain gaps |
| $\geq$50 human checks | Low | Lost ecological validity |

Table 31: Proposed minimum reporting standard for SLM-judge evaluation. All items are cheap to compute and add no overhead to the evaluation itself.

### L.12 Direction 9: Training Methodology for SLM Judges

**Causal gap.** The optimal training methodology for SLM judges at constrained parameter budgets is not established. Current practice includes SFT on evaluation data, DPO/preference optimization, and RL with verifiable rewards. Each has been studied in isolation, but no systematic comparison at the same model size under the same data budget exists.

**Open problems.**

1. **Pretraining selection vs. fine-tuning**: Does the base model matter more than the fine-tuning objective? The M-Prometheus result (Qwen2.5 $>$ Llama for multilingual) suggests pretraining composition is critical.
2. **Reward shaping for consistency**: The J1 protocol (Whitehouse et al., 2025) uses consistency-based rewards. How does the consistency reward weight interact with accuracy rewards at different model scales?
3. **Data composition**: What is the optimal mixture of direct scoring, comparison, language feedback, and rubric application data at different model sizes?
4. **Ensemble distillation**: Can a heterogeneous LLM panel's consensus be distilled into an

SLM student? Does ensemble distillation produce more robust judges than single-teacher distillation?

5. **Multi-format unified training**: The Multitask-J1 result (combining pairwise and pointwise training improves both) suggests systematic study of multi-format training at small scales.

THE INTERACTION BETWEEN BASE MODEL SELECTION AND FINE-TUNING STRATEGY IS THE MOST UNDER-STUDIED VARIABLE IN SLM JUDGE TRAINING. [PRELIMINARY: analytical argument] *Current papers report results on a single base model with a single fine-tuning strategy, making it impossible to disentangle the contribution of the base model's pretraining composition from the fine-tuning objective's effectiveness; a controlled factorial study varying both base model and fine-tuning strategy at a fixed parameter budget would provide the field's most valuable training methodology insight.* The practical stakes are high: if pretraining composition dominates (as M-Prometheus suggests), then the community should focus on base model selection rather than fine-tuning innovation; if fine-tuning strategy dominates, then RL-based training (J1, JudgeLRM) may be categorically superior regardless of base model.

### L.13 Direction 10: Societal Implications and Deployment Ethics

**Causal gap.** The survey literature has focused predominantly on technical reliability with minimal attention to the societal implications of deploying automated judgment at scale. When SLM judges filter AI-generated content, evaluate AI-assisted work, or provide automated feedback in educational settings, their systematic biases propagate at scale.

**High-stakes deployment concerns.**

- **Homogenization of quality standards**: If all generators are trained against the same SLM judge, they converge toward that judge's preferences, homogenizing output style and potentially encoding particular cultural norms at scale.
- **Protected attribute sensitivity**: Bias measurements must extend to evaluation of responses about different protected groups. An SLM judge that rates responses about minority communities more harshly encodes discriminatory evaluation at scale.
- **Educational deployment**: Using SLM judges to evaluate student work raises accountability questions: what recourse exists when the judge is wrong?
- **Transparency requirements**: Disclosure that evaluation is automated becomes ethically necessary and potentially legally mandated.

MONOCULTURE RISK IS THE MOST DANGEROUS SOCIETAL CONSEQUENCE OF WIDESPREAD SLM JUDGE DEPLOYMENT. [PRELIMINARY: proposed mechanism] *If a single dominant SLM judge architecture is used across multiple RLHF pipelines, all trained generators converge toward that judge's quality standards, creating a monoculture where diversity of output style, reasoning approach, and cultural perspective is systematically suppressed.* This is analogous to monoculture risk in agriculture: a single point of failure (the judge's biases) propagates across the entire ecosystem. The mitigation is deliberate diversity: different RLHF pipelines should use different judges from different model families, ensuring that no single quality standard dominates. This recommendation aligns with the cross-family diversity principle established throughout this survey (Appendix E).

### L.14 Research Agenda Summary

Table 32 provides a consolidated overview of the ten research directions, their causal gaps, time horizons, and highest-leverage interventions.

**Closing note.** The ten directions above share a common thread: SLM-as-a-Judge is not a solved problem reduced to engineering implementation, but a field with deep open theoretical and empirical

| # | Direction | Causal Gap | Hor. | Intervention |
|---|---|---|---|---|
| 1 | Repr.-based eval. | Gen. bottleneck | Near | Probe atlas |
| 2 | Adaptive compute | Fixed budget | Near | Difficulty-pred. |
| 3 | Calib. uncert. | No confidence | Near | Conformal pred. |
| 4 | Scalable ovsght. | Capacity ceiling | Med | W2SG $\times$ verif. |
| 5 | Agentic eval. | Static mismatch | Med | PRM $\rightarrow$ traject. |
| 6 | Multilingual | English-centric | Med | Native + verify |
| 7 | Continual learn. | Judge staleness | Med | Drift monitoring |
| 8 | Std. protocols | Inconsist. rep. | Near | Report standard |
| 9 | Training meth. | No comparison | Near | Factorial study |
| 10 | Societal impact | Bias at scale | Long | Monocult. audit |

Table 32: Extended research agenda: directions, time horizon, and tractability assessment. "Horizon" indicates estimated time to meaningful progress.

questions. The efficiency advantages of SLMs, substantial cost reduction relative to frontier LLM judges, feasibility of private on-device deployment, and reduced latency for real-time evaluation, make this field practically important beyond academic interest. Closing the gaps identified above would make SLM judges not just economically attractive but reliably trustworthy, the prerequisite for high-stakes deployment at scale.

# M Survey Methodology

This section documents the search strategy, inclusion criteria, and corpus statistics for this survey, addressing the reproducibility of our literature selection.

## M.1 Search Strategy

We searched Semantic Scholar, arXiv (cs.CL, cs.AI, cs.LG), ACL Anthology, and Google Scholar between November 2024 and July 2026 using the following query families:

1. **Core judging:** `("LLM-as-a-judge" OR "LLM judge" OR "language model evaluator" OR "automated evaluation")` combined with `("small" OR "7B" OR "8B" OR "3B" OR "13B" OR "14B" OR "efficient")`.
2. **Training for evaluation:** `("reward model" OR "process reward" OR "GRPO" OR "DPO") AND ("judge" OR "evaluator" OR "scoring")`.
3. **Multi-agent evaluation:** `("multi-agent debate" OR "ensemble judge" OR "panel of judges" OR "jury evaluation")`.
4. **Reasoning budget:** `("overthinking" OR "token budget" OR "test-time compute" OR "chain-of-thought efficiency") AND ("evaluation" OR "judge")`.
5. **Bias and robustness:** `("position bias" OR "self-preference" OR "adversarial" OR "prompt sensitivity") AND ("judge" OR "evaluator")`.

We also followed citation chains from three anchor surveys (Gu et al., 2024; Li et al., 2025a, 2024a) and tracked the RewardBench leaderboard (Lambert et al., 2025) for newly submitted judge systems.

## M.2 Inclusion and Exclusion Criteria

**Inclusion.** A paper was included if it met at least one of:

- It proposes, trains, or evaluates an SLM ($\leq$14B parameters) in an evaluative or scoring role.
- It introduces a benchmark, dataset, or metric used to evaluate language-model judges.
- It provides general-purpose methodology (multi-agent debate, ensemble voting, bias mitigation) that has been applied to or is directly applicable to SLM judges.
- It reports results for LLM judges ($>$14B) on benchmarks also used by SLM judge papers, enabling direct comparison. Such systems serve only as reference points and never count toward an insight's evidence grade.

**Exclusion.** We excluded:

- Papers where language models are used purely as data annotators (e.g., labeling NER datasets) rather than as quality judges.
- Papers focused exclusively on reward modeling for RLHF without evaluation of the model as a standalone judge.
- Non-peer-reviewed blog posts, unless they describe a released system with reproducible results (e.g., model cards on HuggingFace with benchmark numbers).
- Preprints whose central claims could not be verified from the manuscript or released artifacts.

## M.3 Selection Accounting

Screening followed a two-stage procedure for literature surveys. Records were first screened on title and abstract against the criteria above, and the remaining candidates were read in full before a final inclusion decision. Works found through citation chasing went through the same full-text check. Figure 15 summarizes the selection process.

The largest source of exclusion at the full-text stage was scope mismatch: papers about reward modeling for RLHF that did not evaluate the model as a judge, and papers about LLM evaluation that studied only models above 70B without reporting results on smaller systems.

## M.4 Corpus Statistics

We characterize the surveyed literature along four dimensions.

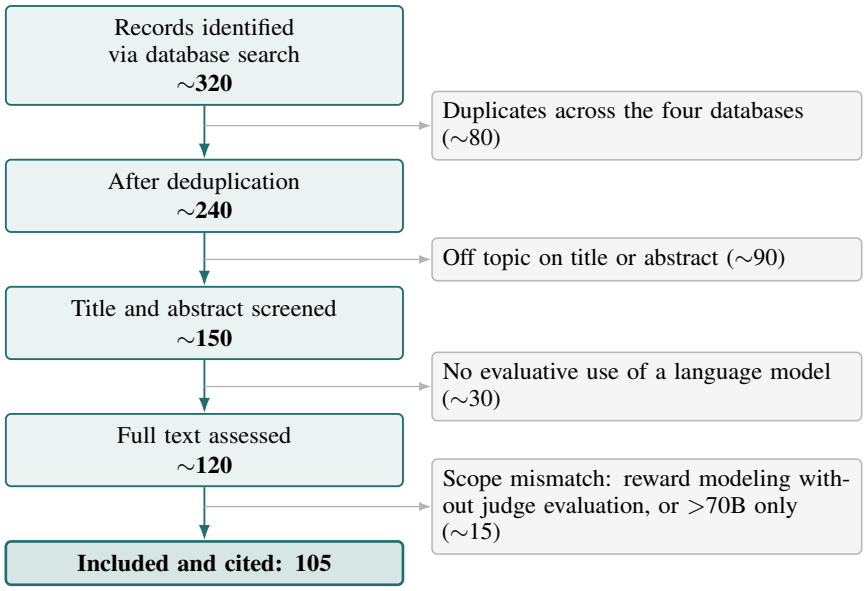



Figure 15: Literature selection flow. Left column gives records retained at each stage; right column gives the dominant exclusion reason. Only the final count is exact and can be checked against the reference list; upstream counts are recorded from the search logs and are approximate, since the same work was often retrieved by several query families.

**By training paradigm.** Table 9 groups the 19 judge systems into five paradigms: off-the-shelf prompting (1), supervised fine-tuning from distillation, human labels, or rubrics (8), self-supervised or preference optimization (1), reinforcement learning with verifiable rewards (7), and representation-based probing (2). Table 11 decomposes the same space more finely, separating the three supervised variants and listing preference optimization on its own. The RL systems all come from 2025–2026 work while the supervised systems cluster in 2023–2024, which is the decreasing-supervision gradient developed in Appendix C.

**By base model family.** Qwen backbones (Qwen2.5, Qwen 3) support 8 of the 19 systems and LLaMA variants support 5, making these the two dominant bases. Mistral, Phi-3.5-mini, Vicuna, PaLM 2, and R1-Distill each support one. One system, J1, reports on both Qwen and LLaMA backbones and is counted under each, so the family totals exceed 19; the two representation-based systems probe a frozen model without committing to a family. The concentration on Qwen and LLaMA likely reflects permissive licensing and strong instruction-following baselines rather than any architectural advantage for judging.

**By evaluation domain.** Nine of the 19 systems report on general-purpose or broad-coverage evaluation. Five target reasoning or verifiable tasks, and the remaining five each target a single setting: rubric-conditioned assessment, multilingual evaluation, mathematical process supervision, structured tasks, and fine-grained scoring. No system in the table is dedicated to safety, creative, or medical evaluation; those domains enter the survey only through benchmark and analysis papers (Appendix J). The imbalance is itself a finding: evidence is dense where answers can be checked and thin everywhere else (Insight I5).

**By publication year.** Of the 105 cited works, 21 are from 2023 or earlier, 26 from 2024, 44 from 2025, and 14 from 2026. The subfield is young: every judge system in Table 9 carries a bibliography date of 2024 or later, and the two earliest of them, PandaLM and JudgeLM, first circulated as 2023 preprints.

## N Per-Insight Evidence Tables

This section collects the quantitative evidence behind each of the survey's five insights. Each table aggregates every relevant comparison from the surveyed literature, with evidence-level grading (con-

sensus, replicated, or preliminary) applied to each row.

### N.1 Evidence Summary for Insight I1

Table 33 collects every case in the surveyed literature where a trained judge at $\leq$14B was compared against a larger model on a named benchmark. Eight papers report a trained judge next to its own base model under one protocol, and training improves the base model in every one of them, by 15% to 63% in relative terms. Beyond that block, the in-scope systems that reach or pass a larger baseline form five independent lines of evidence, and the table as a whole spans three training paradigms (SFT, DPO, GRPO) and four base families (LLaMA, Phi, Qwen, Mistral). No single system, recipe, or benchmark drives the finding.

In the lower blocks, the judges were fine-tuned for evaluation while the baselines were prompted. That is the nature of I1's claim, which concerns what training achieves at small scale rather than inherent capacity. The upper block removes the asymmetry: each paper compares its judge with the very model it was trained from, under the same protocol, so the gain isolates the training variable and cannot be attributed to scale, architecture, or pretraining data.

The benchmarks also differ in what they reward. RewardBench tests preference selection, FLASK measures fine-grained scoring correlation, and PandaLM uses human-annotated pairwise labels. A system strong on one need not transfer, which is why the counter-evidence block matters: Prometheus 2 sits below GPT-4o-mini on FLASK even though trained judges of similar size clear the same baseline on RewardBench. The crossover is a property of the task as much as of the training.

What the table shows, taken as a whole: the crossover between trained SLMs and larger zero-shot judges is not a fluke of one benchmark or one model family. It holds on RewardBench (Selene Mini, J1), on FLASK (GLIDER), on human-annotated data (JudgeLRM on PandaLM), and on step-level verification (GenPRM on ProcessBench). The margin shrinks on harder benchmarks: under JudgeBench's original protocol, J1-8B reaches 42% position-consistent accuracy and no SLM reaches Claude-3.5-Sonnet's 64.3%. The claim is not that small judges are universally superior, but that evaluation-specific training closes most of the gap on most benchmarks, and inverts the ranking on some.

### N.2 Evidence Summary for Insight I2

Table 34 collects findings on how reasoning budget (token count allocated to chain-of-thought) affects judge and verifier accuracy.

The collective picture is non-monotonic. Giving a judge room to reason helps when the model can finish its chain of thought, and the $\sim$10% gain from thinking mode in small Qwen-3 judges (Jayarao et al., 2025) is real. But past the point where reasoning completes, extra tokens add noise rather than signal. Ma et al. (2025) showed this directly: at low token budgets, skipping the thinking trace entirely beats truncated reasoning by a wide margin. Chen et al. (2025) found the same from the training side: an RL reward that incentivizes longer outputs costs 3% F1, because the model pads rather than reasons.

For small judges, this means setting the budget per task. Thinking mode lifts 0.6–4B judges by about 10 points at 1.3–2.9$\times$ the compute (Jayarao et al., 2025), which pays off on reasoning-heavy comparisons, while easy comparisons are settled before any reasoning is needed. The default for deployed SLM judges is therefore a direct verdict on routine pairs and a bounded thinking budget on hard ones.

### N.3 Evidence Summary for Insight I3

The evidence for I3 is real but thinner than for I1 or I2. PoLL (Verga et al., 2024) is the strongest result: a panel of cheap, diverse models beats a single expensive judge on human agreement at 7$\times$ lower cost. COSMosFL (Cho et al., 2025) reproduces the pattern in code fault localization and JudgeBoard (Bi et al., 2026) on mathematics, both with open-weight panels.

The mechanism matches what the Condorcet argument needs. Distinct pretraining corpora and architectures produce weakly correlated errors, so majority voting cancels them. Every result above is consistent with it, and a matched comparison of same-family and cross-family panels would put a precise number on the effect.

### N.4 Evidence Summary for Insight I4

Table 36 sorts the debate literature by how much information the agents actually hold apart from one another. The ordering is the finding: every study reporting a gain runs agents that see different evidence, hold different roles, or call different tools, and every study reporting a loss runs agents on shared context. Round count separates none of

| Judge system | Size | Benchmark | Result | Baseline | Baseline | Control |
|---|---|---|---|---|---|---|
| ***Same backbone, same protocol: trained judge vs. its own base model*** | | | | | | |
| Selene Mini | 8B | 11-benchmark average | 75.3 | LLaMA-3.1-8B-Instruct | 65.3 | Same backbone (+15%) |
| GLIDER | 3.8B | Pearson, 7 human-rated sets | 0.534 | Phi-3.5-mini-instruct | 0.328 | Same backbone (+63%) |
| JudgeLRM | 3B | PandaLM F1 | 72.1 | Qwen2.5-3B | 51.6 | Same backbone (+40%) |
| JudgeLRM | 7B | PandaLM F1 | 75.1 | Qwen2.5-7B | 59.8 | Same backbone (+26%) |
| J1 | 8B | 5-benchmark overall | 61.9 | LLaMA-3.1-8B | 48.3 | Same backbone (+28%) |
| Prometheus 2 | 7B | Direct-assessment Pearson | 62.4 | Mistral-7B | 41.7 | Same backbone (+50%) |
| CompassJudger-2 | 7B | 4-benchmark average | 72.1 | Qwen2.5-7B | 57.3 | Same backbone (+26%) |
| J4R | 7B | Overall | 46.0 | Qwen2.5-7B | 35.9 | Same backbone (+28%) |
| TIR-Judge | 8B | PPE average | 75.7 | Qwen3-8B | 65.5 | Same backbone (+16%) |
| TIR-Judge | 4B | PPE average | 76.3 | Qwen3-4B | 60.4 | Same backbone (+26%) |
| ***Trained judge at or above a larger or proprietary baseline*** | | | | | | |
| Selene Mini | 8B | RewardBench | 0.891 | GPT-4o-mini | 0.801 | Standard benchmark |
| Selene Mini | 8B | EvalBiasBench | Beats GPT-4o | GPT-4o | n/a | Out-of-distribution suite |
| GLIDER | 3.8B | FLASK | $\approx$0.615 | GPT-4o-mini | 0.565 | Outside training distribution |
| JudgeLRM | 3B | PandaLM | Exceeds | GPT-4 | n/a | Human-annotated labels |
| JudgeLRM | 7B | PandaLM | +2.6 F1 | DeepSeek-R1 | n/a | Human-annotated labels |
| J1 (Pairwise) | 8B | RewardBench | 85.7 | Llama-3.3-70B (prompted) | 85.4 | Same paper, 9$\times$ larger baseline |
| J4R | 7B | Overall‡ | +6.7% | GPT-4o | n/a | Standard benchmarks |
| GenPRM | 7B | ProcessBench | Exceeds | Qwen2.5-72B PRM | n/a | Test-time scaling |
| FLAMe-RM | 24B* | RewardBench | 0.878 | GPT-4-0125 | 0.859 | Reference point only |
| ***Counter-evidence and weakly controlled comparisons*** | | | | | | |
| Prometheus 2 | 7B | FLASK | 0.470 | GPT-4o-mini | 0.615 | Same evaluation run |
| JudgeLM | 7B | JudgeLM-test | >90% agree | GPT-4, its own teacher | n/a | None; measures imitation |
| INSPECTOR | 1.7B | GSM8K, MATH, GPQA | > prompted SLM | Same model, prompted | n/a | No larger-model baseline |

*FLAMe-RM-24B exceeds the $\leq$14B scope; listed as the largest open reference point and not counted toward the grade. ‡J4R's overall score across PPE, JudgeBench, and ReasoningJudgeBench. Each paper in the upper block reports its scores on its own scale, so the relative gain in brackets is the cleanest way to compare them. The two GPT-4o-mini FLASK baselines differ (0.565 and 0.615) because each row quotes the run its own source performed.

Table 33: Evidence for Insight I1, reported exactly as stated by each source. The upper block holds the backbone fixed: each of the eight papers reports its trained judge next to its own base model under one protocol, and training improves the base model every time. The **Control** column records what each other comparison holds fixed. Aggregate evidence level: CONSENSUS, on five independent lines of evidence (Selene Mini and J1, which share a backbone and benchmark and count once; GLIDER; JudgeLRM; J4R; GenPRM) spanning three training paradigms and four base families (LLaMA, Phi, Qwen, Mistral). The lower block is listed rather than omitted: Prometheus 2 falls below the baseline on fine-grained rubric correlation, and the JudgeLM and INSPECTOR comparisons do not involve a larger model.

them.

The practical consequence for SLM panels is narrow. Debate is worth its cost when agents can be given genuinely different inputs, whether that means different retrieved evidence, an execution trace, or an assigned adversarial role. On shared context it spends several times the tokens of a single judge to arrive at a worse verdict, and the mechanism Cemri et al. (2026) document, agents converging on an early confident answer, applies whenever one voice dominates the shared context.

### N.5 Evidence Summary for Insight I5

The gradient is clear in the data: SLM judges perform well where answers can be checked (math, code, factual QA) and poorly where they cannot (creative writing, medical, safety). This pattern is not unique to SLMs, since LLM judges show the same trend, but the drop-off is sharper for smaller models because they have less world knowledge to compensate when verification is impossible.

The practical takeaway is that deployment recommendations must be domain-specific. An SLM judge can replace a frontier model for math grading or code correctness checking with minimal loss. It cannot replace a human for medical or legal evaluation. The middle ground of safety, summarization, and multilingual evaluation is where the cost-quality tradeoff is most interesting, and where ensemble methods (I3) or cascade architectures (C1) add the most value.

## O Figure Data Provenance

This section documents where each plotted element in Figure 2 comes from. For panels (a) and (c) we list the reported numbers, and for panels (b) and (d) the studies and the outcome each one reports.

The four panels in Figure 2 synthesize trends from the surveyed literature. Panels (a) and (c) plot reported numbers, and panels (b) and (d) place

| Source | Model size | Setting | Key finding | Evid. |
|---|---|---|---|---|
| Jayarao et al. (2025) | 0.6–4B | Thinking vs. non-thinking mode | ∼10% pairwise accuracy gain with thinking | R |
| Ma et al. (2025) | 1.5–32B | NoThinking vs. Thinking | NoThinking wins at low budgets (51.3 vs. 28.9% on ACM) | R |
| Srivastava et al. (2025b) | Multiple | Budget sweep | Non-monotonic: sweet spot at intermediate lengths | R |
| Shojaee et al. (2026) | Frontier | Problem complexity | Accuracy collapse beyond complexity threshold (∼36% drop) | R |
| Aggarwal et al. (2025) | Multiple | Overthinking + underthinking | OptimalThinkingBench: both failure modes documented | P |
| Chen et al. (2025) | 3–14B | Length reward in RL | Incentivizing longer output degrades F1 by ∼3% | P |
| Ghosal et al. (2026) | Multiple | Test-time scaling | Mirage of scaling: extra compute does not always help | R |

Table 34: Evidence for Insight I2: reasoning budget effects. **Setting** names what was varied and **Key finding** the reported relationship between budget and accuracy. The **Evid.** column grades each row on the three levels of §3: C = consensus (four or more independent sources), R = replicated (two or three sources, or one study with strong controls), P = preliminary (a single study or an untested mechanism). Aggregate evidence level: CONSENSUS, on seven independent studies that jointly establish the non-monotone relationship between reasoning budget and judge accuracy, spanning prompted, trained, and frontier models; individual rows report a gain, a decline, or an explicit sweet spot depending on the budget regime each one probed.

| Source | Panel composition | Key finding | Cost | All open-weight? |
|---|---|---|---|---|
| PoLL (Verga et al., 2024) | Command R, GPT-3.5, Claude Haiku | Better human correlation than single GPT-4 | 7× cheaper | No; mixed providers |
| COSMosFL (Cho et al., 2025) | SLM ensemble, fault localization | Pareto-optimal on cost and accuracy | Lower compute | Yes |
| JudgeBoard (Bi et al., 2026) | Cross-family SLM panel | Matches standalone LLM judges on MATH | Fraction of frontier | Yes |
| ***Related but distinct: same-model resampling*** | | | | |
| Self-consistency (Wang et al., 2022) | One model, $k$ samples | Majority vote reduces variance | $k\times$ single call | Yes |

Table 35: Evidence for Insight I3: cross-family panels. Aggregate evidence level: REPLICATED, on three independent studies, the weakest of the five insights. Self-consistency is listed separately because it resamples one model and therefore reduces variance without decorrelating systematic bias (Appendix D); it is the baseline that cross-family panels are meant to improve on, not evidence for them.

each study by the outcome it reports, so this appendix documents which source each plotted element traces to. The cross-system comparison in Table 10 and the evidence tables in Appendix N provide the underlying raw numbers.

**Panel (a): Training beats scale (I1).** This panel carries no normalization. The $y$-axis is RewardBench accuracy exactly as reported by each system's source paper, restricted to a single benchmark so that the vertical positions are directly comparable. Systems that do not report a RewardBench number are absent from the panel and appear instead in Table 33. Table 38 lists every plotted marker with its source.

**Panel (b): Reasoning-budget effects (I2).** Each marker is one study from Table 34, placed in the region of the effect it reports for more reasoning: a gain (Jayarao et al., 2025), a non-monotone effect with the best budget in between (Srivastava et al., 2025b; Aggarwal et al., 2025; Ghosal et al., 2026), or a loss (Ma et al., 2025; Chen et al., 2025; Shojaee et al., 2026). Studies that evaluate judges sit in the upper half with filled markers, and problem-solving studies, whose results we transfer to judging, sit in the lower half with open markers. The label next to each marker gives the model size the study reports. Four of the seven studies report several sizes or frontier models only, so size is a label rather than an axis, and the panel draws no size trend.

**Panel (c): Cross-family panel (I3).** Both axes carry numbers reported by Verga et al. (2024). The vertical axis is Cohen's $\kappa$ with human labels on three single-hop QA datasets from KILT (their Table 1). The PoLL panel of Command R, GPT-3.5-Turbo, and Claude Haiku reaches 0.763, 0.906, and 0.867 on NQ, TriviaQA, and HotpotQA, against 0.627, 0.841, and 0.830 for GPT-4. The horizontal axis is cost relative to the panel. At list prices the panel costs \$1.25 per million input tokens and \$4.25 per million output tokens, against \$10 and \$30 for GPT-4 Turbo, which makes it seven to eight times cheaper depending on the mix of input and output tokens, so GPT-4 is drawn across that range. The vertical bar is GPT-4's $\kappa$ on NQ across six prompt variants (0.518 to 0.725, their Table 3), and the dashed line marks $\kappa = 0.8$, a common threshold for strong agreement. The panel members are not plotted, because PoLL reports their agreement but not their individual cost. Panel studies built from open SLMs, JudgeBoard and COSMosFL, are listed in Appendix N.3. No surveyed study

| Source | Setting | Information asymmetry | Outcome | Key finding |
|---|---|---|---|---|
| Lin et al. (2025) | Safety evaluation, adversarial roles | High | + | Adversarial roles surface harms a single pass misses |
| Liang et al. (2024) | Open-ended evaluation, role-diverse | Medium | + | Divergent roles counter premature convergence |
| Du et al. (2024) | Factuality, distinct evidence fragments | Low but non-zero | + | Debate improves factual accuracy |
| Wynn et al. (2025) | Binary correctness, homogeneous agents | None | − | Debate underperforms the single-agent baseline |
| Zhang et al. (2025a) | General judging, homogeneous agents | None | − | Gains vanish under iso-compute baselines |
| Cemri et al. (2026) | 1,600+ traces, 7 frameworks | Varied | ± | 36.9% of failures are inter-agent misalignment |

Table 36: Evidence for Insight I4: debate outcomes ordered by information asymmetry. Outcome is positive (+) when debate beats a single-agent baseline, negative (−) when it loses, and mixed (±) for the failure-mode analysis. Aggregate evidence level: CONSENSUS, on five independent outcome studies that split cleanly along the asymmetry axis, with Cemri et al. (2026) supplying the failure-mode mechanism. Zhang et al. (2025a) additionally show that part of the reported MAD advantage is a baseline artifact, since debate consumes several times the tokens of the single-agent comparison.

| Domain | Verifiability | System(s) | SLM performance | Key gap | Evid. |
|---|---|---|---|---|---|
| Math (exact answer) | Fully | JudgeLRM, GenPRM, StepWiser | Match/exceed LLMs | Format matching | C |
| Code (executable) | Fully | TIR-Judge, COSMosFL | Strong with tool use | Non-functional req. | R |
| Factual QA | Mostly | JudgeLM, Auto-J, Selene Mini | Good agreement | Hallucination det. | C |
| Safety | Partially | FLAMe | Moderate | Harm is context-dependent | R |
| Multilingual | Partially | M-Prometheus | SOTA among open judges | Translation artifacts | P |
| Summarization | Subjective | Prometheus 2 | Moderate correlations | Rubric dependence | R |
| Creative writing | Subjective | Wei et al. (2025) | Principle-guided judge reward works | No ground truth | P |
| Medical | Subjective | Selene Mini (CRAFT-MD) | Gains over its base model | Domain expertise | P |

Table 37: Evidence for Insight I5: performance across the verifiability gradient. **Verifiability** uses the four levels defined in Appendix J: *Fully* (a deterministic oracle decides correctness), *Mostly* (strong structural constraints with a subjective residual), *Partially* (domain knowledge constrains but judgement remains), and *Subjective* (no external oracle). Rows are ordered by that level and match the ordering plotted in Figure 2(d). Systems are grouped by the domain where they report results. The **Evid.** column grades each row as in Table 34: C = consensus, R = replicated, P = preliminary. Aggregate evidence level: CONSENSUS, across the eight domains listed, which cover all four verifiability levels and agree on the direction of the gradient even where individual rows rest on a single study.

compares a same-family panel with a cross-family one under one protocol, which is the open question recorded there.

**Panel (d): Verifiability gradient (I5).** Each marker is one system from Table 37, placed in its domain's column and at the quality level the surveyed studies report against an LLM judge: near parity for math and code, good agreement for factual QA, moderate for safety, multilingual evaluation, and summarization, and a wide gap for creative writing and medical evaluation. The levels are categories because the eight domains are not measured on a common metric. Columns follow the four verifiability levels of Appendix J. Medical sits at the far end rather than creative writing because clinical correctness is contextual and no deterministic oracle can adjudicate it, so the judge is left relying on domain knowledge that a small model does not have. The open marker is FLAMe, which exceeds the 14B scope. The sources behind each position are:

- **Math**: SLM judges match LLM judges on answer verification (Chen et al., 2025; Bi et al., 2026).
- **Code**: strong once execution is delegated to a tool (Jiang et al., 2025; Xu et al., 2026).
- **Factual QA**: good agreement, with hallucination detection the residual gap (Zhu et al., 2025; Alexandru et al., 2025).
- **Safety**: standardized axes exist (Lin et al., 2022; Wang et al., 2023a), but harm judgements stay context-dependent, which is what places safety in the partially verifiable band.
- **Multilingual**: M-Prometheus leads open judges while coverage remains thin (Pombal et al., 2025).
- **Summarization**: moderate correlation that depends heavily on rubric design (Kim et al., 2024b).
- **Creative writing**: no ground truth, although a principle-guided judge reward still improves a 7B writer (Wei et al., 2025).

| Marker | Size | RewardBench | Training | Source |
|---|---|---|---|---|
| GLIDER | 3.8B | 0.784 | SFT | Deshpande et al. (2024) |
| Prometheus 2 | 7B | 0.720 | SFT | Kim et al. (2024b) |
| J1 (Pairwise) | 8B | 0.857 | RL (GRPO) | Whitehouse et al. (2025) |
| Selene Mini | 8B | 0.891 | SFT+DPO | Alexandru et al. (2025) |
| FLAMe-RM | 24B* | 0.878 | SFT | Vu et al. (2024) |
| LLaMA-3.1-8B-Instruct | 8B | 0.750 | prompted | Alexandru et al. (2025) |
| GPT-4-0125 (reference line) | n/a | 0.859 | prompted | Vu et al. (2024) |
| GPT-4o-mini (reference line) | n/a | 0.801 | prompted | Alexandru et al. (2025) |

*FLAMe-RM exceeds the $\leq$14B scope and is plotted as the largest available open reference point. All values are reported RewardBench accuracies under each source's own protocol; the axis applies no rescaling. Systems without a published RewardBench number (INSPECTOR, JudgeLRM, JudgeLM, J4R, GenPRM) are not plotted and appear in Table 33 instead.

Table 38: Provenance for Figure 2(a). Every marker is a published RewardBench accuracy, so the vertical axis is directly readable and no normalization is involved. The annotated pair at 8B is Selene Mini against LLaMA-3.1-8B-Instruct: same base model, same parameter count, differing only in whether evaluation-specific training was applied.

- **Medical**: Selene Mini improves over its own base model on the clinical benchmark CRAFT-MD yet stays well short of expert agreement (Alexandru et al., 2025).

The levels summarize the sources listed rather than a single reported metric, and only their order should be read from the panel.